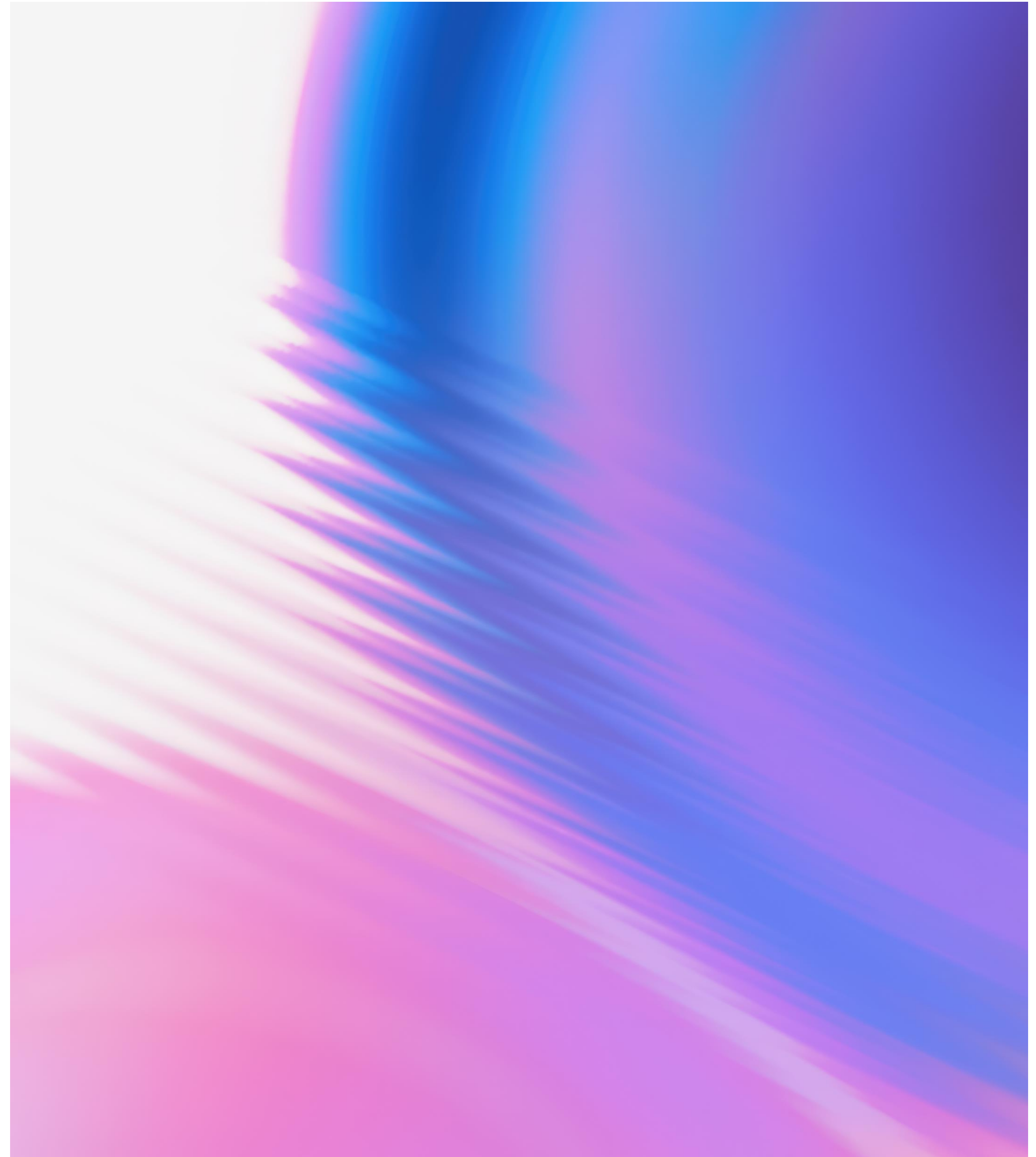
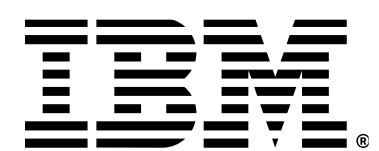


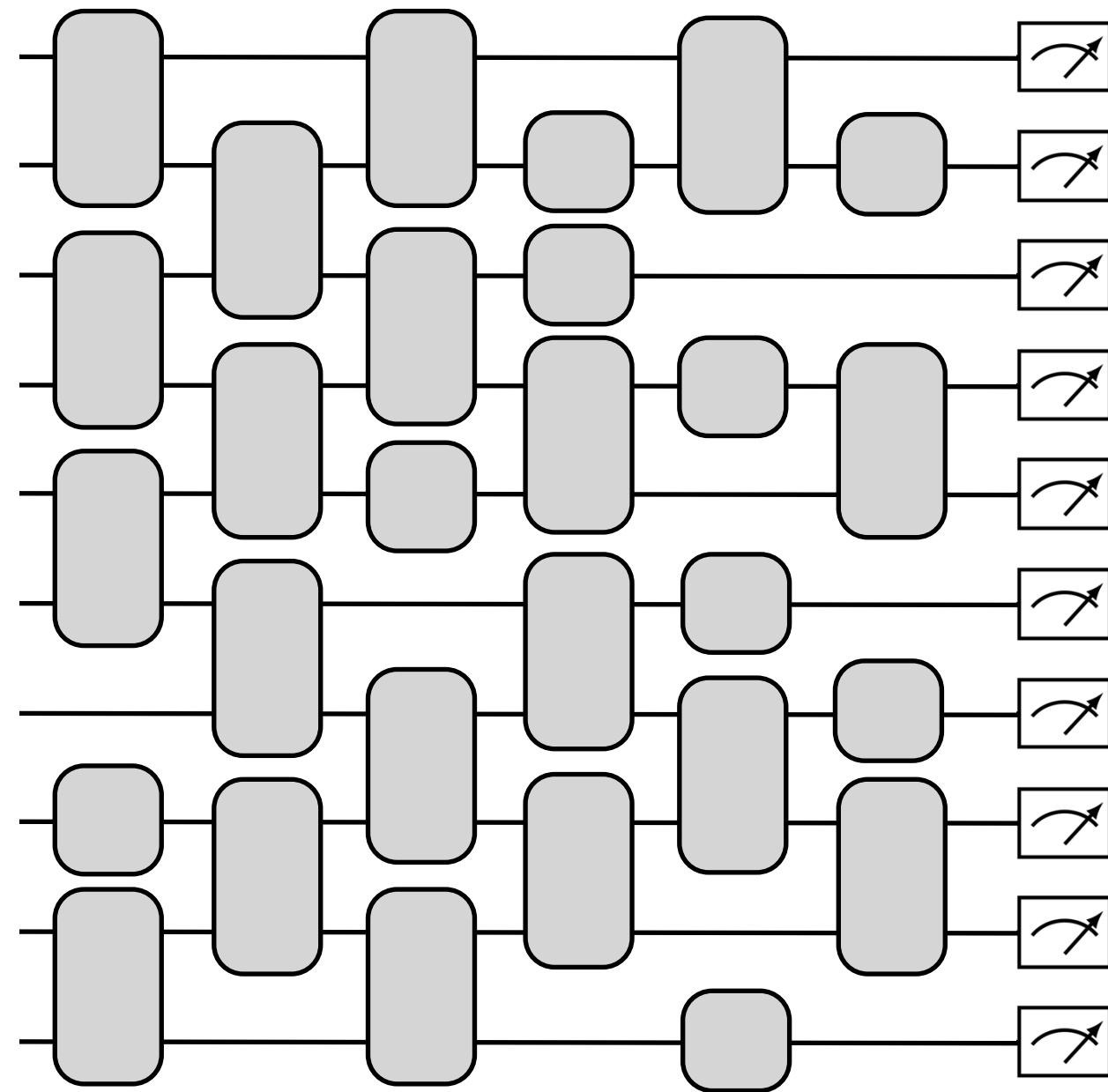
Utility scale dynamic circuits

Mirko Amico
Technical Lead – Quantum Algorithm Engineering
IBM Quantum

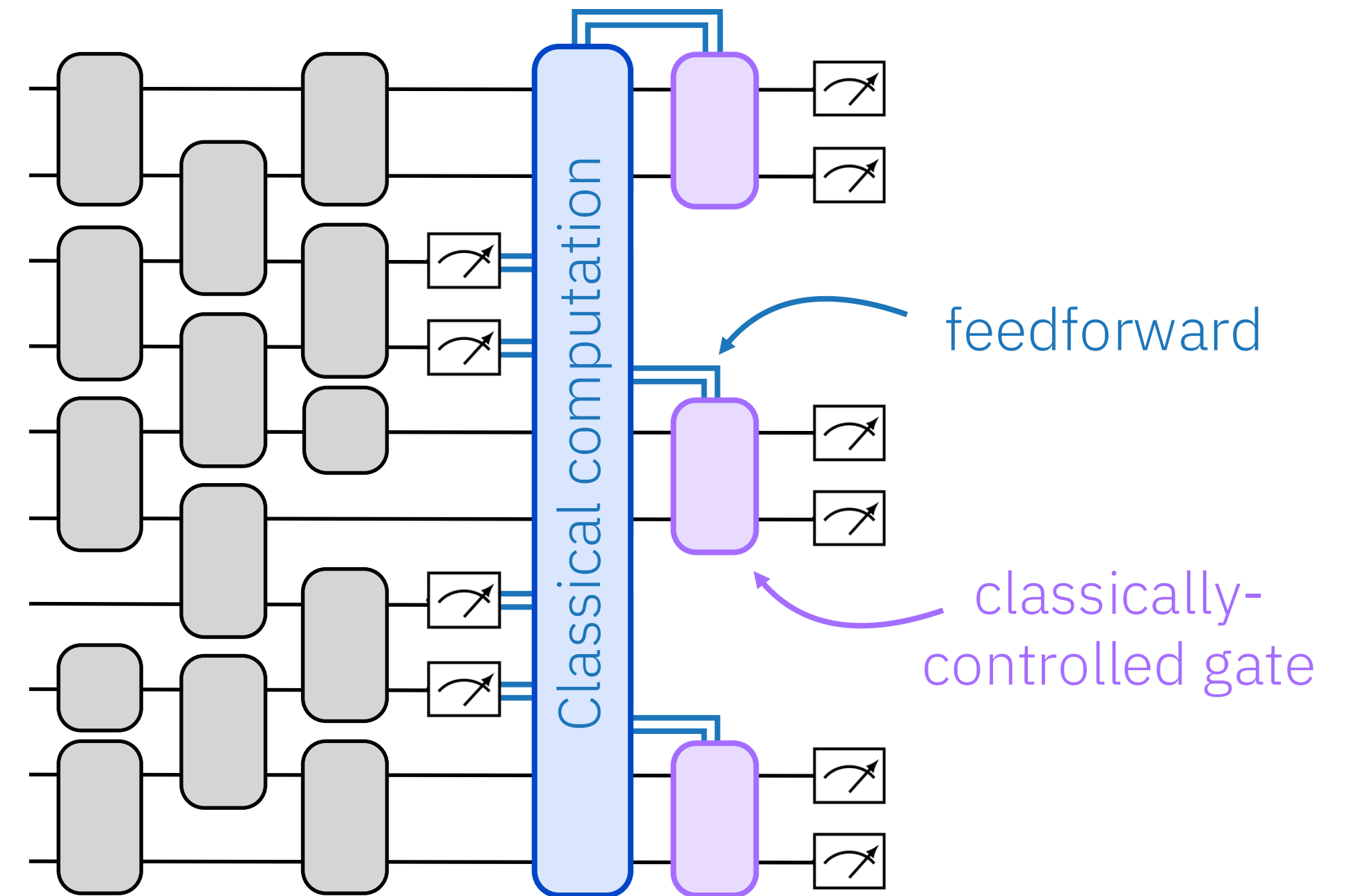


What is dynamic circuit?

Unitary quantum circuit

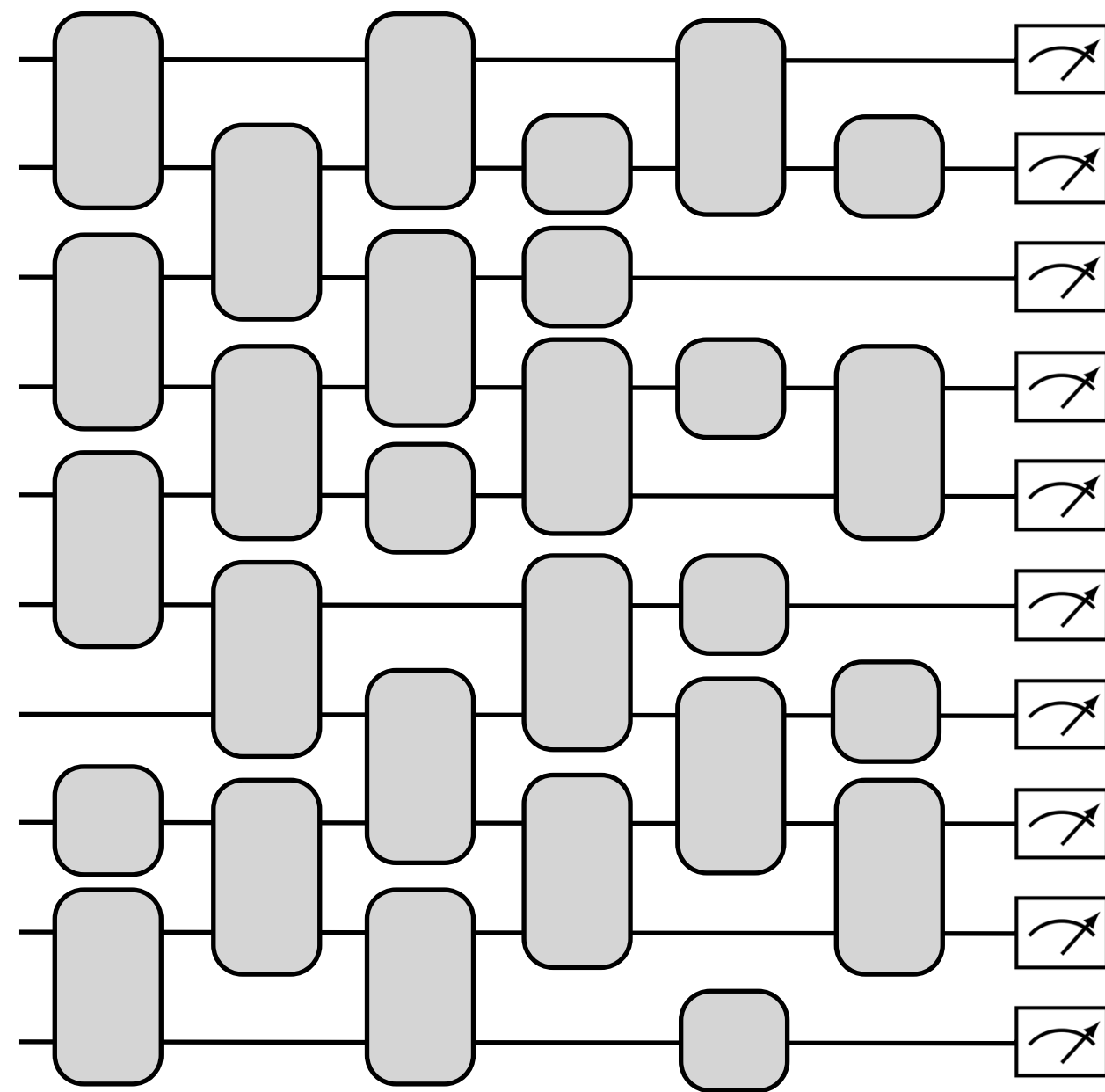


Dynamic quantum circuit

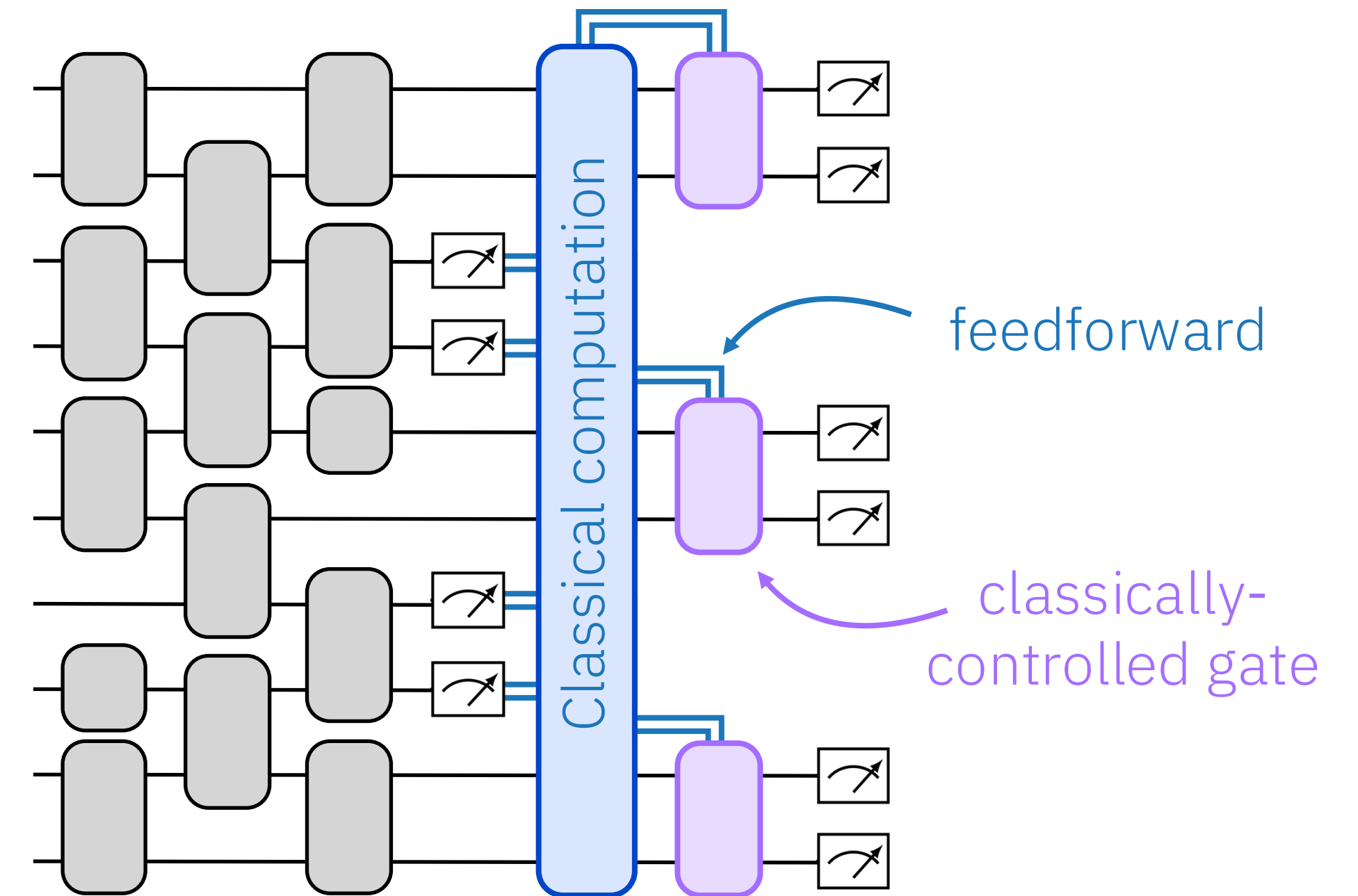


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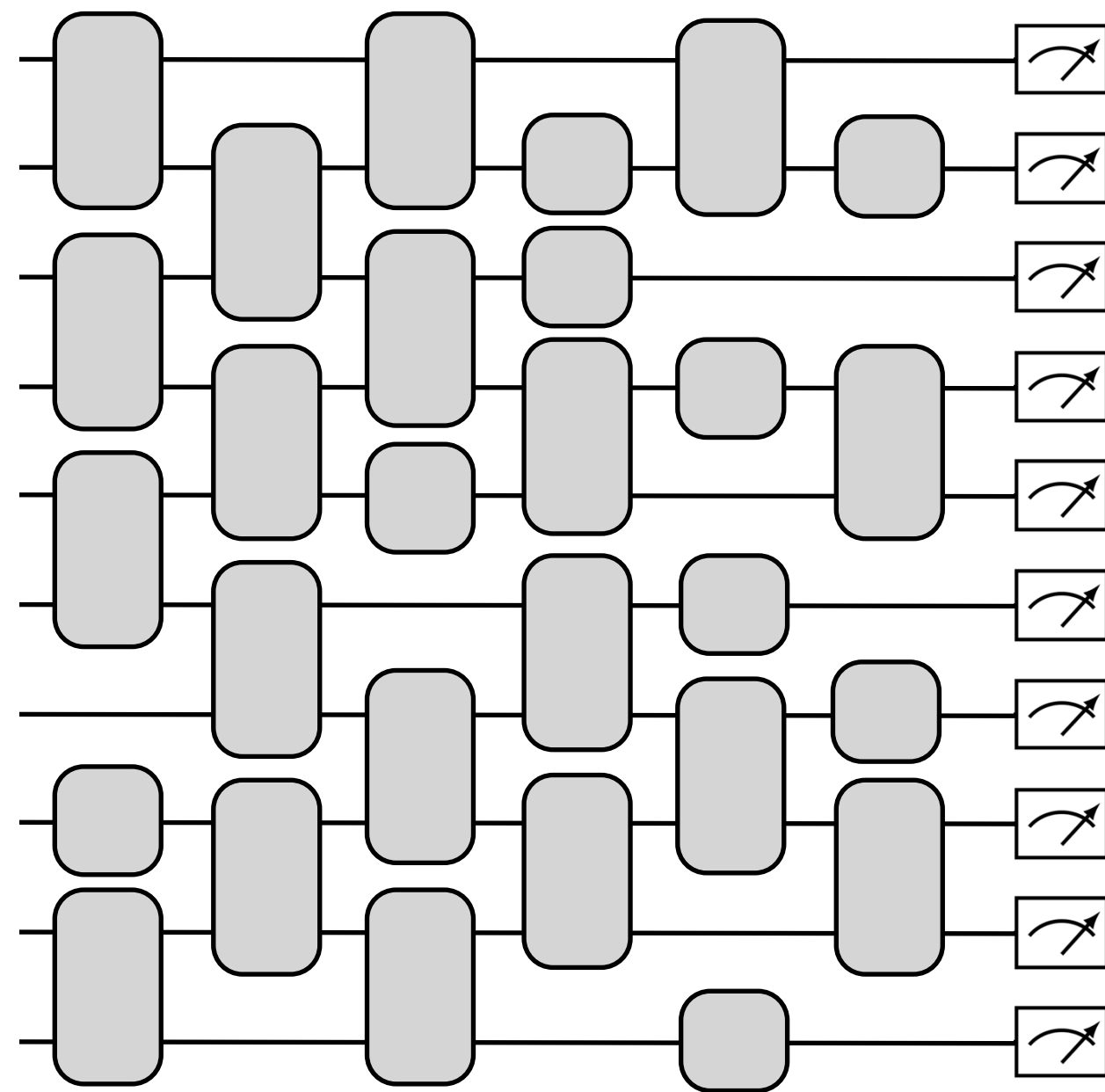
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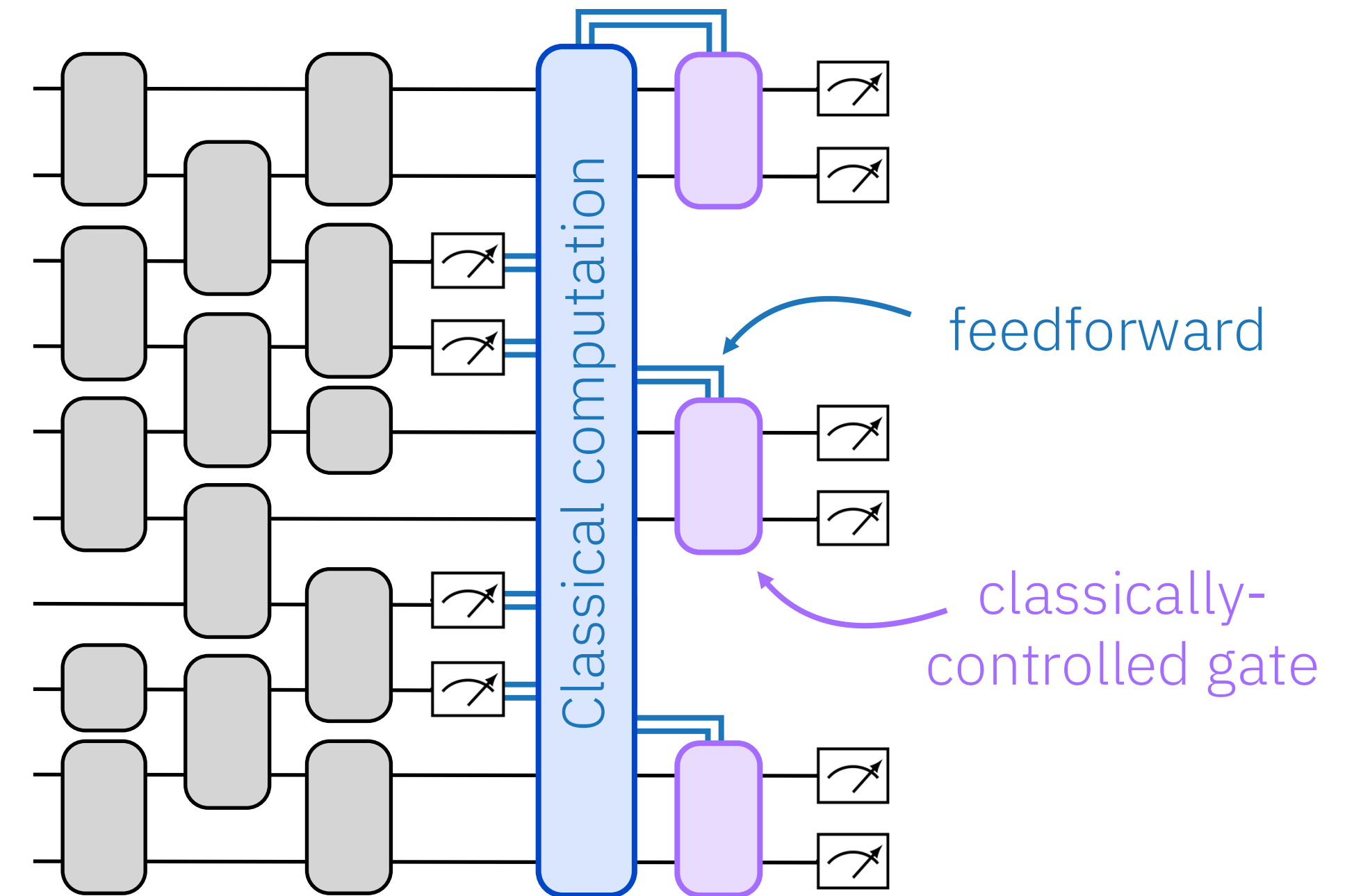
Dynamic circuits → can implement *any* allowed quantum process, not just unitary evolution.

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Dynamic quantum circuit

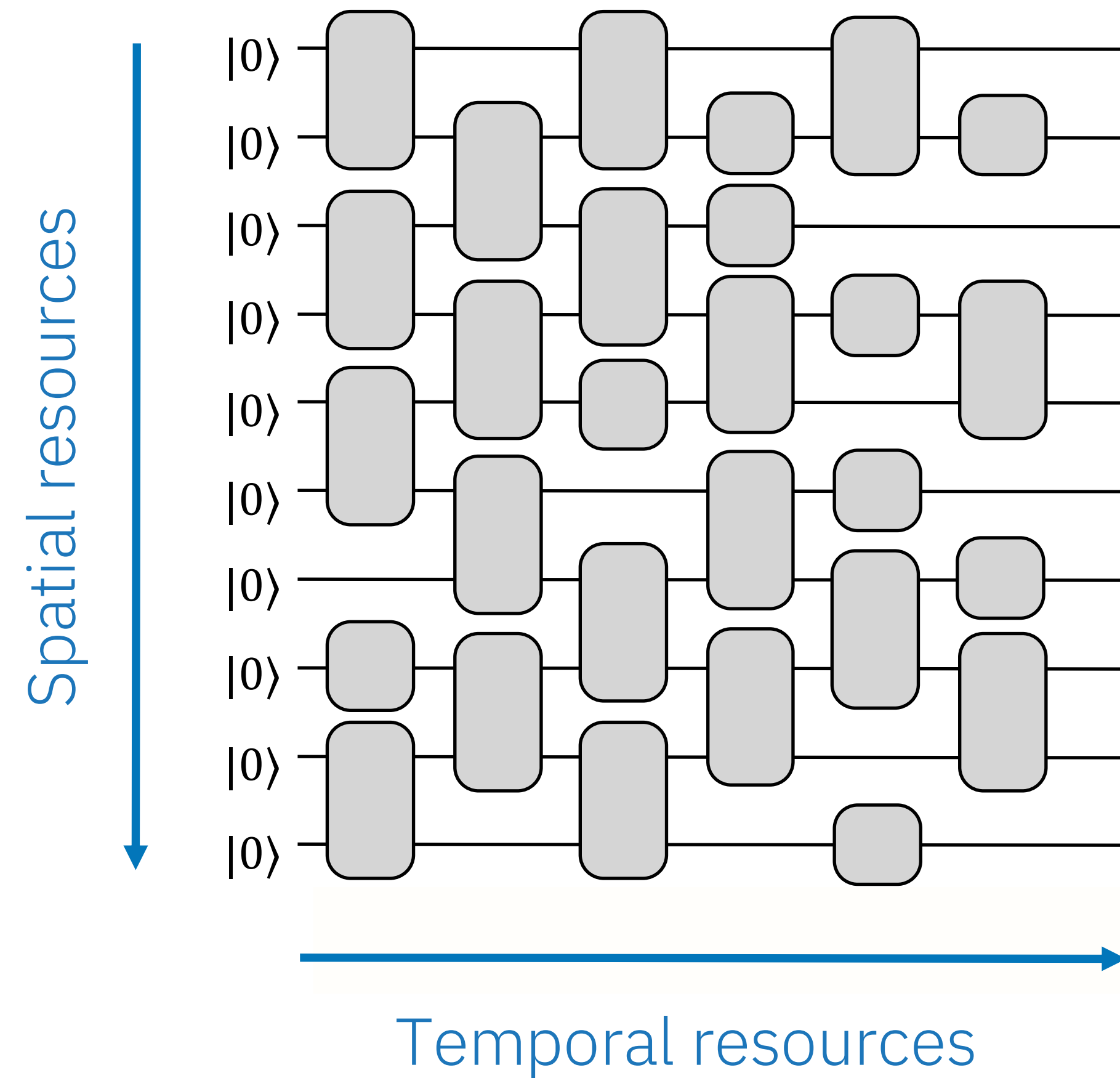


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Measurements and feedforward → new quantum algorithms?

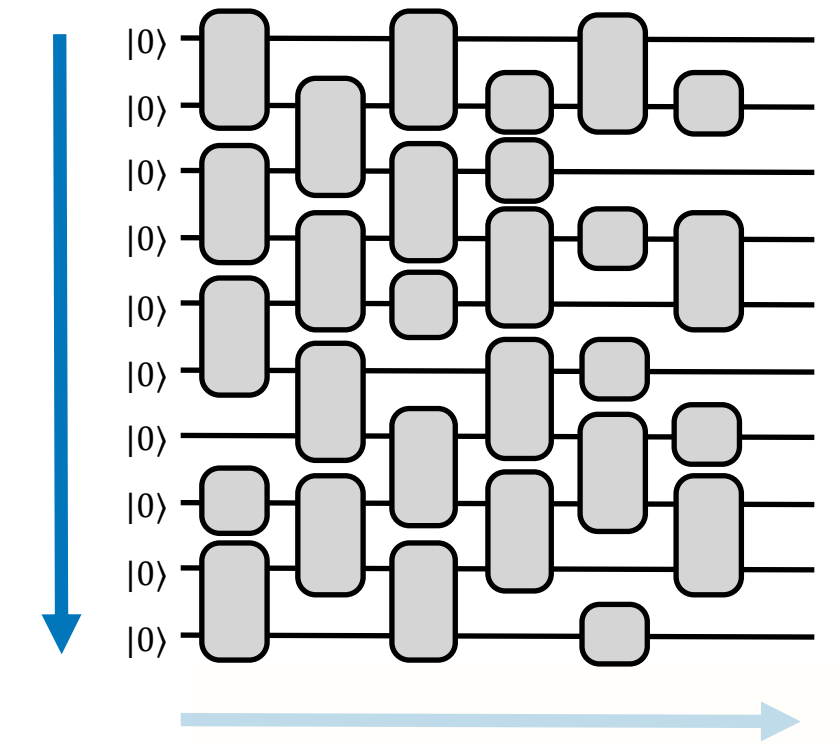
What makes dynamic circuits powerful?

A spacetime view:



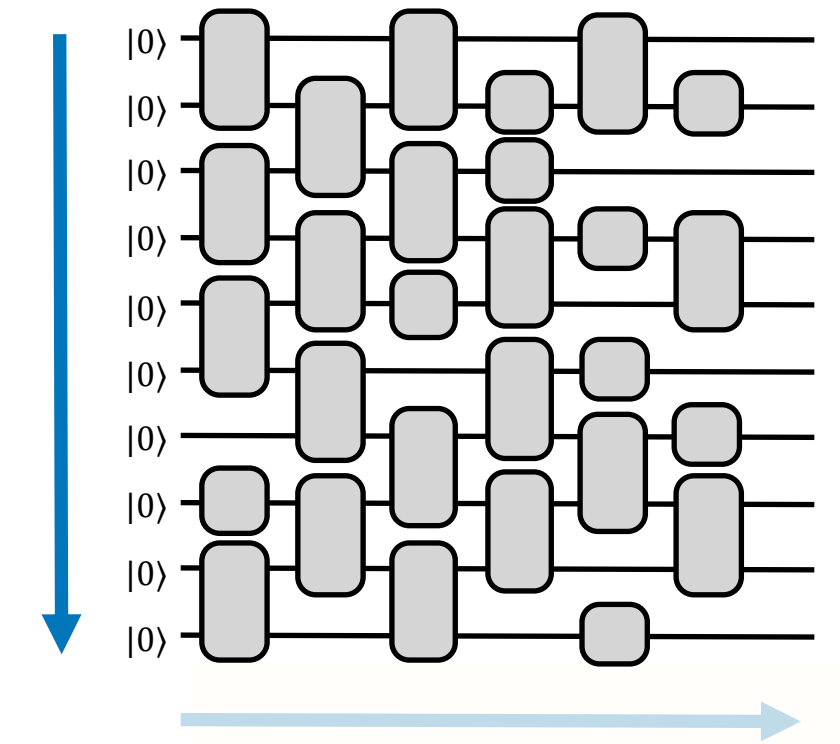
What makes dynamic circuits powerful?

1. The ability to **reset** & **reuse** qubits can reduce **spatial** overhead

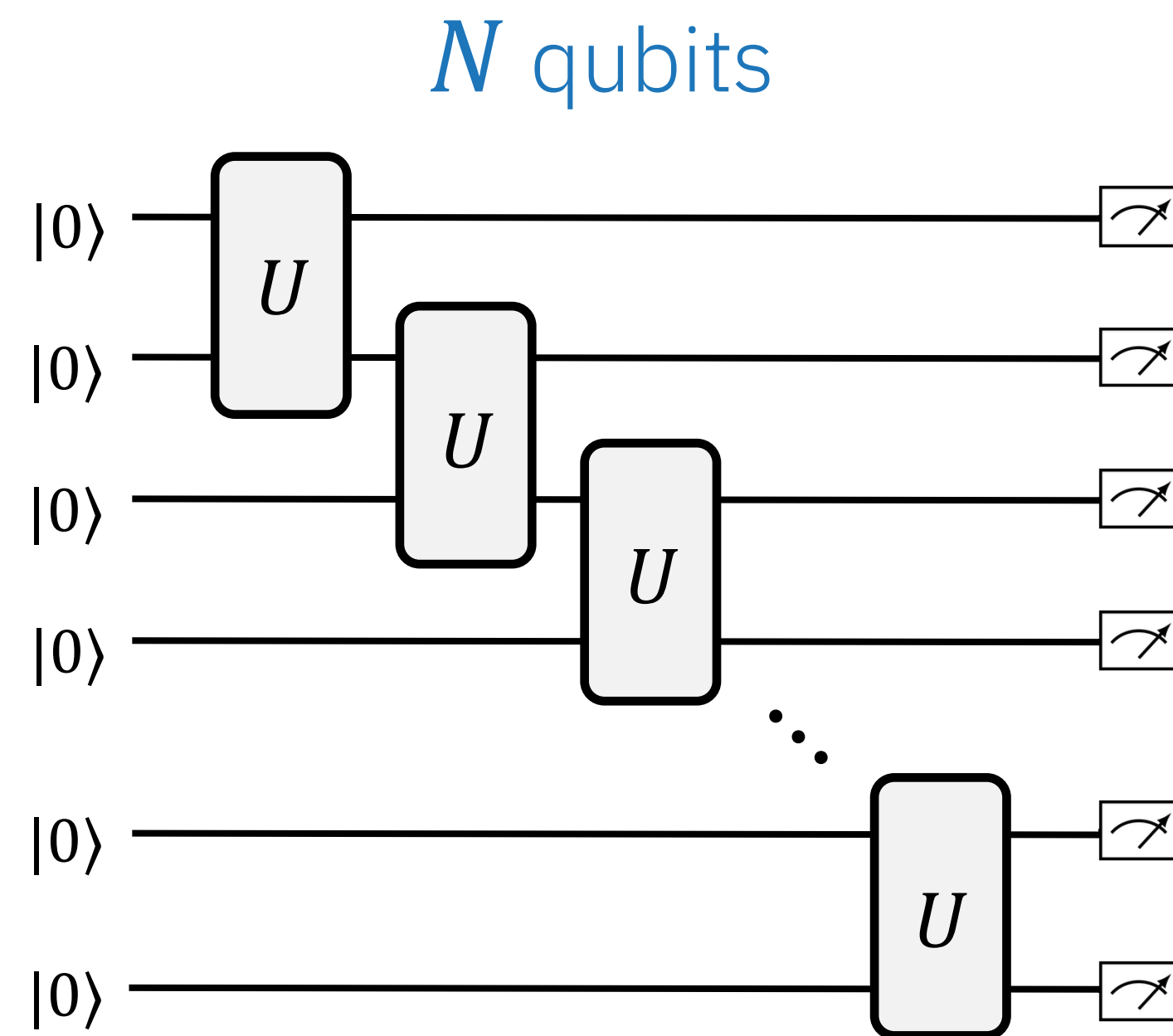


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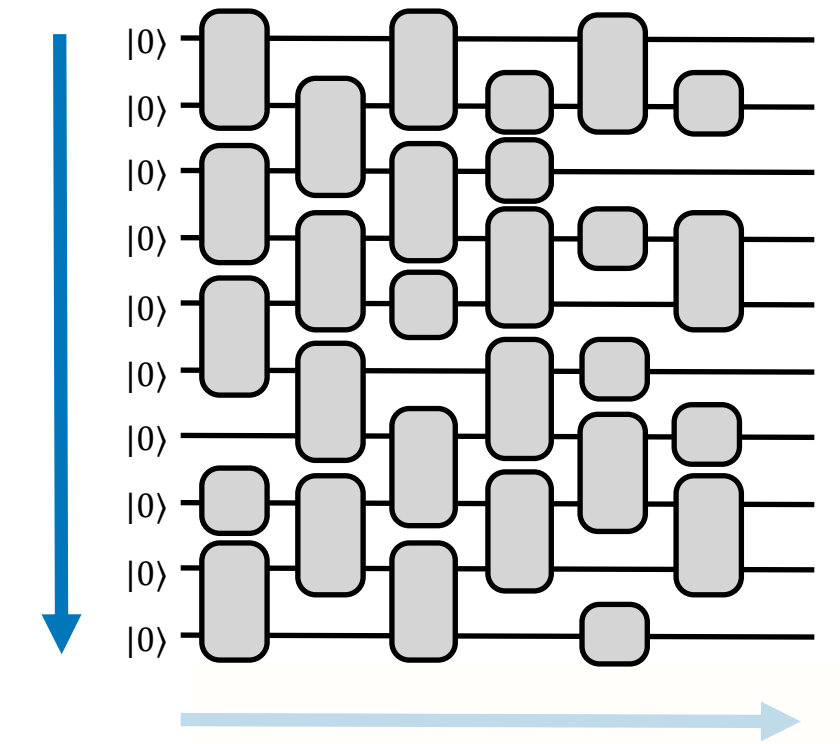


Example: staircase circuit

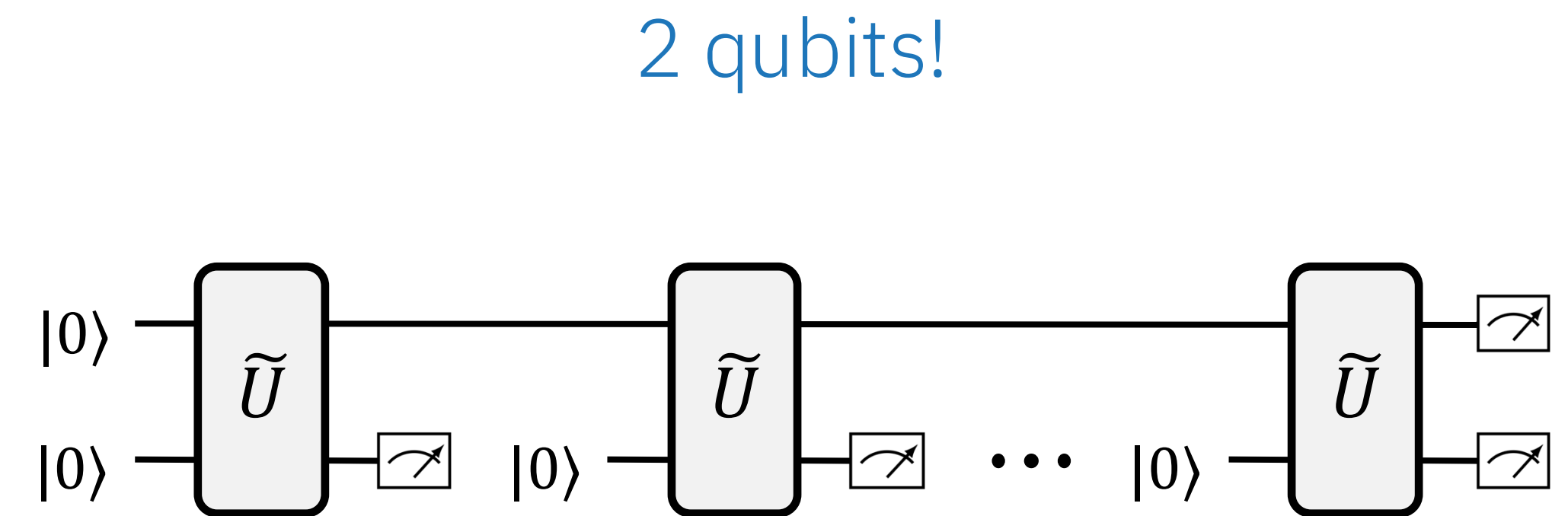
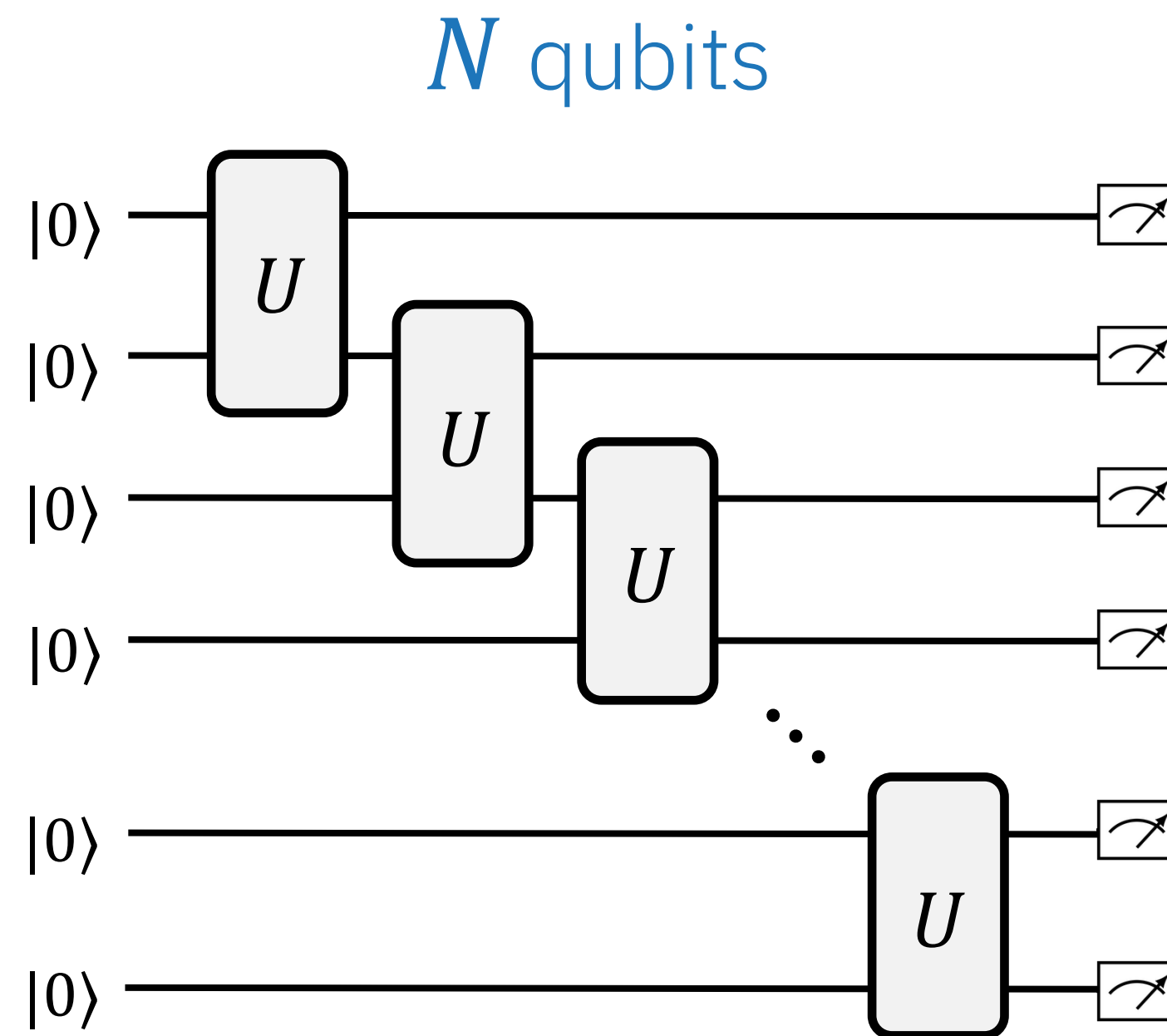


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Example: staircase circuit



See related work on so-called *holographic* algorithms

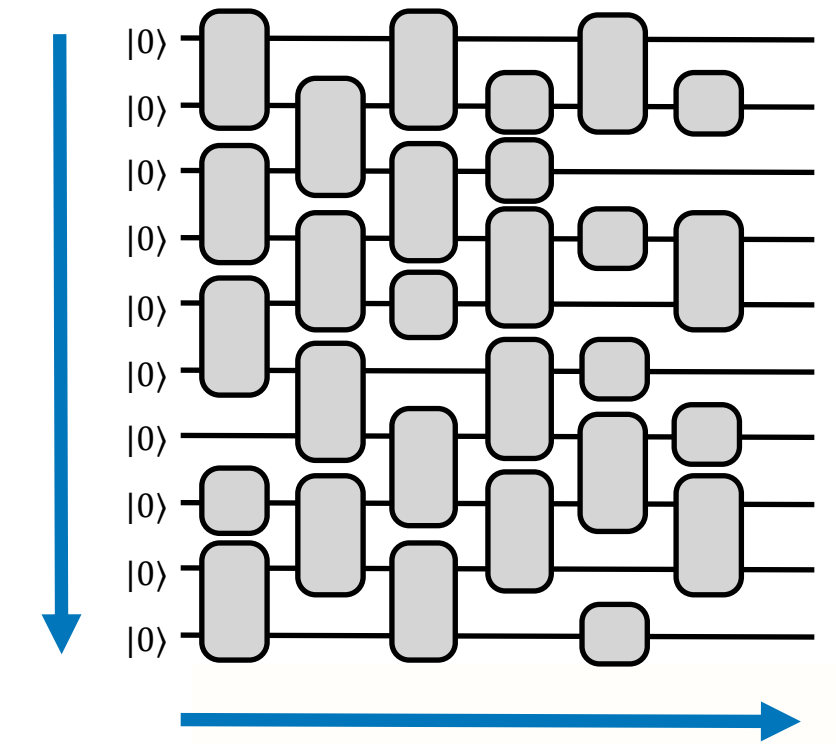
Foss-Feig et al., Phys. Rev. Research (2022)

Chertkov et al., Nat. Phys (2022)

Decross et al., PRX (2023)

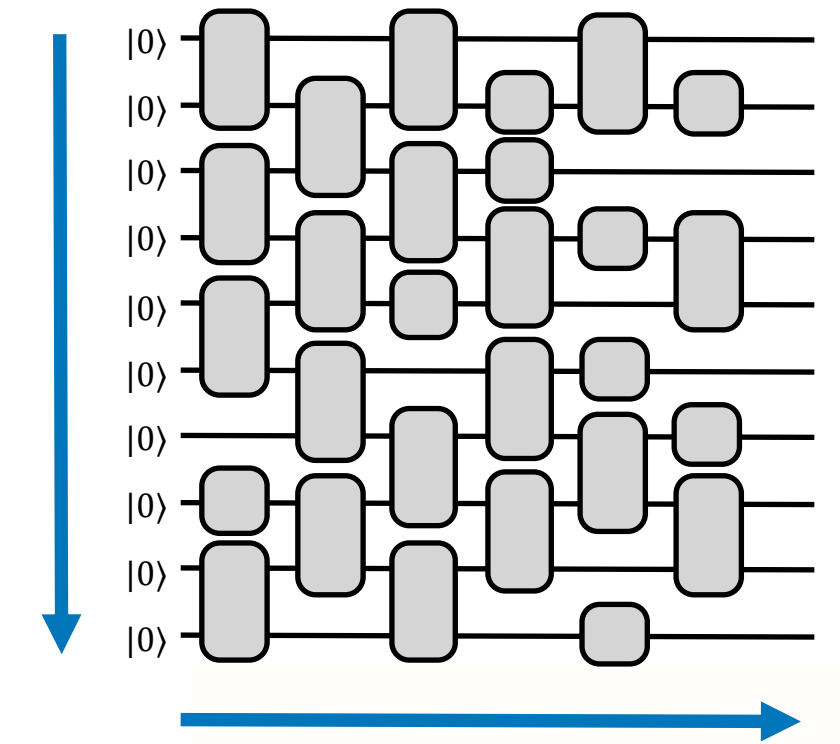
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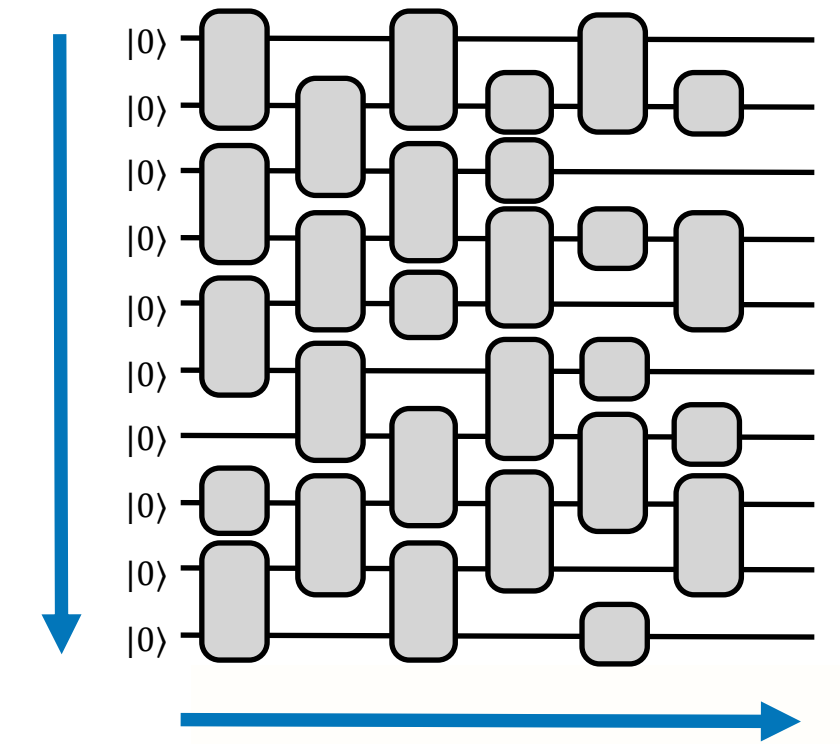


Example: GHZ state preparation

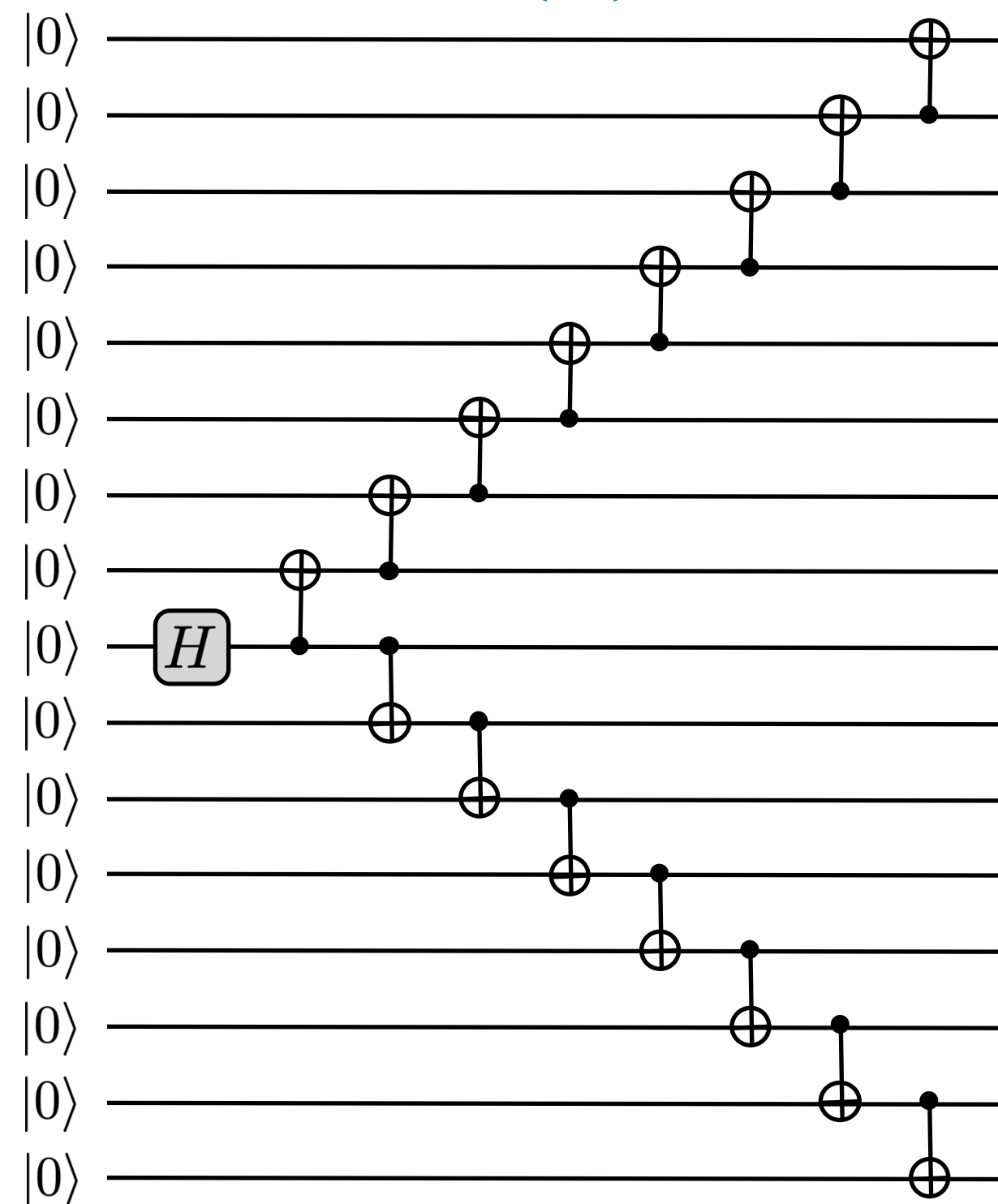
$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|000\dots 0\rangle + |111\dots 1\rangle)$$

What makes dynamic circuits powerful?

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Unitary: $O(N)$ depth

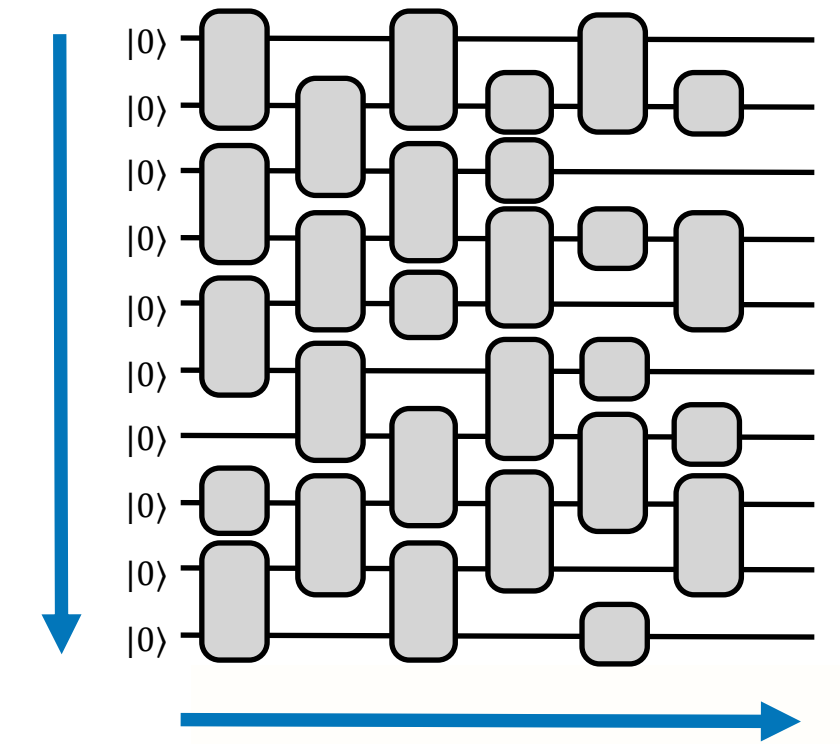


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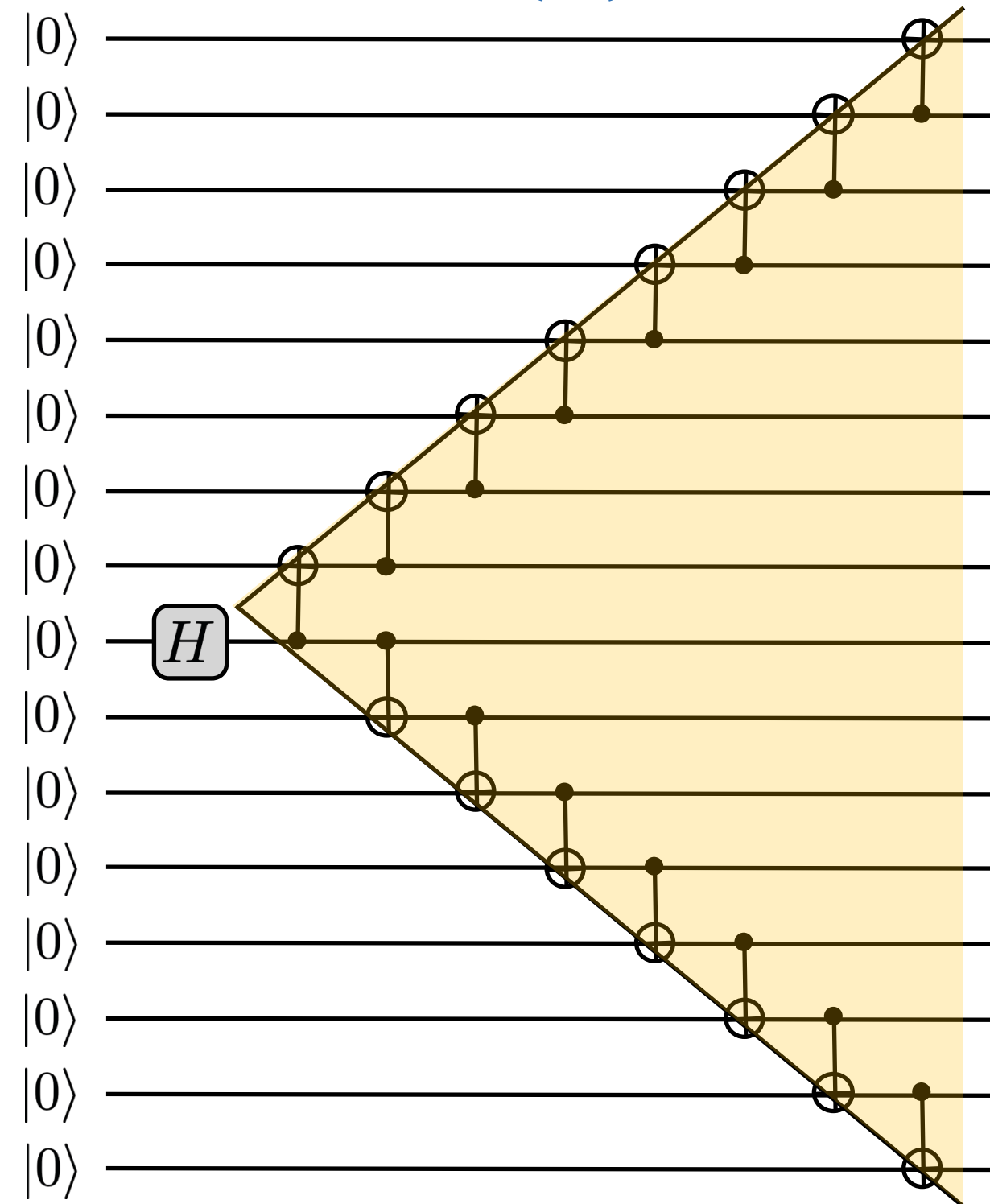
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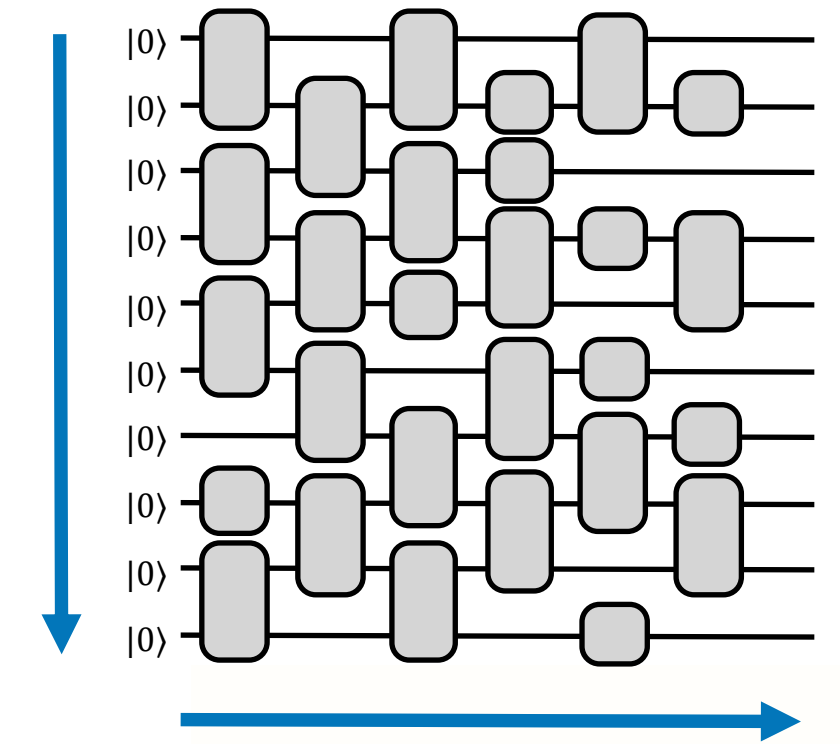
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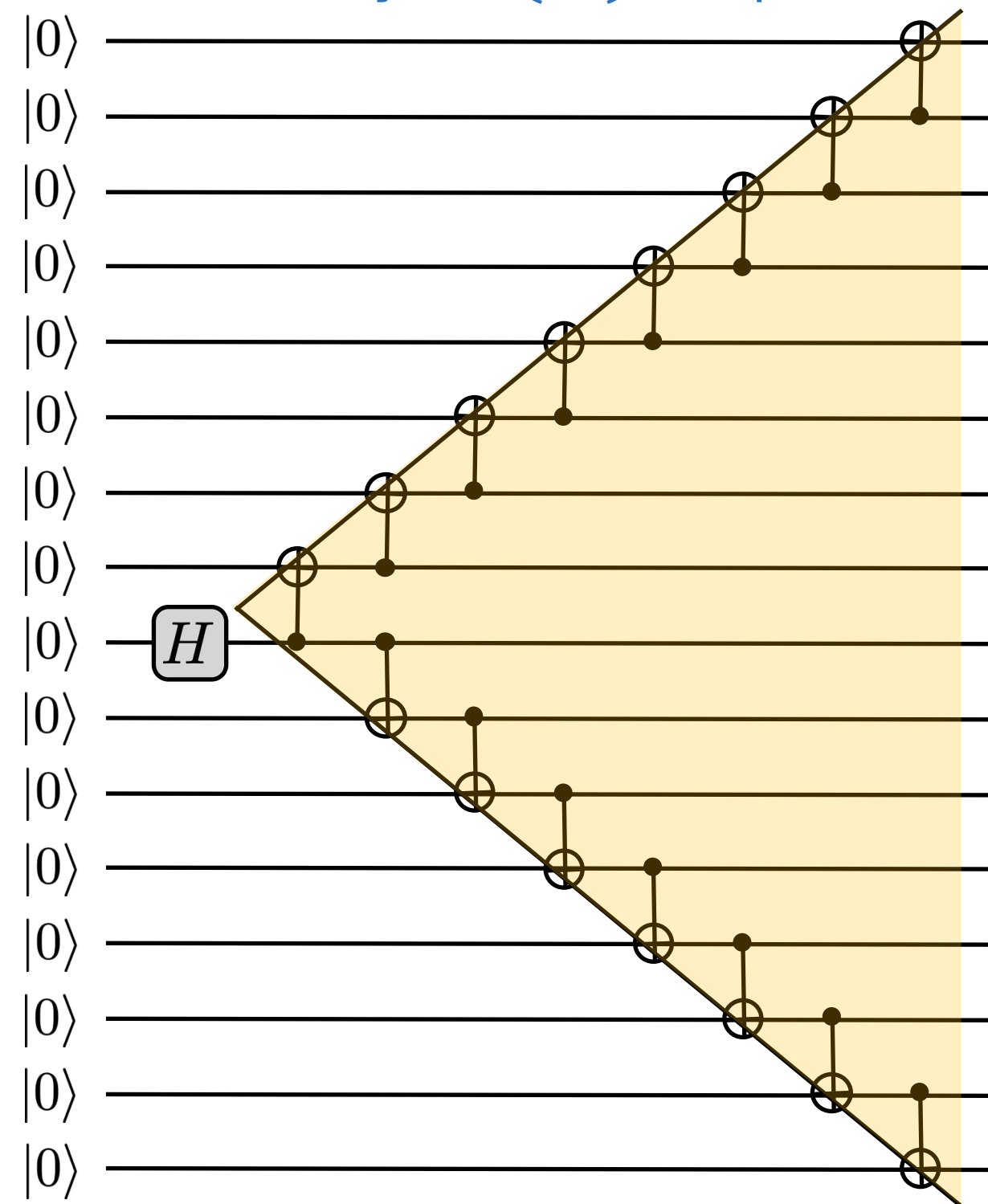
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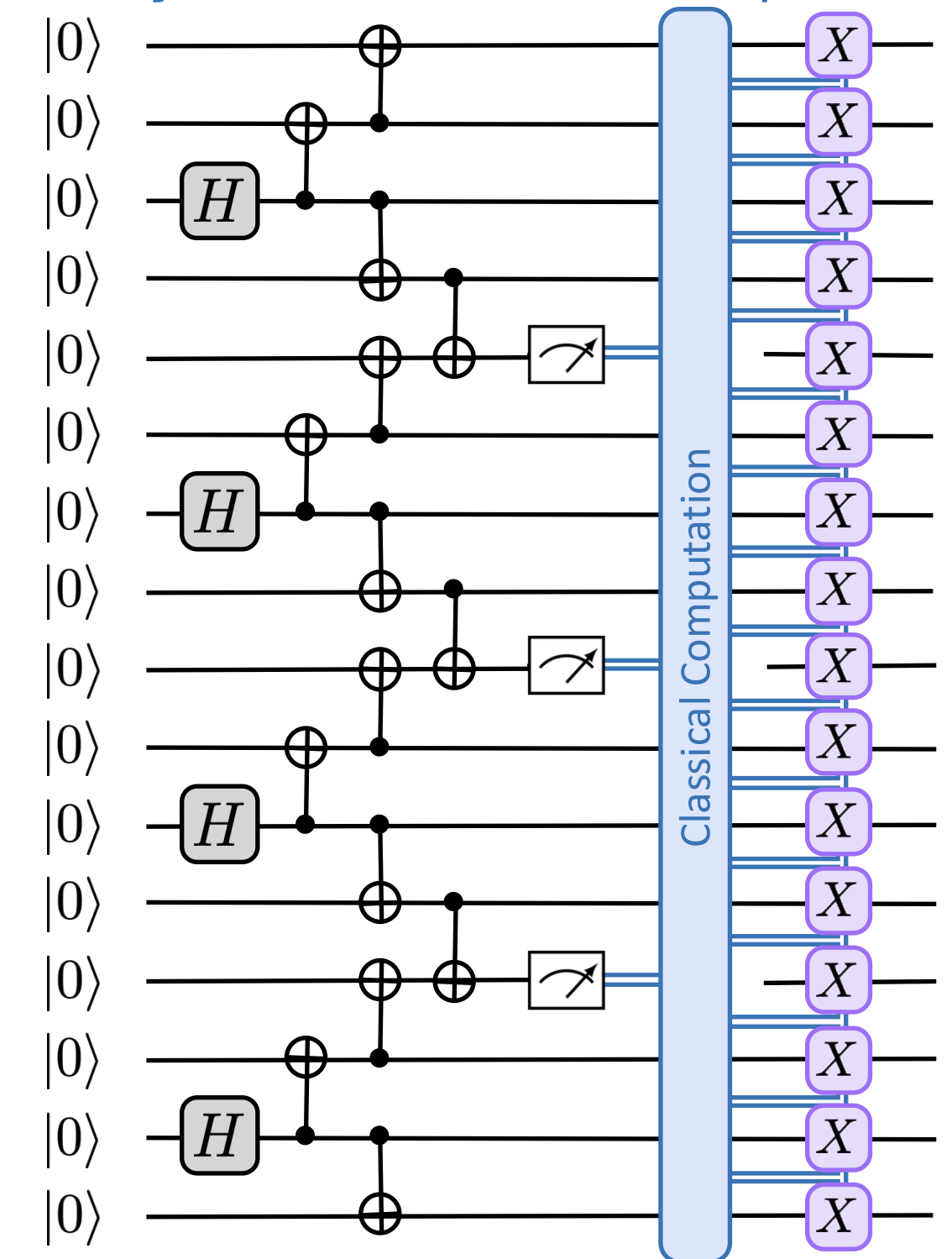
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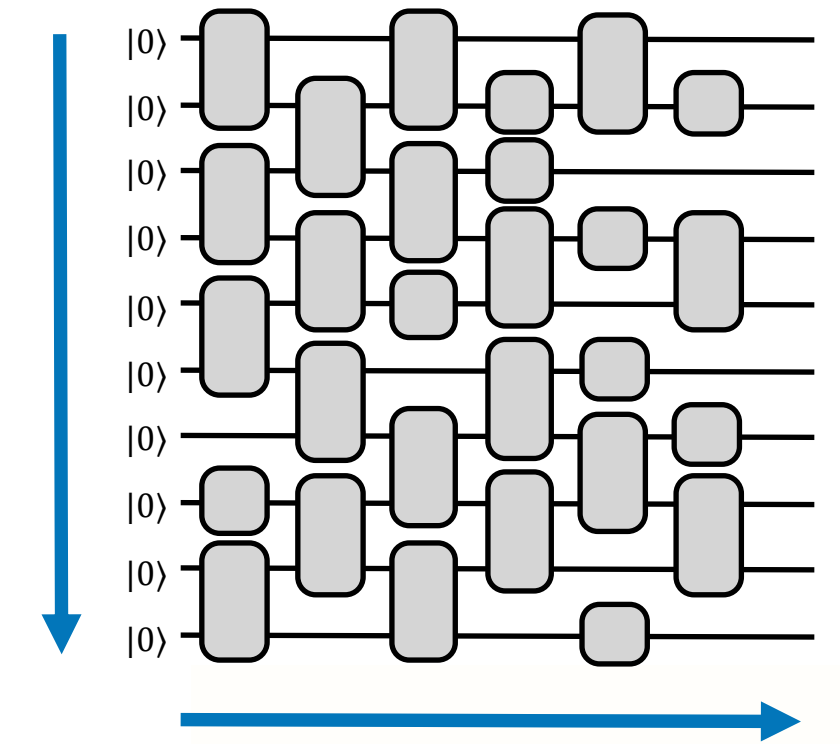
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Dynamic: $O(1)$ depth

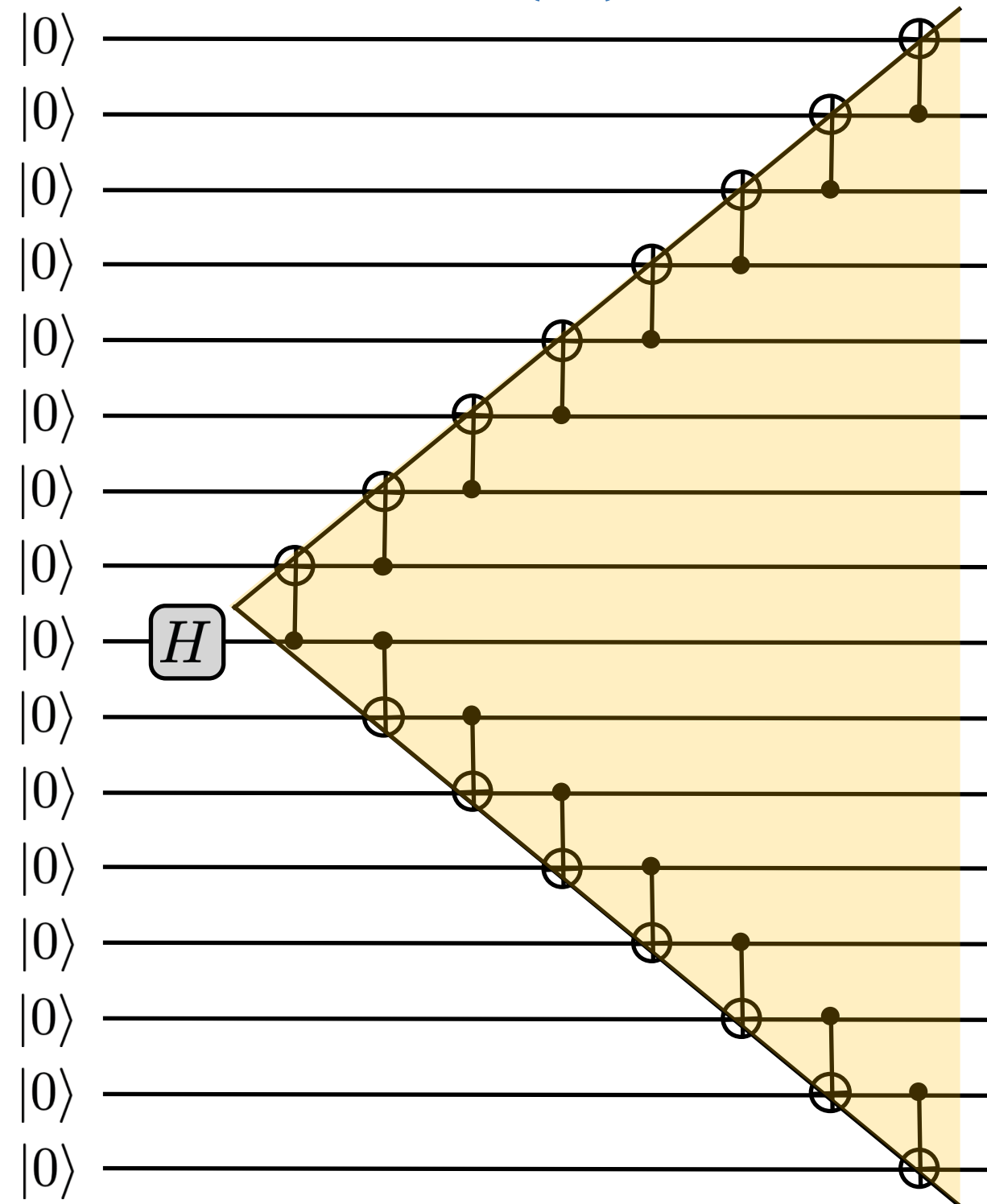


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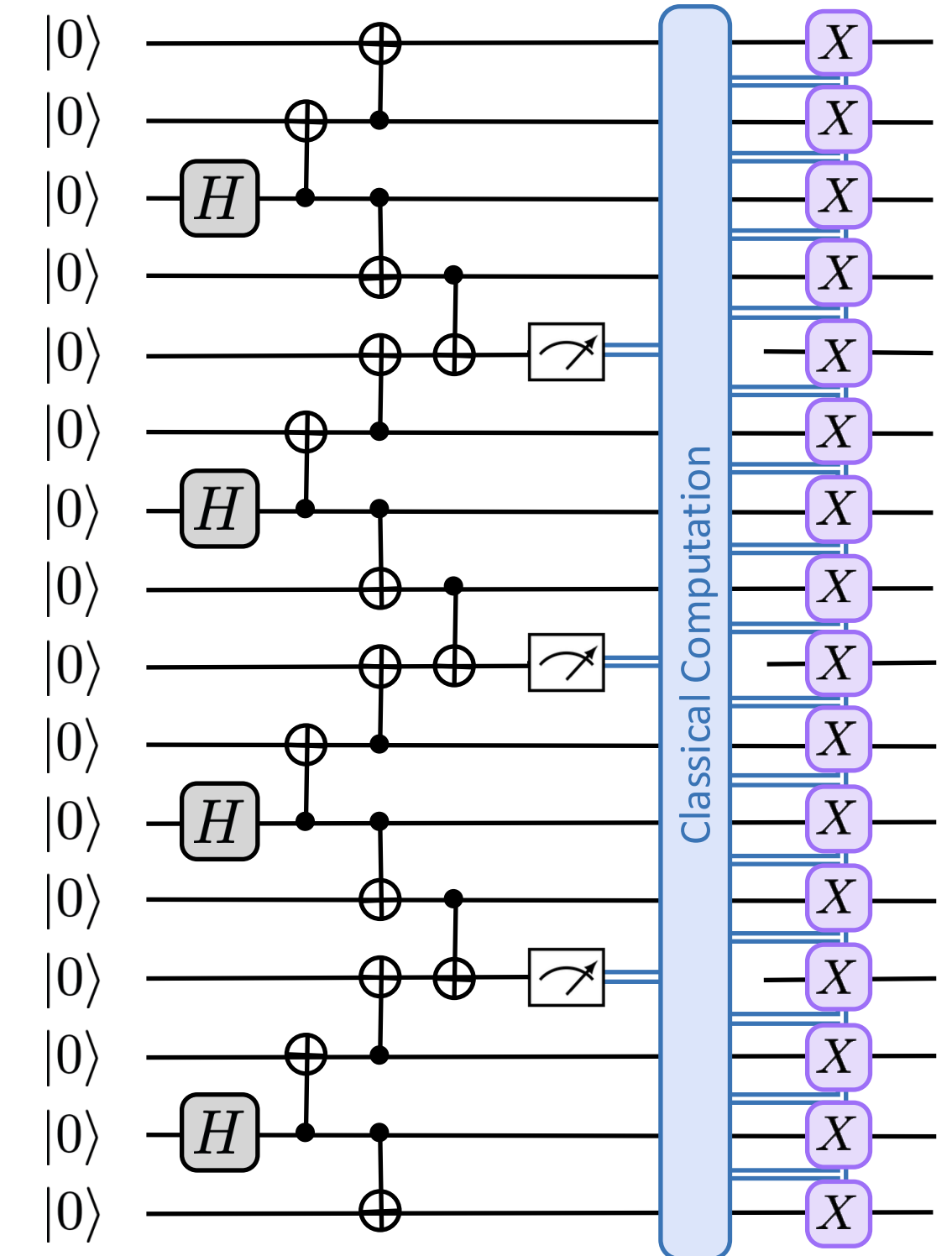
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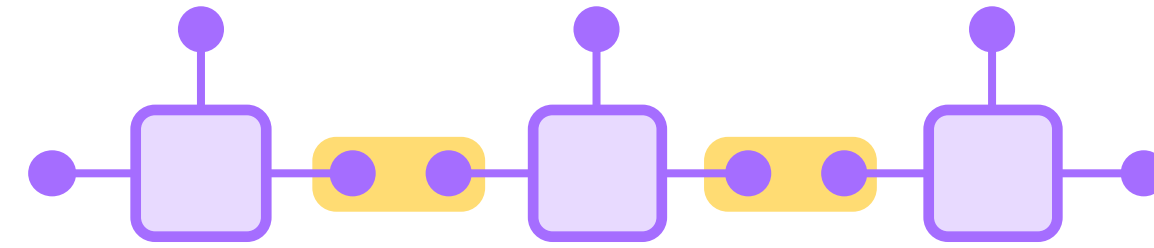
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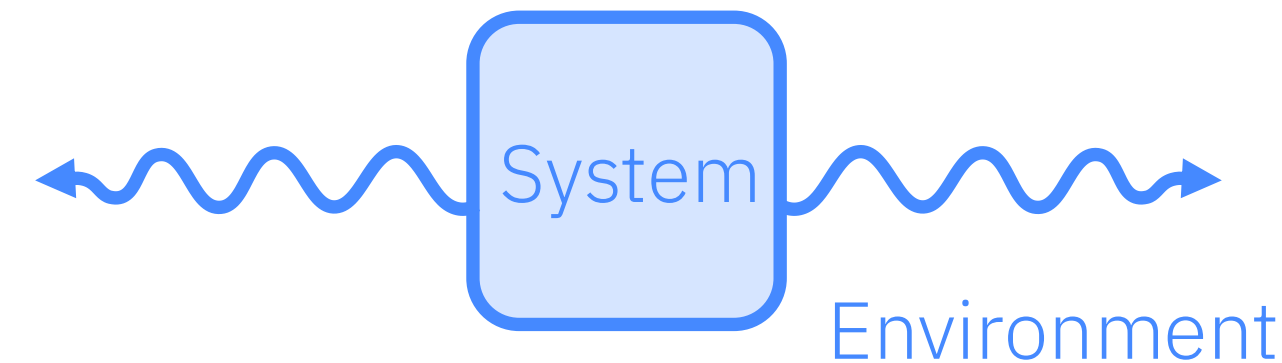
Applications of dynamic circuits

(beyond error correction)



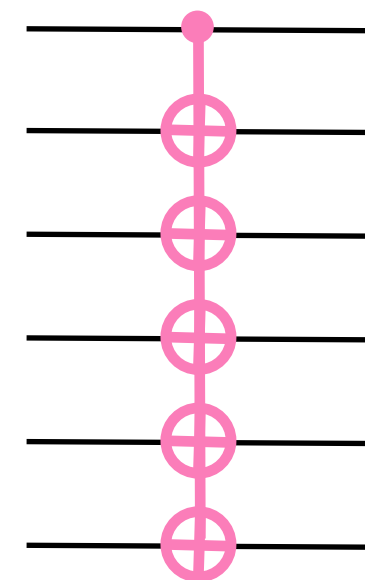
State preparation

- Long-range entangled states [1]
- Matrix product states [2]
- Topologically ordered states [3]
- Gibbs states [4]



Non-unitary processes

- Driven-dissipative dynamic
- Measurement-induced phase transitions [5]
- Nonlinear transformations [6]

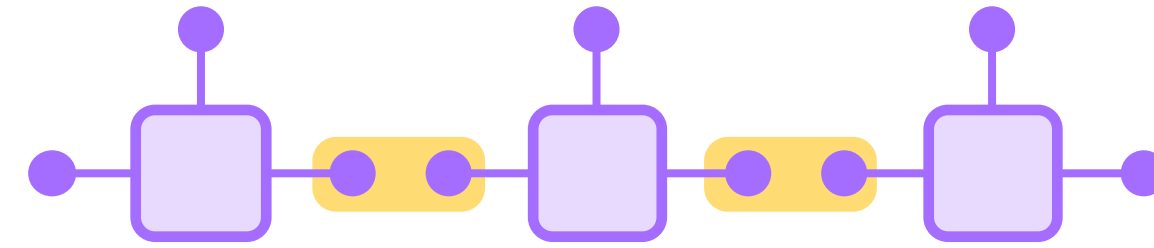


Unitary compilation

- Long-range entangling gates [1]
- Many-qubit gates (e.g. fanout) [7]
- Distributed quantum computing

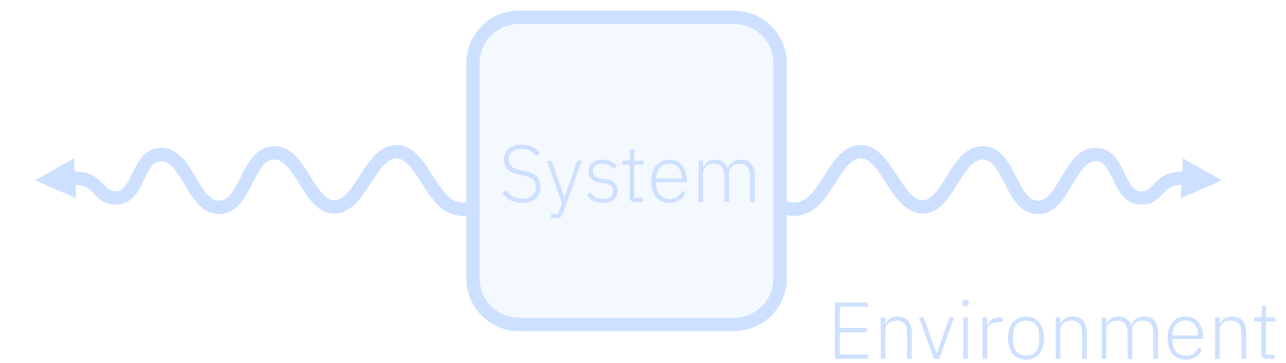
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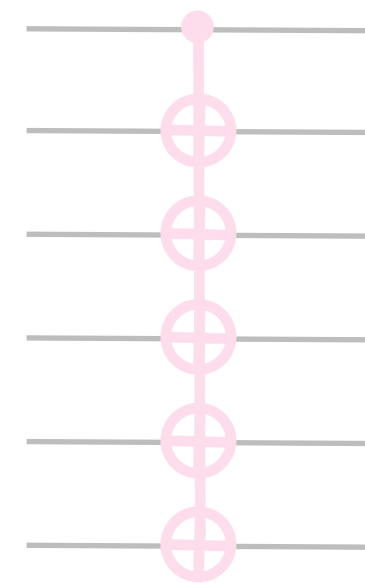
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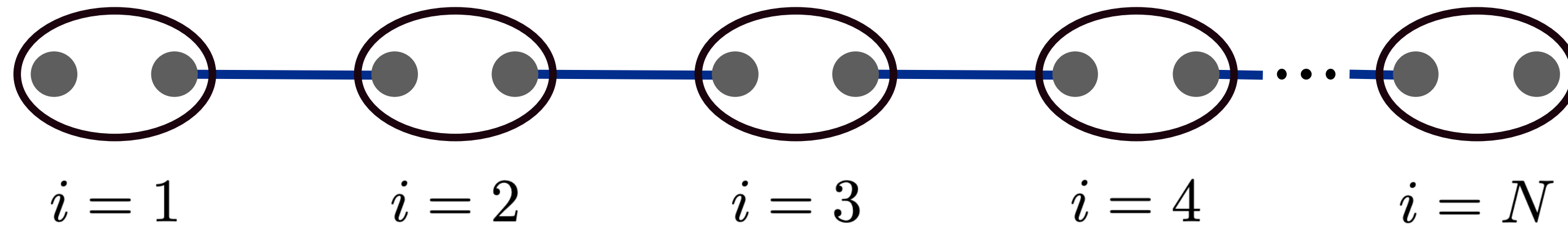


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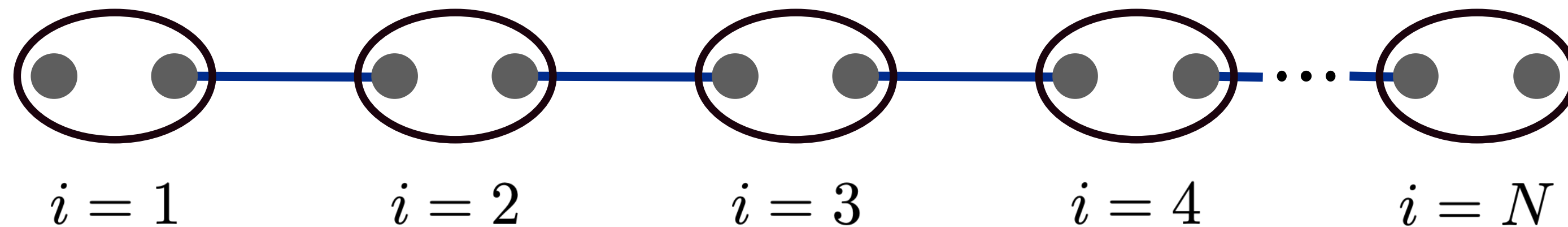
Affleck-Kennedy-Lieb-Tasaki (AKLT) state

Spin-1 chain: encode each spin-1 site with a pair of qubits



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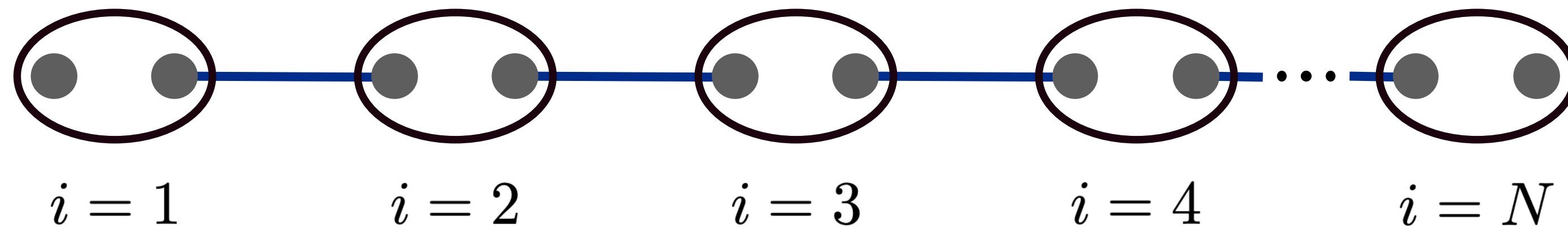
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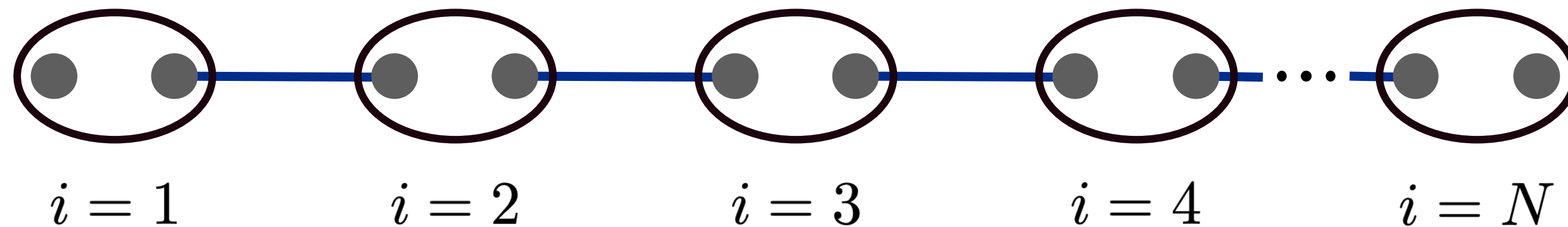
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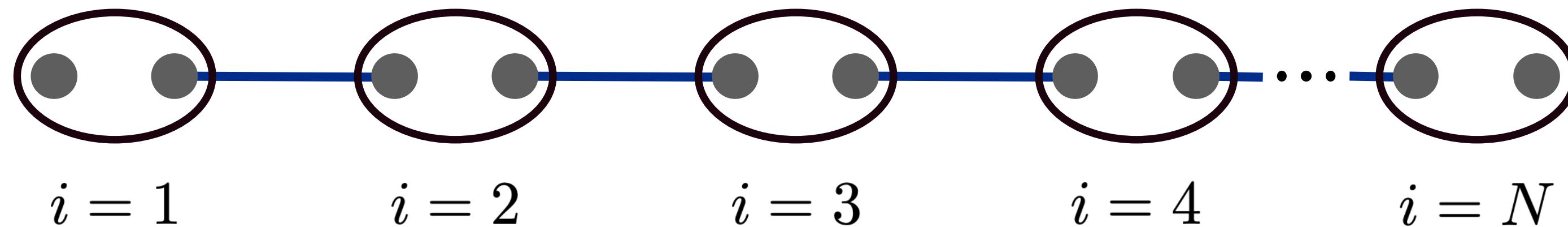
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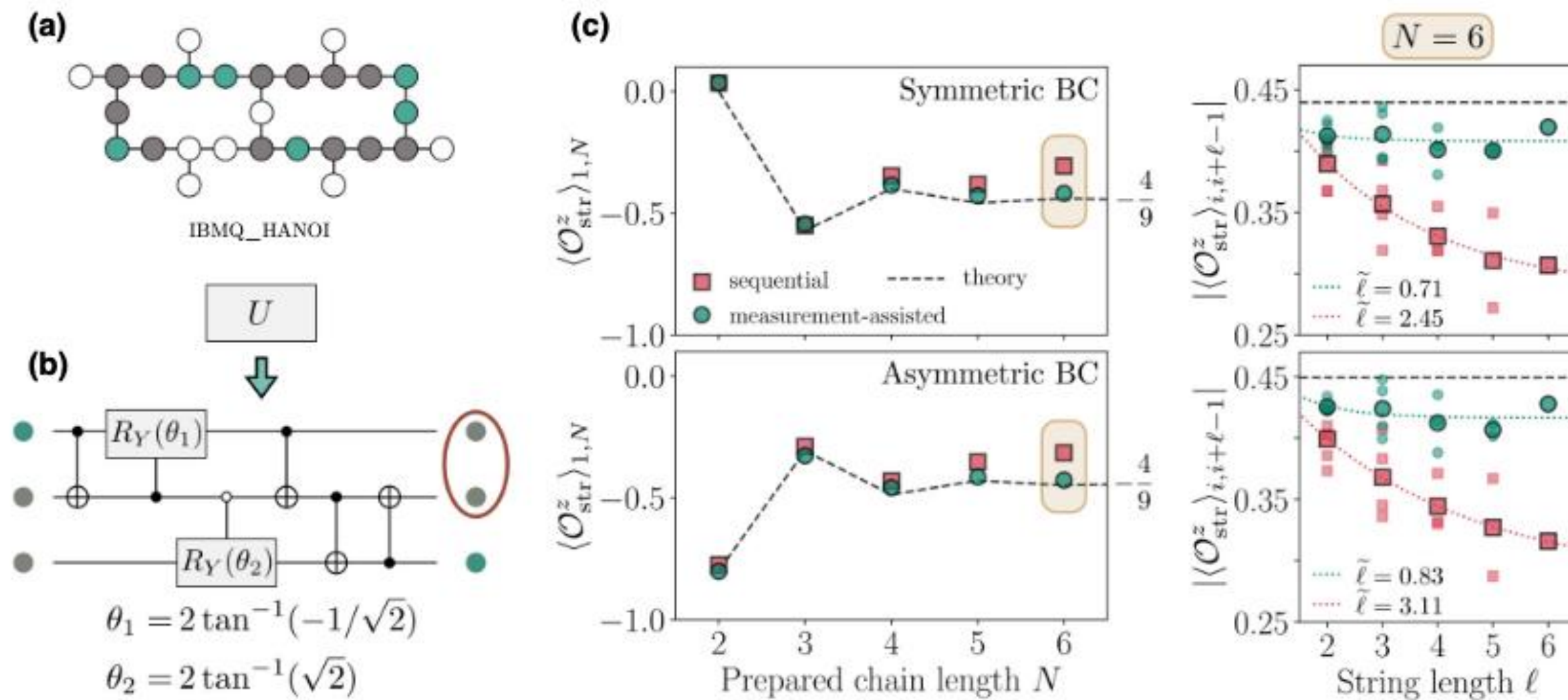


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- Preparation: $O(N)$ depth unitary circuit $\rightarrow O(1)$ depth dynamic circuit [4, 5]

Dynamic circuits (initial release)

SMITH, CRANE, WIEBE, and GIRVIN

PRX QUANTUM 4, 020315 (2023)



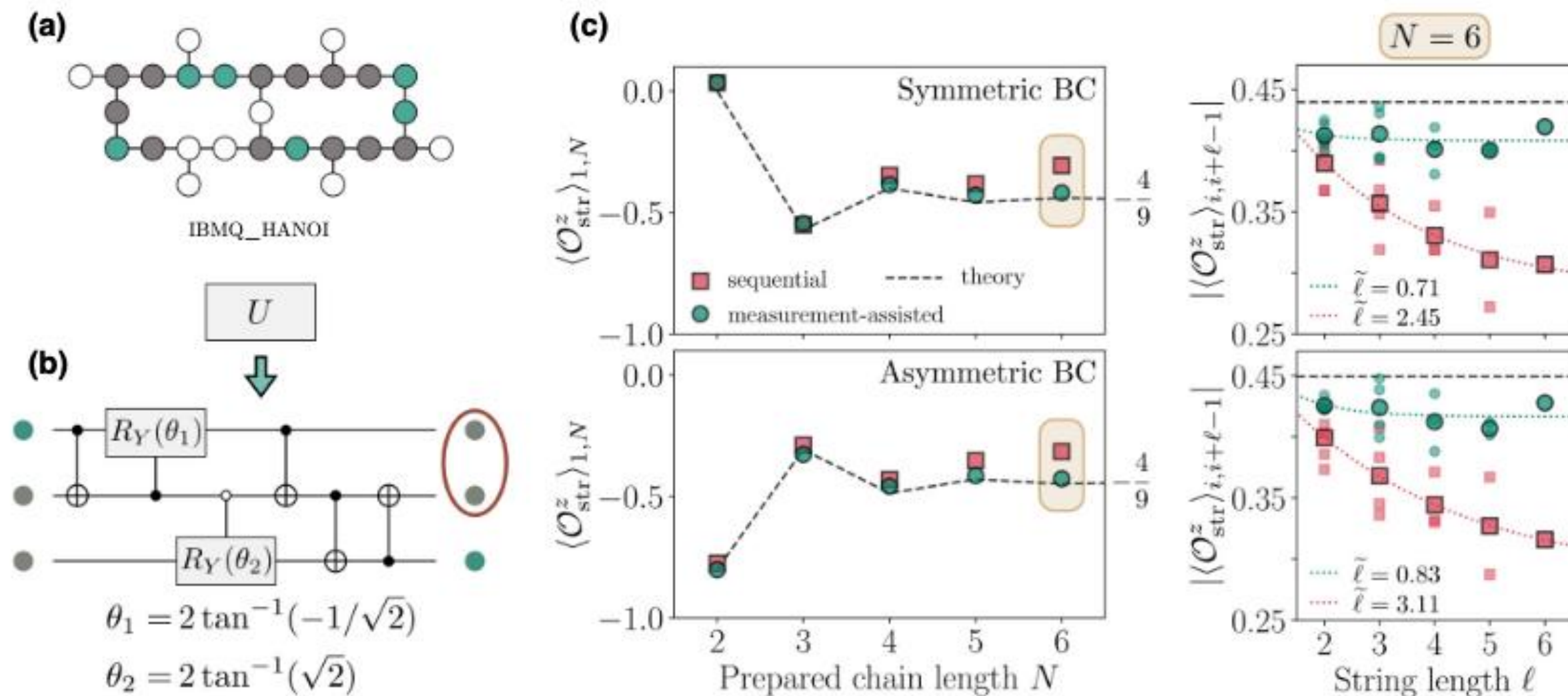
Presented at IBM Quantum Summit (2022)

- Up to $N_{\text{sites}} = 6$ AKLT state preparation
 - 14 qubits (unitary)
 - 18 qubits (measurement based)
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- Post-process only (no ffwd)

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Dynamic circuits capability introduced at the summit in 2022 enabled this experiment to run with **real time feed-forward!**

But with some challenges and limitations...

Re-introduction of Dynamic Circuits

Previous version

- Parallel feed forward operations limited to 10 bit input using Switch
 - Up to 12 site AKLT state

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- Parallel ifs / feed forward operations
 - Demonstrate up to 30 site AKLT state using 90 qubits

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- Scheduled circuits are returned (beta) to help with debugging and optimization

AKLT state preparation with utility scale dynamic circuits

- Different methods to prepare the AKLT states:
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 2. Measurement based without dynamic circuits (post-process)
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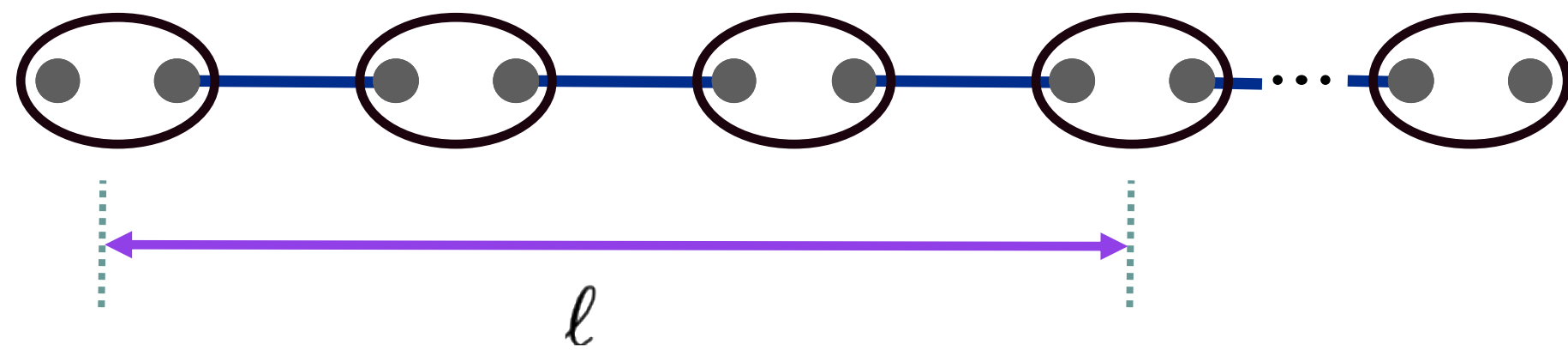
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Validation of AKLT state preparation

String order

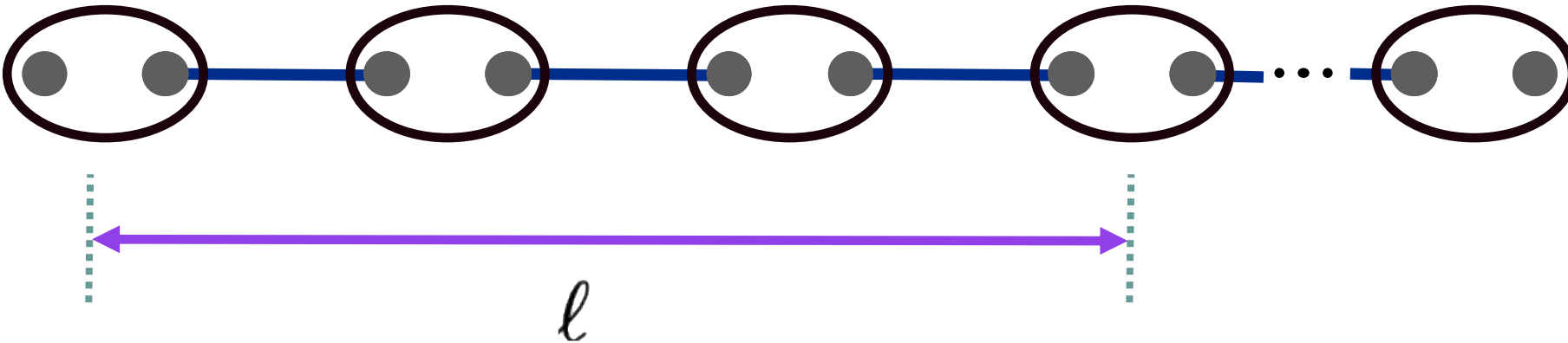
$$\langle O_{\text{str}}^z \rangle_{\ell,i} = \langle S_i^z \prod_{k=1}^{\ell-2} e^{i\pi S_{i+k}^z} S_{i+\ell-1}^z \rangle$$



Non-zero value: fingerprint of symmetry-protected topological (SPT) order

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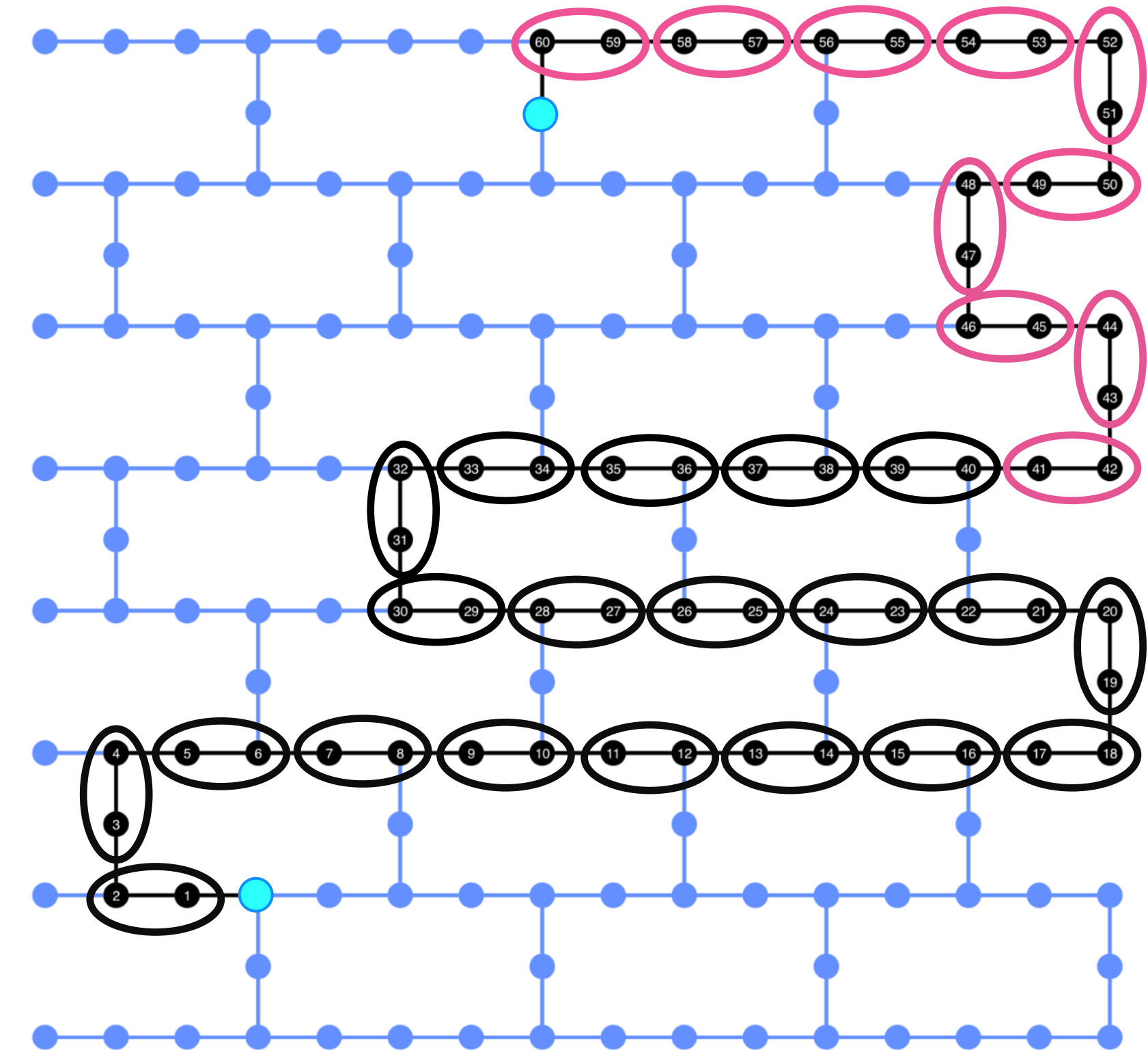
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Ex) $l = 10$

$N_{\text{sites}} = 30$ on Heron processor

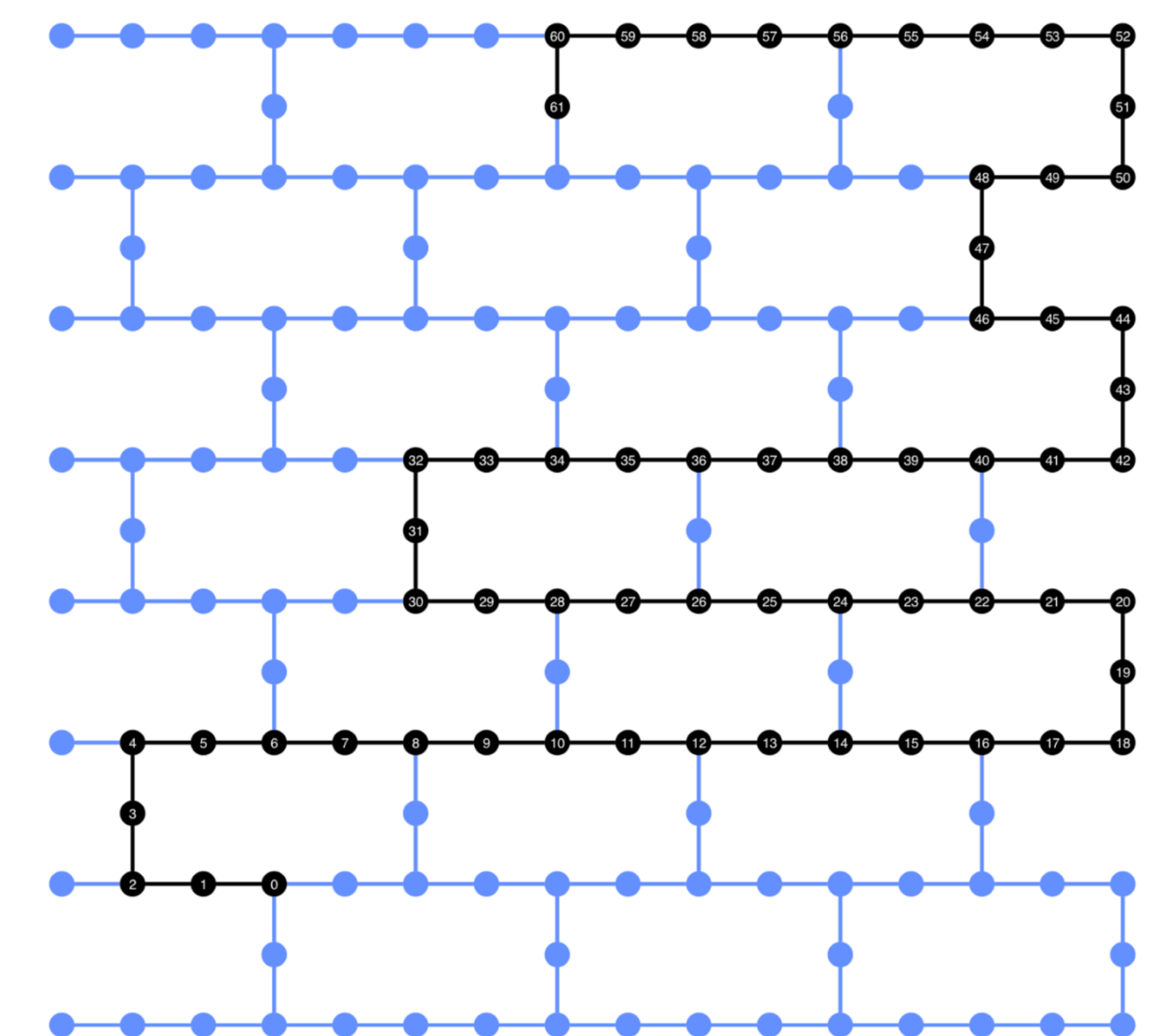


21 total $length\ 10$ chains

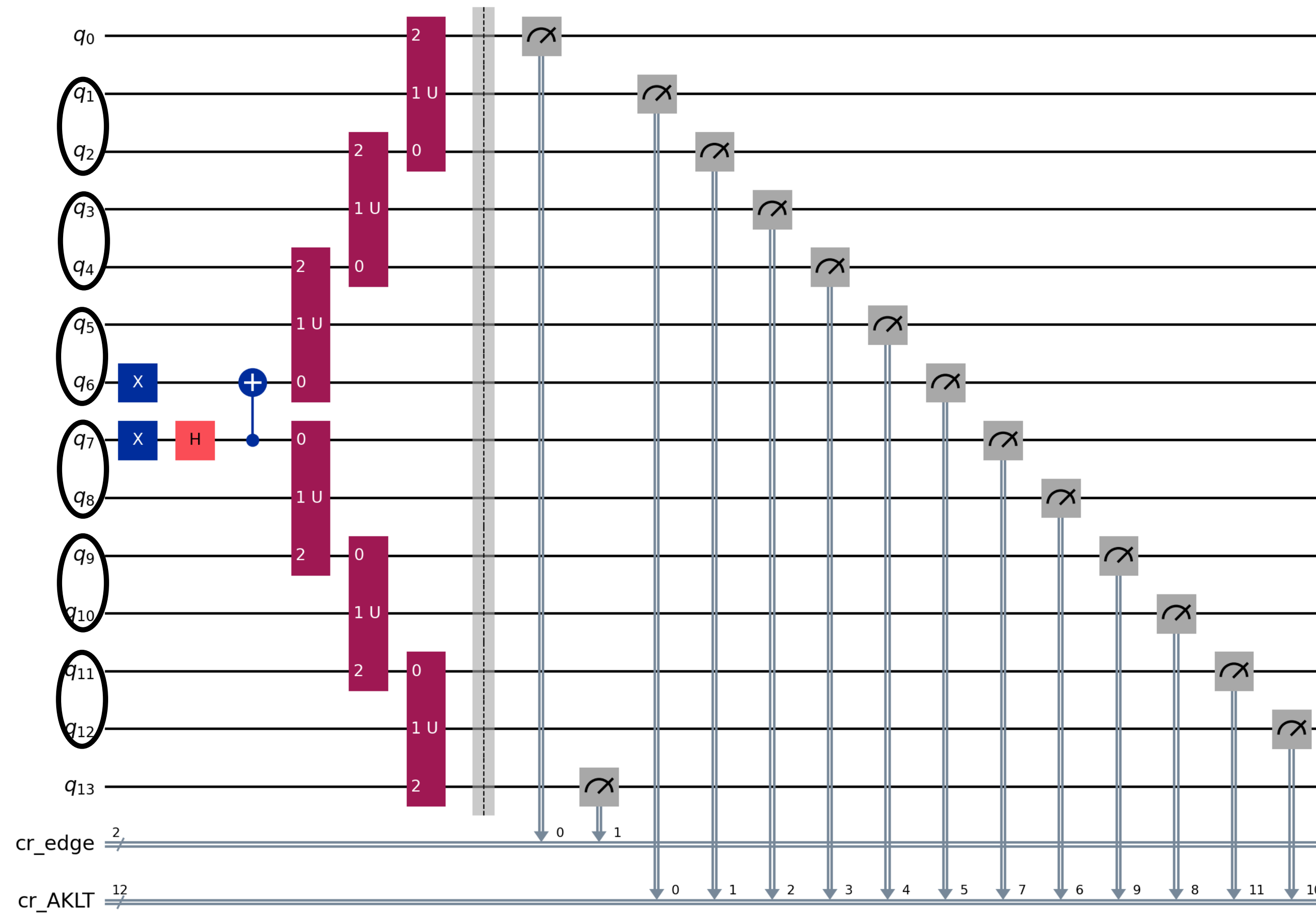
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62 qubits for $N_{\text{sites}} = 30$



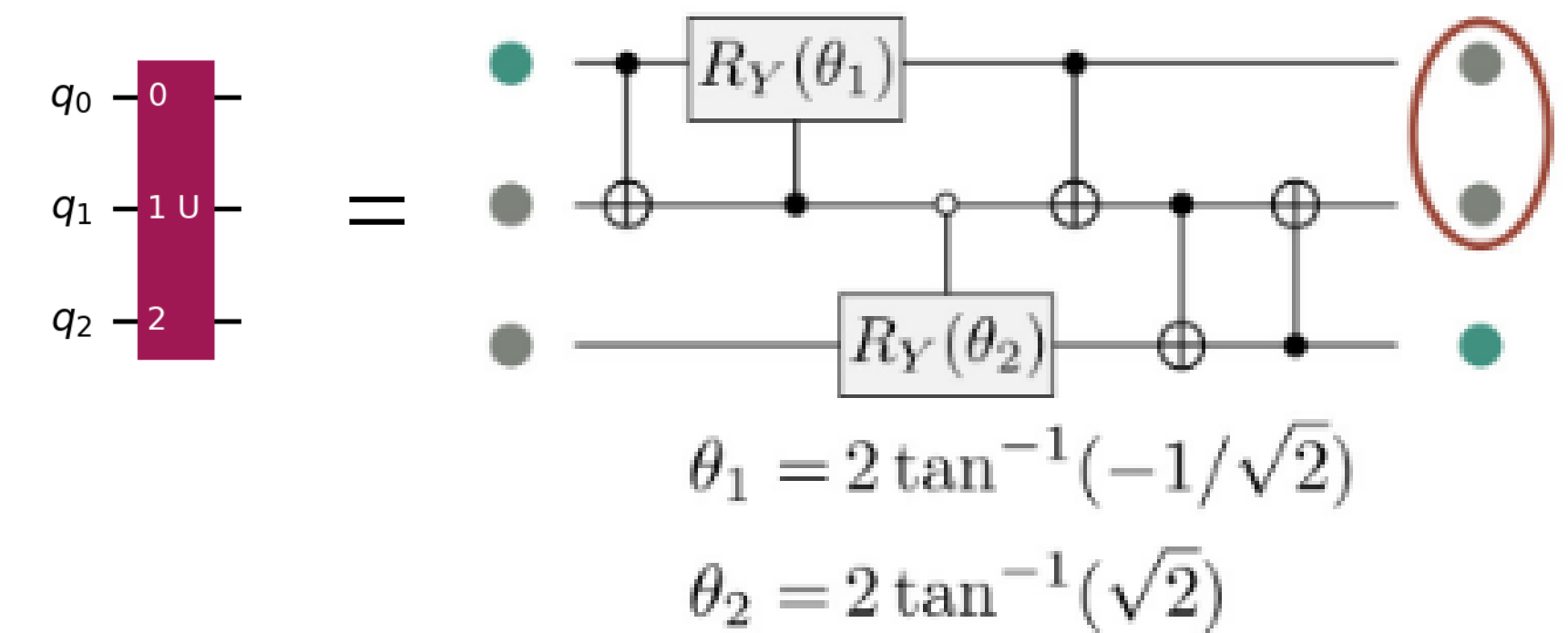
Unitary AKLT state preparation ($N_{\text{sites}} = 6$)



Total number of qubits needed

$$= N_{\text{sites}} \times 2 + \text{ancillas}$$

$$= N_{\text{sites}} \times 2 + 2 = 14$$



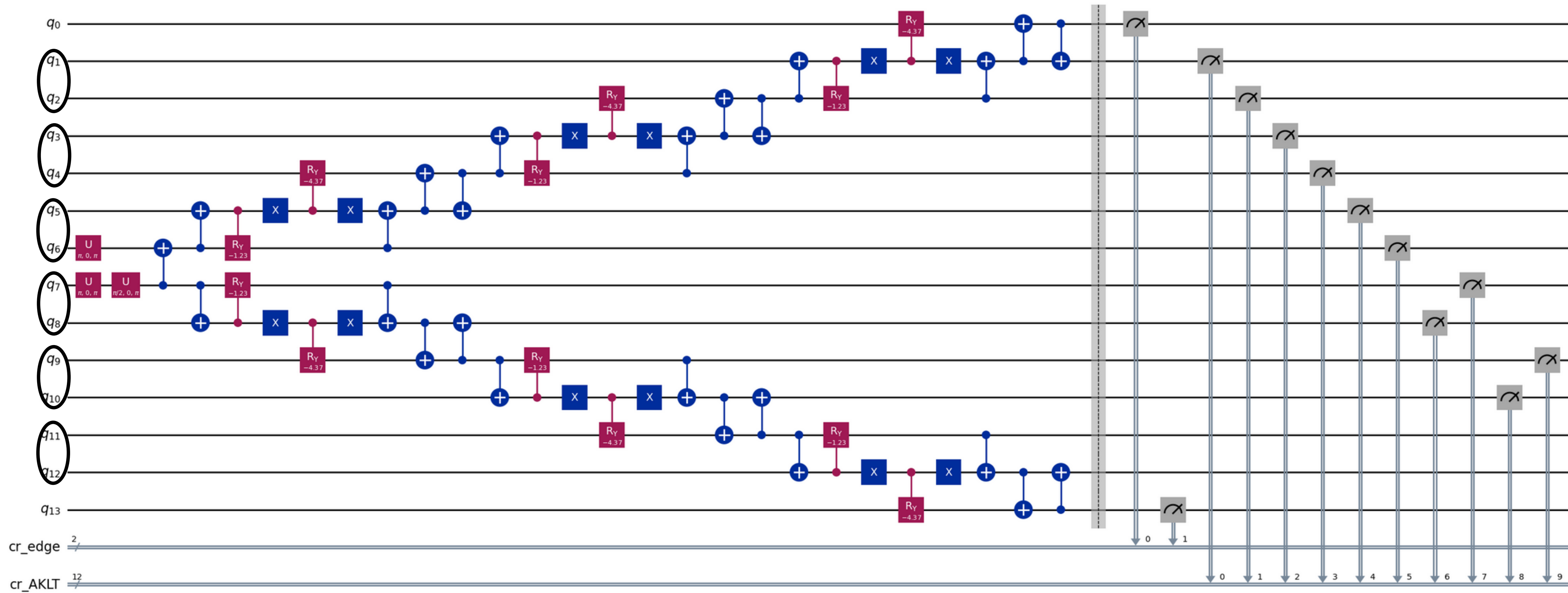
On IBM Herons, transpiles to $R_z(\theta)$'s plus:

$$7CZ + 10Sx$$

per 2 N_{sites}

Unitary AKLT state preparation ($N_{\text{sites}} = 6$)

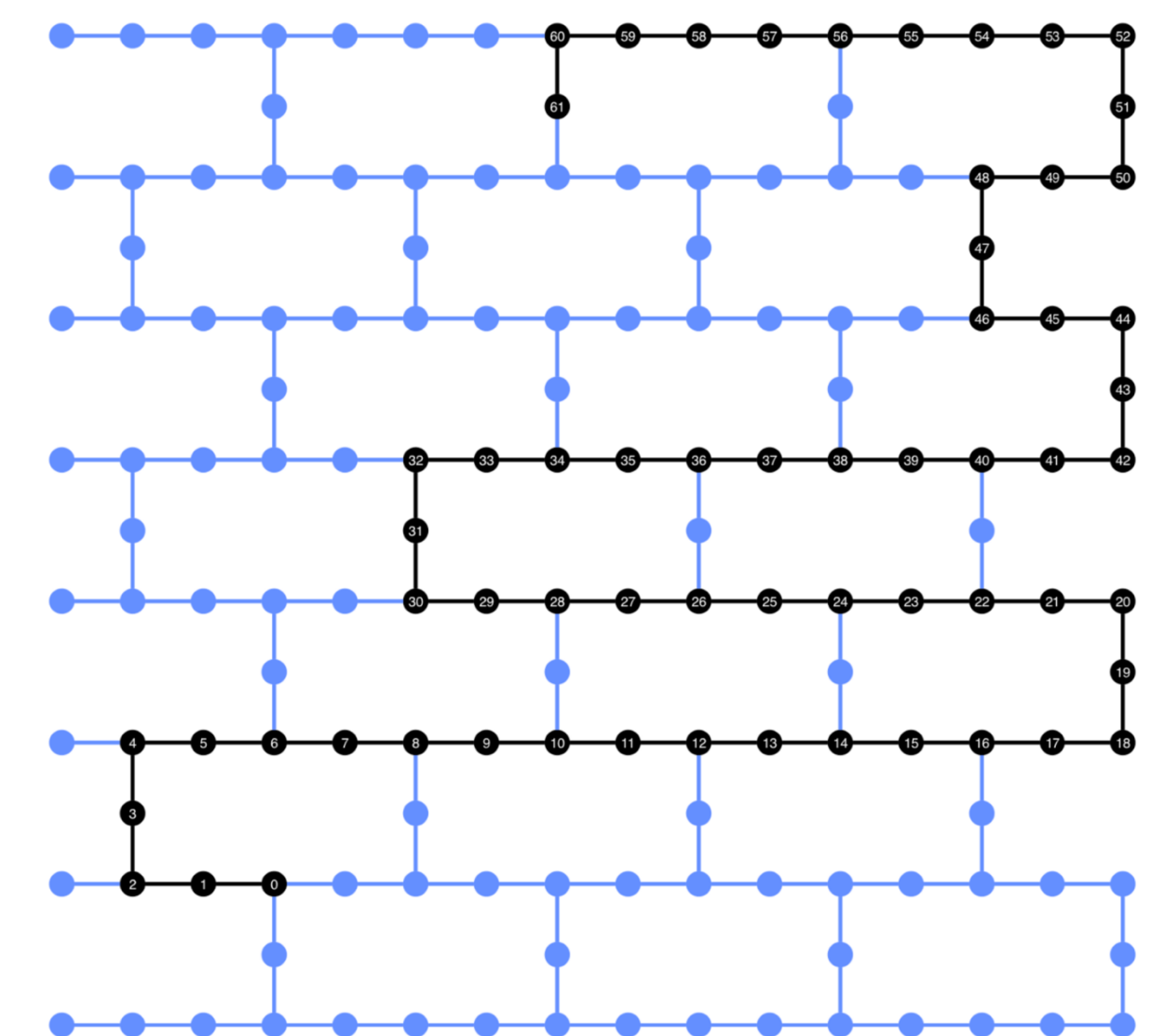
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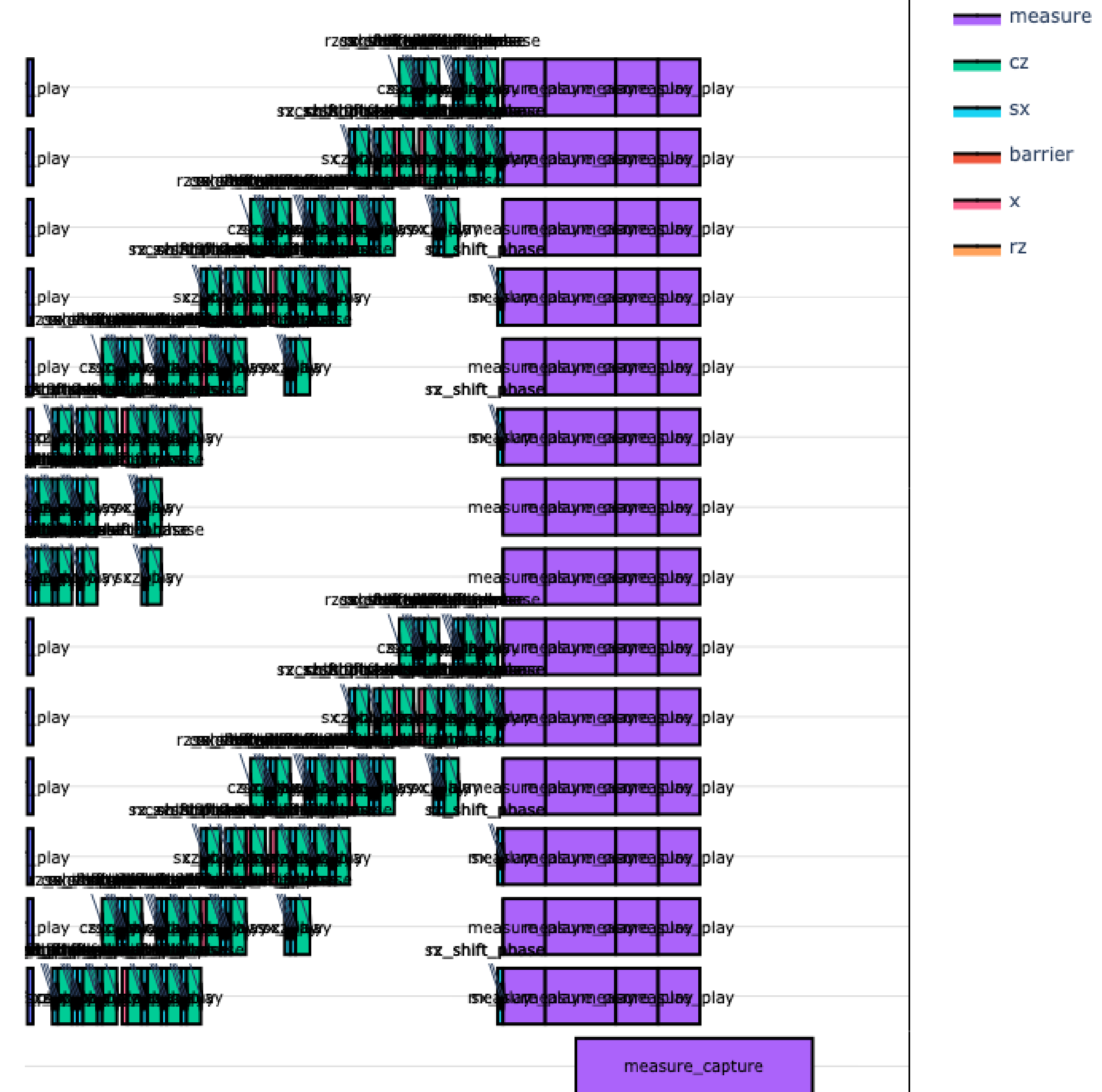
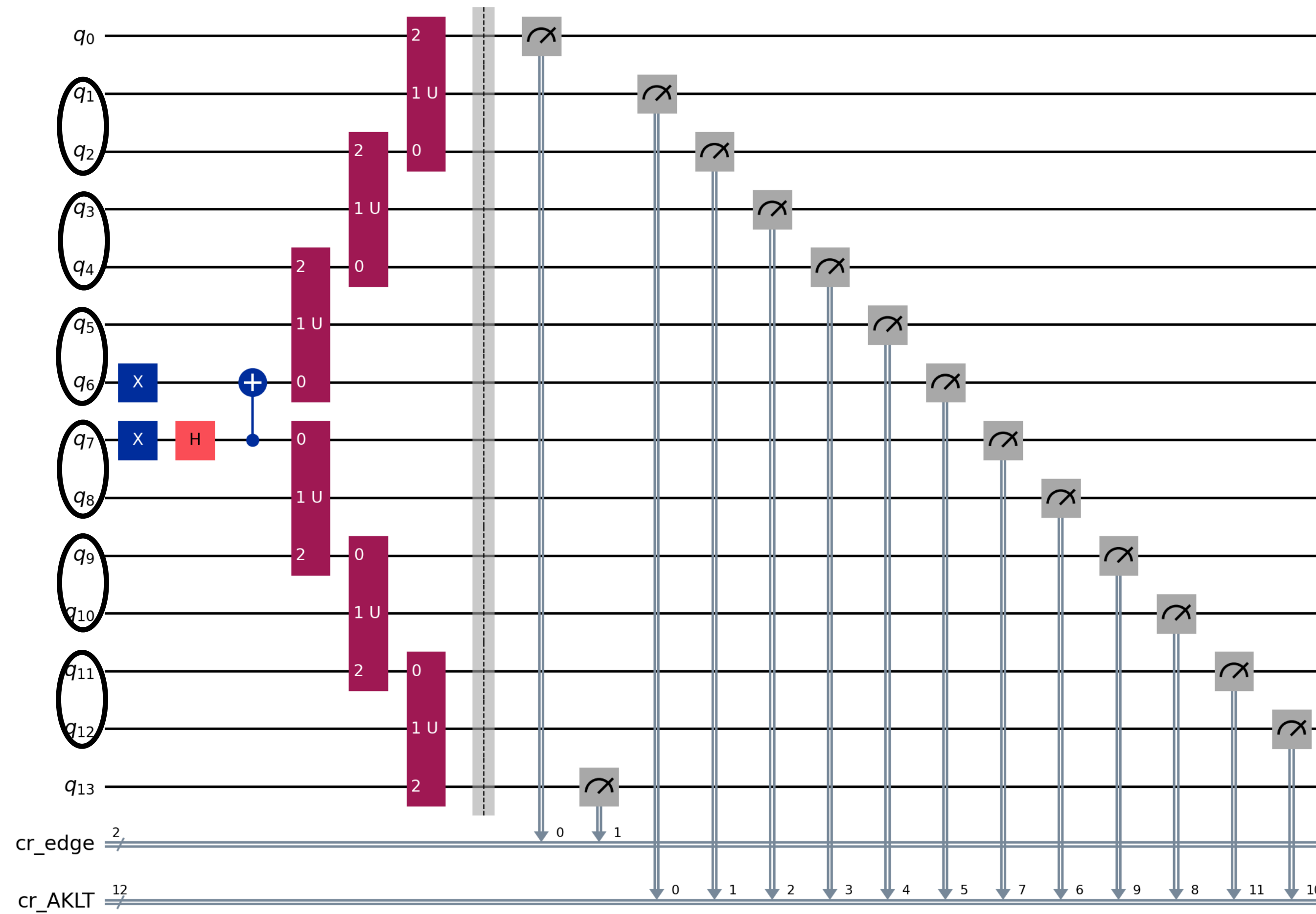
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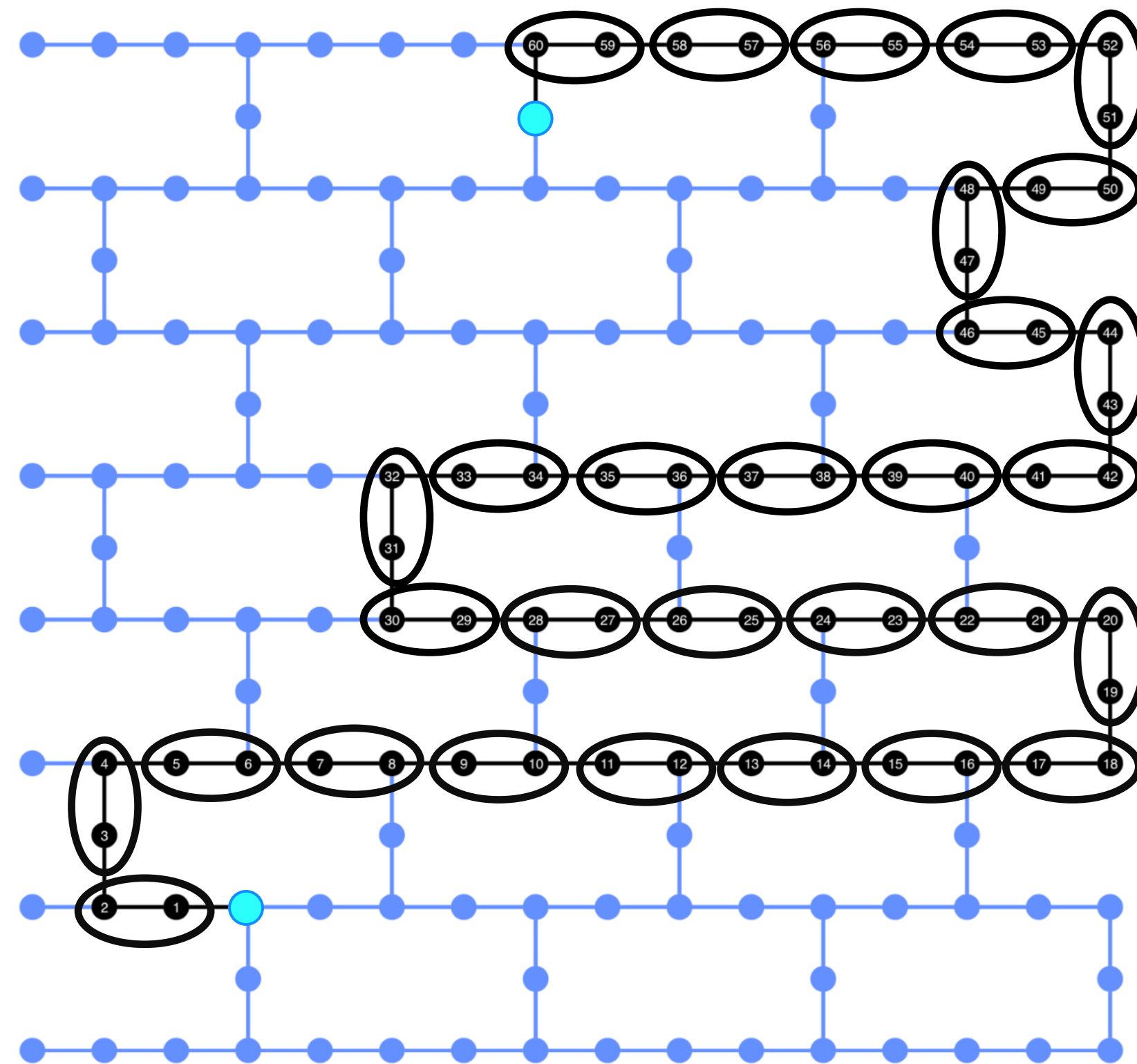
Unitary scheduled circuit ($N_{\text{sites}} = 6$)



`job.result().metadata["compilation"]["scheduler_timing"]["timing"]`

Validation of AKLT state preparation ($N_{\text{sites}} = 30$)

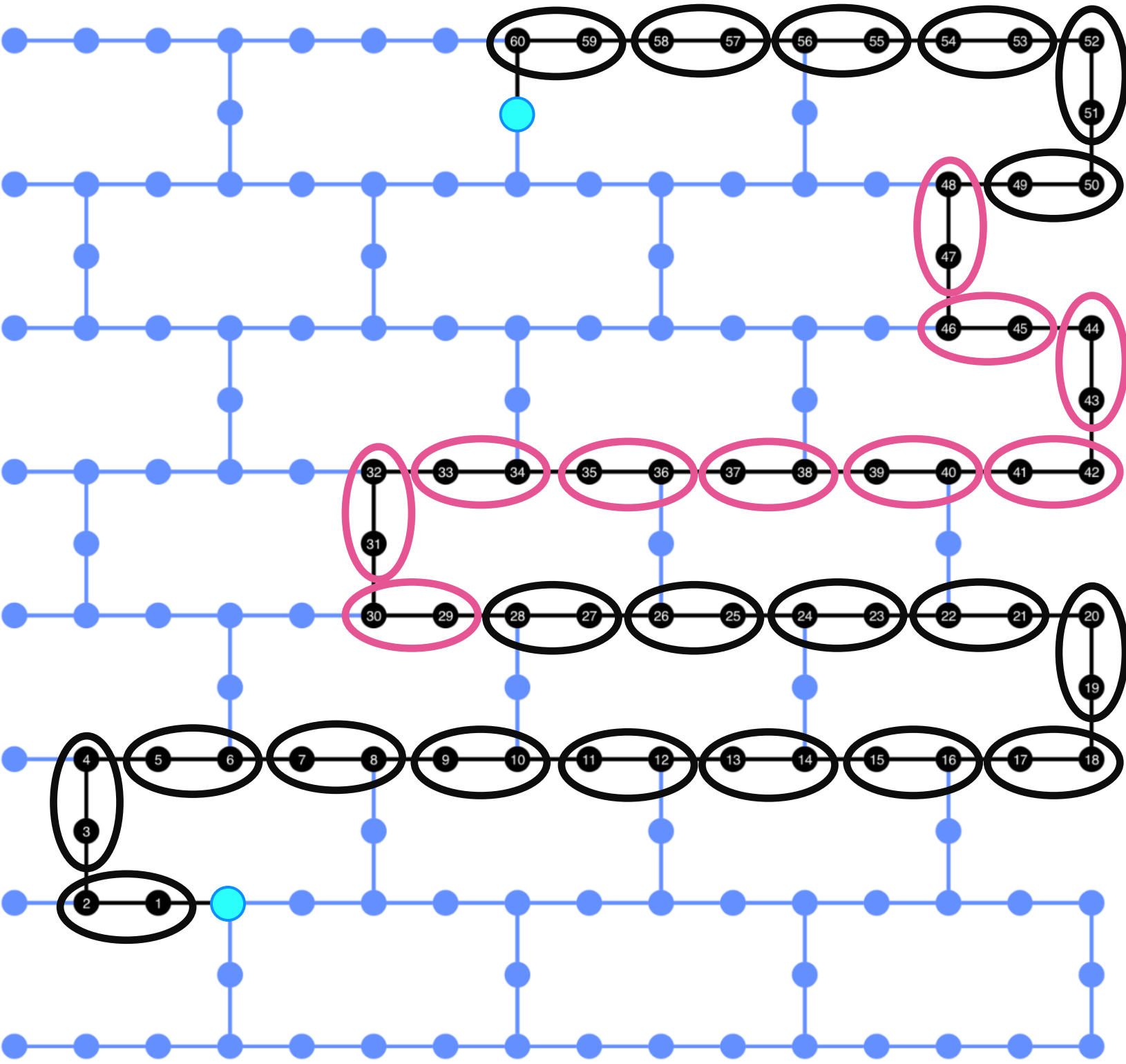
Unitary (62 qubits)



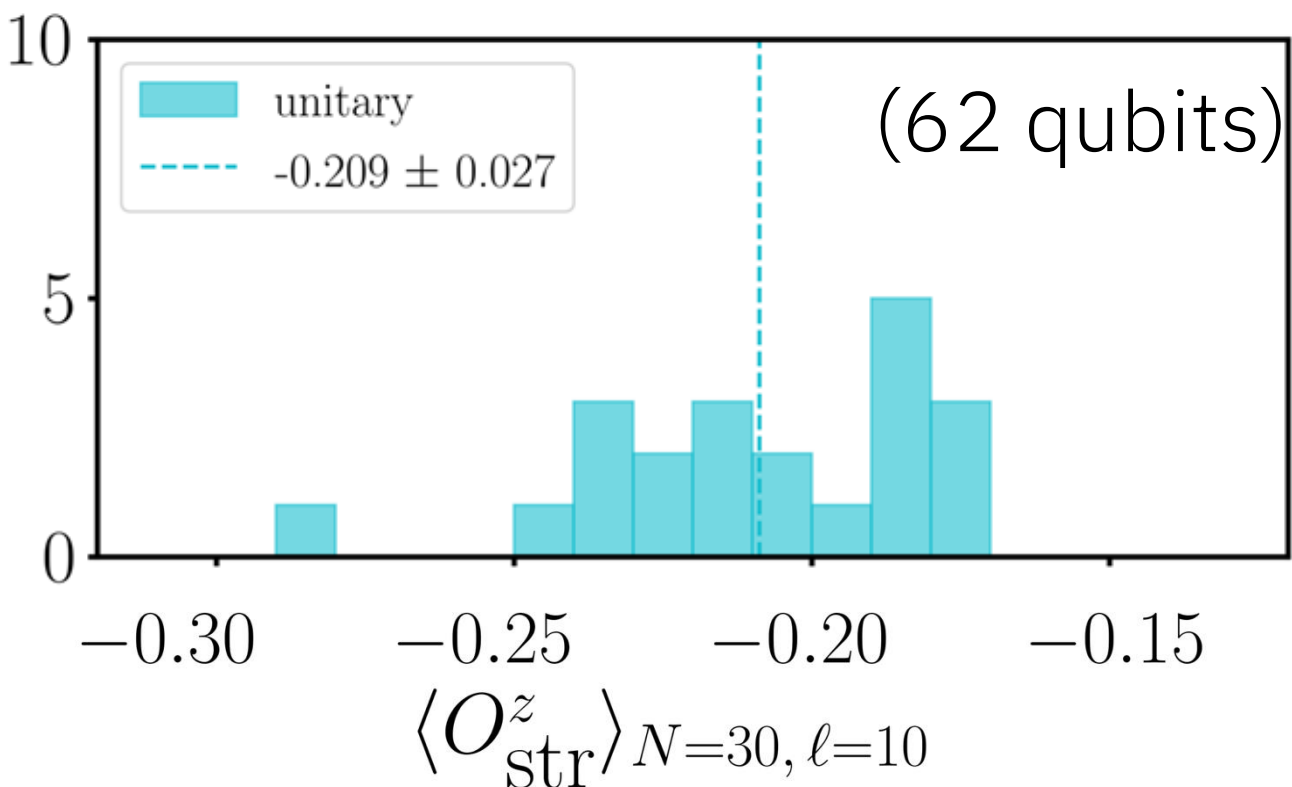
Circuit duration: $\sim 16.3\mu s$

Validation of AKLT state preparation ($N_{\text{sites}} = 30$)

Unitary (62 qubits) $l = 10$



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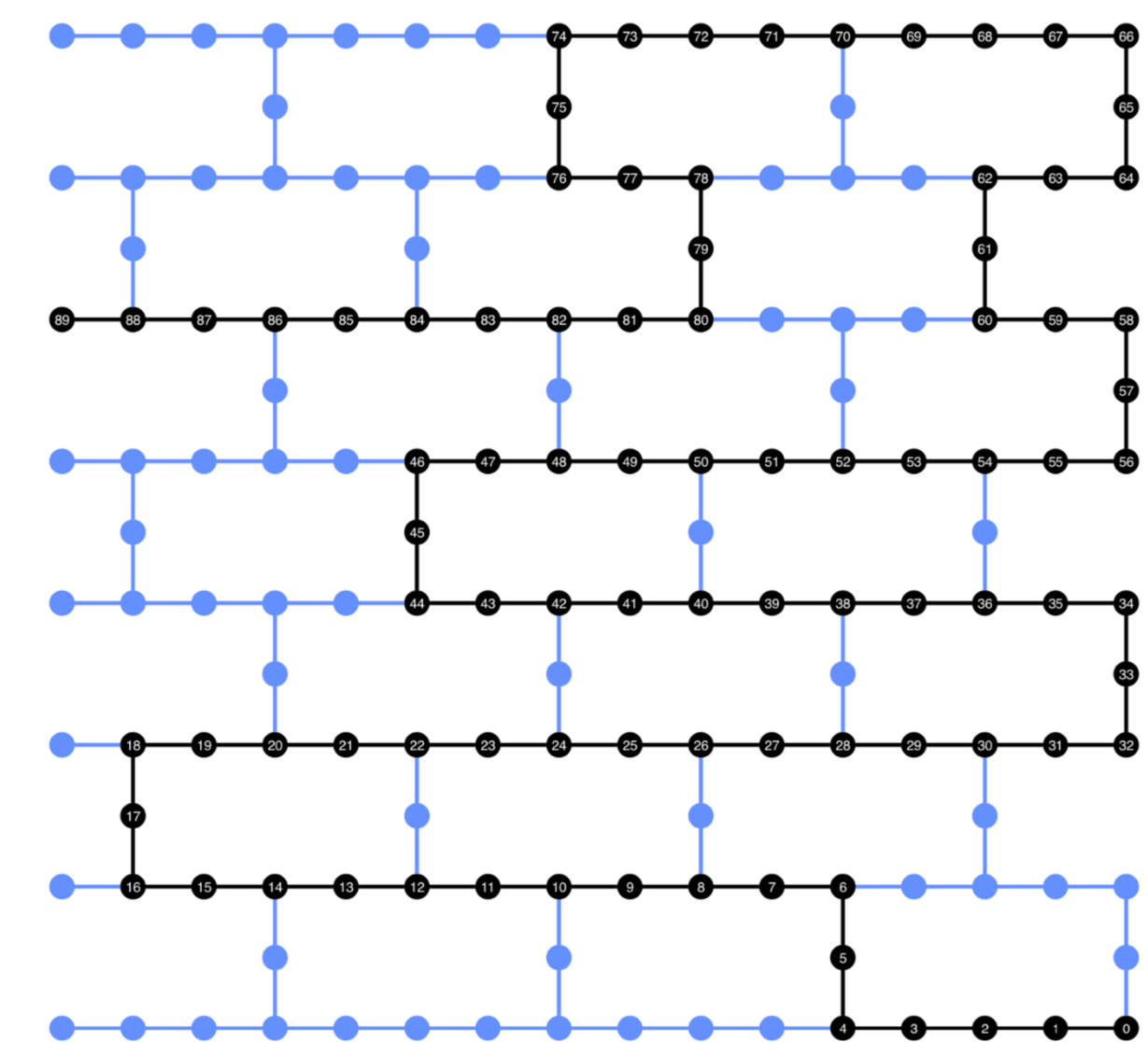


21 x length-10 chains

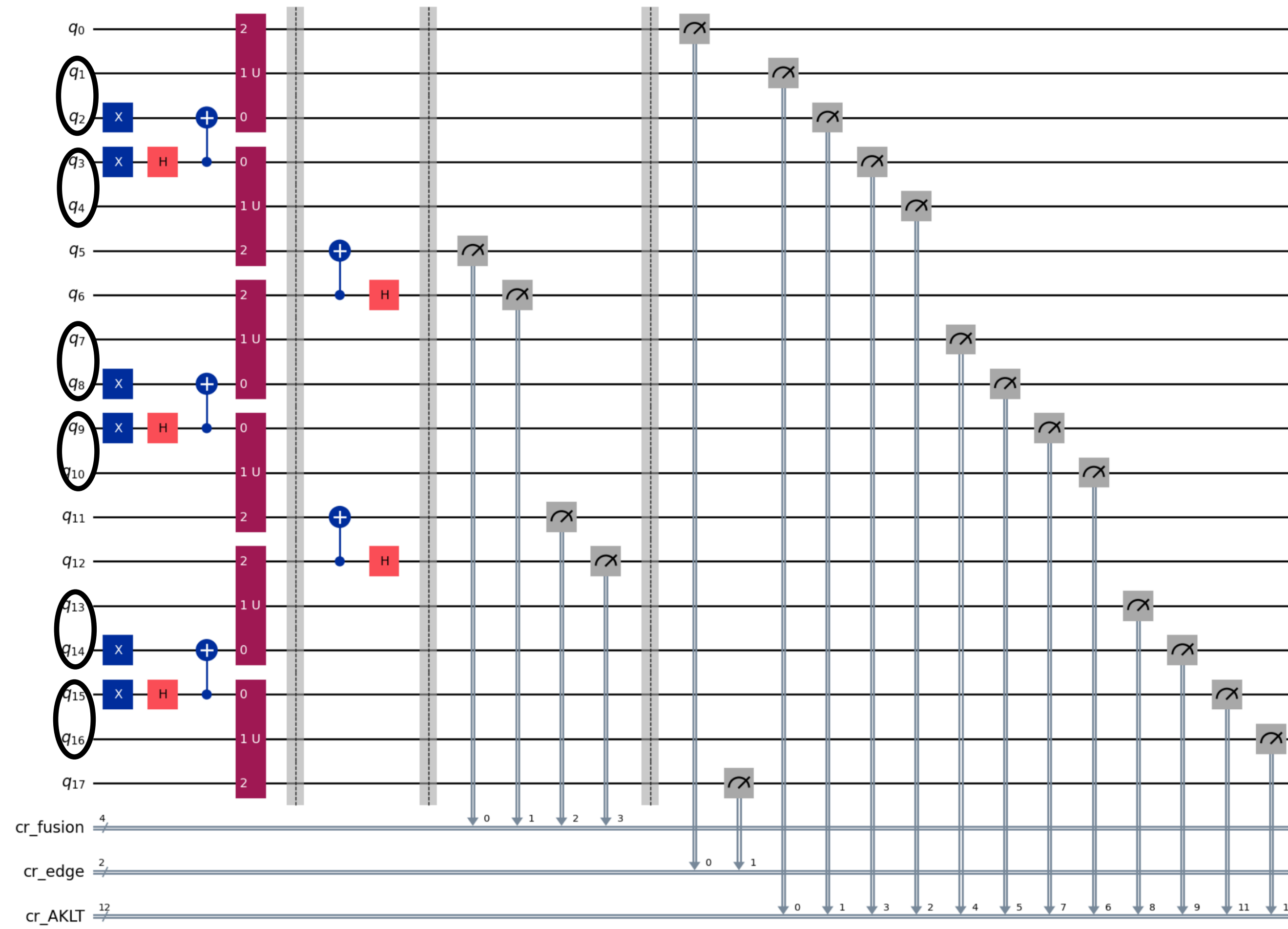
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 1. Unitary ($\sim 16.3\mu s$)
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 4. Visualization (beta) for optimization and debugging

90 qubits for $N_{\text{sites}} = 30$

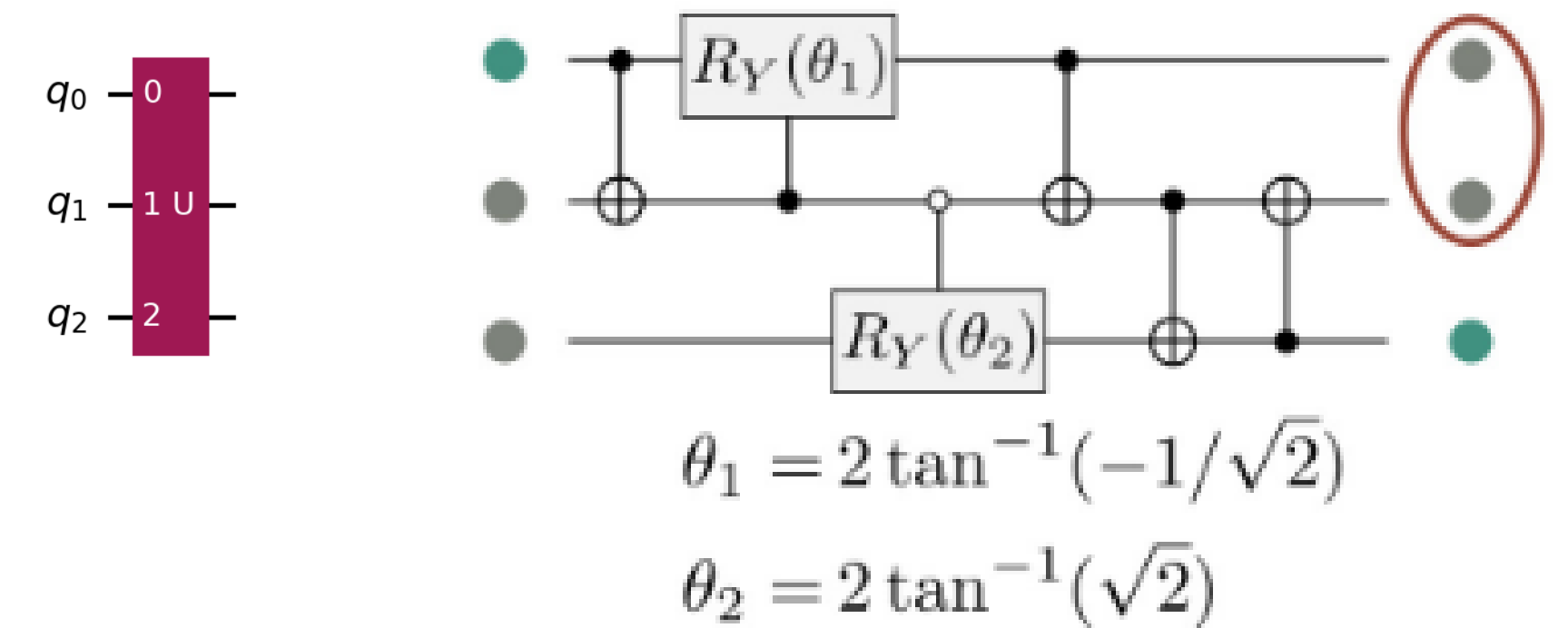


Measurement based AKLT state preparation ($N_{\text{sites}} = 6$)



$Z(\theta)$'s plus: $7 \times CZ + 10 \times Sx$

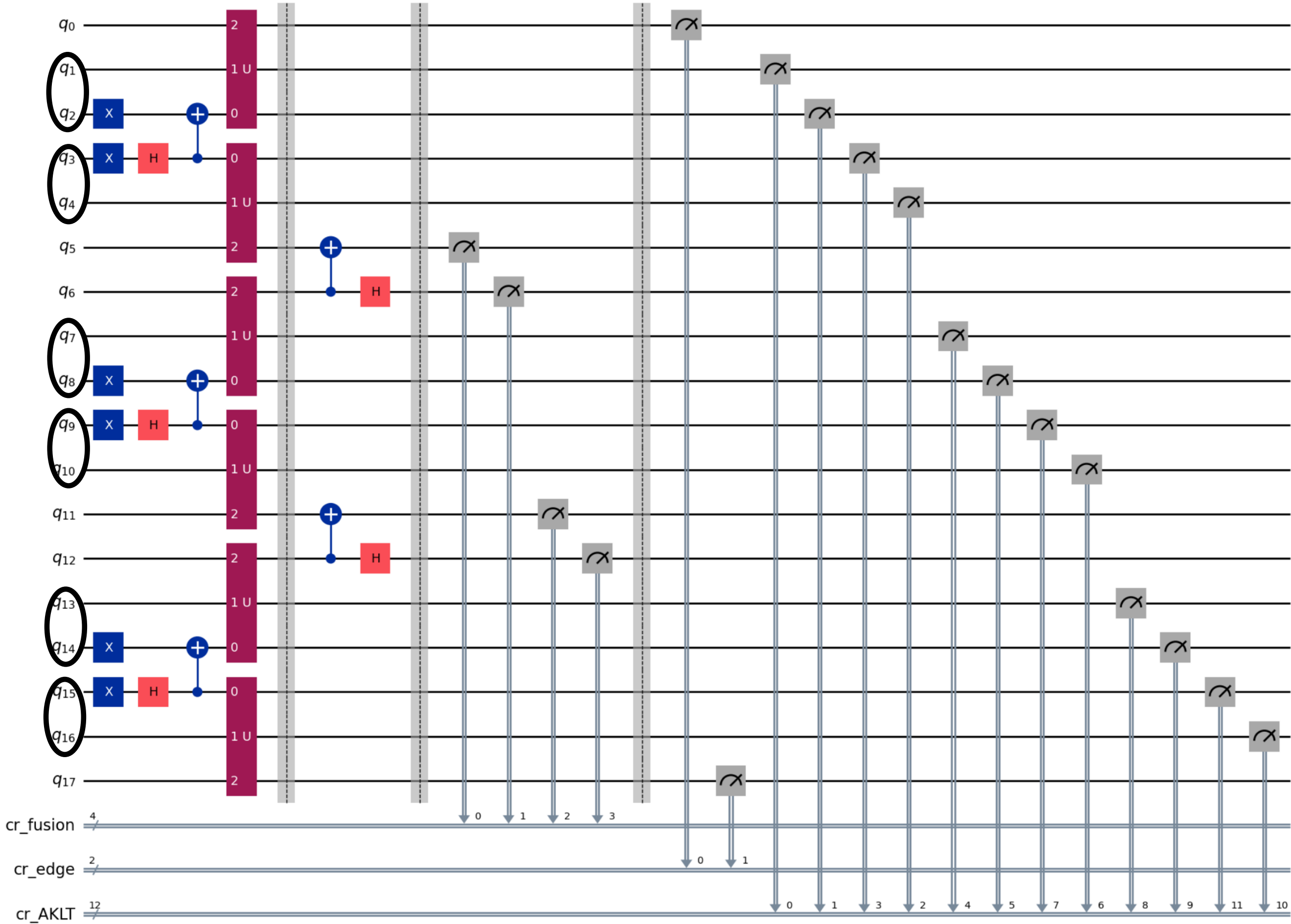
Total number of qubits needed
 $= N_{\text{sites}} \times 2 + \text{ancillas}$
 $= N_{\text{sites}} \times 3 = 18$



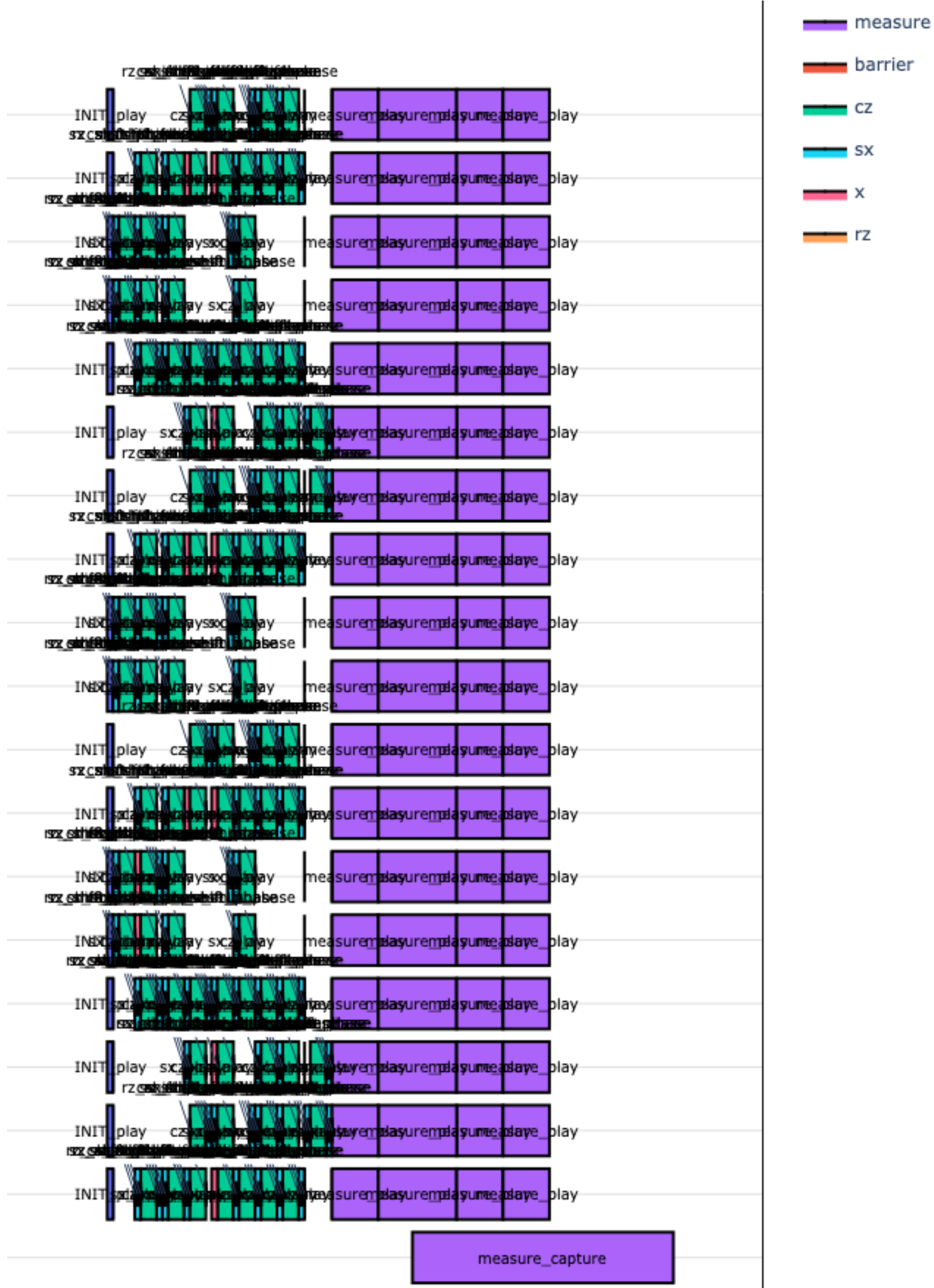
On IBM Herons, transpiles to
 $Rz(\theta)$'s plus:

$7CZ + 10Sx$
 for any N_{sites}

Measurement based scheduled circuit ($N_{\text{sites}} = 6$)

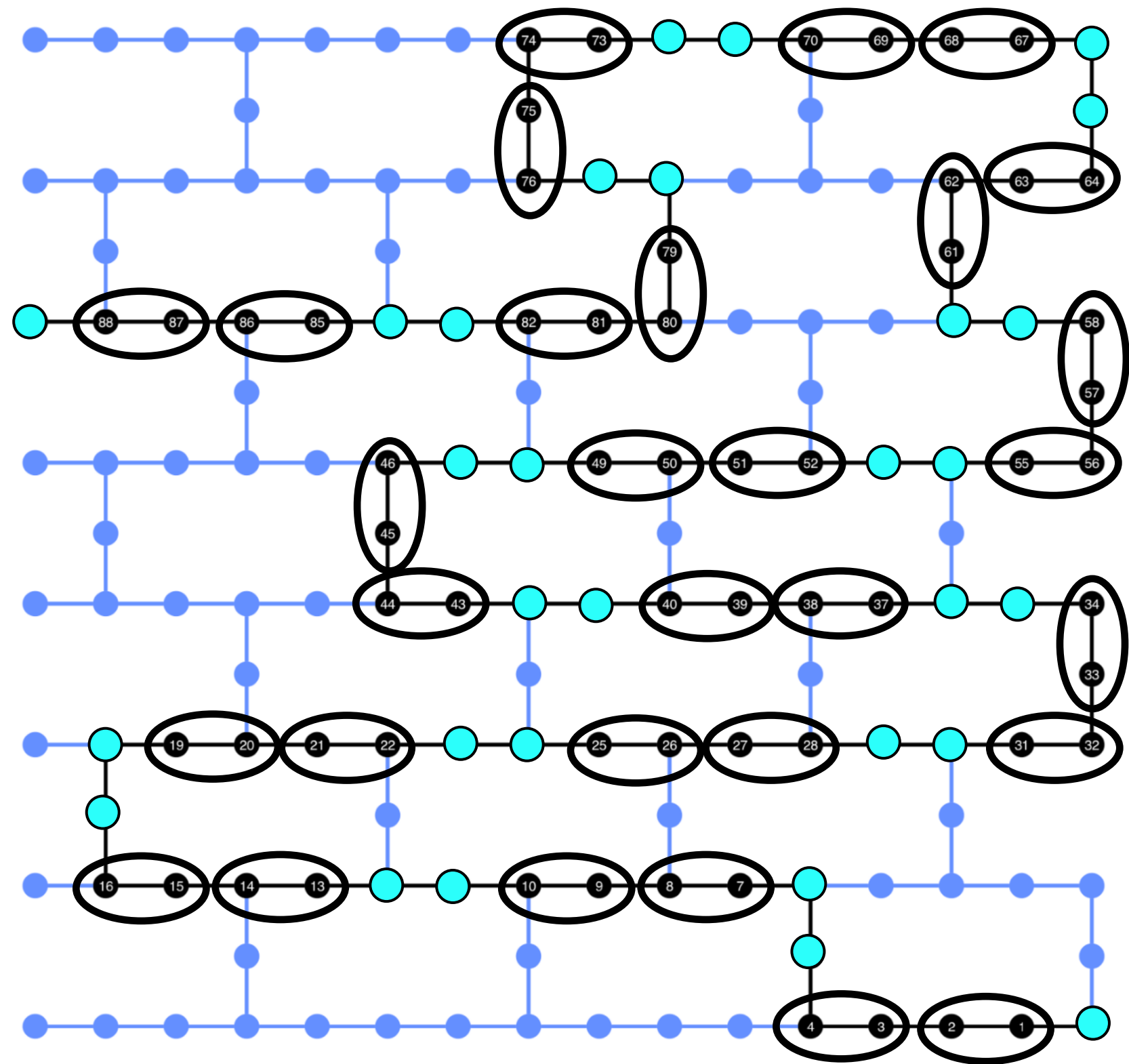


$Z(\theta)$'s plus: $7CZ + 10Sx$

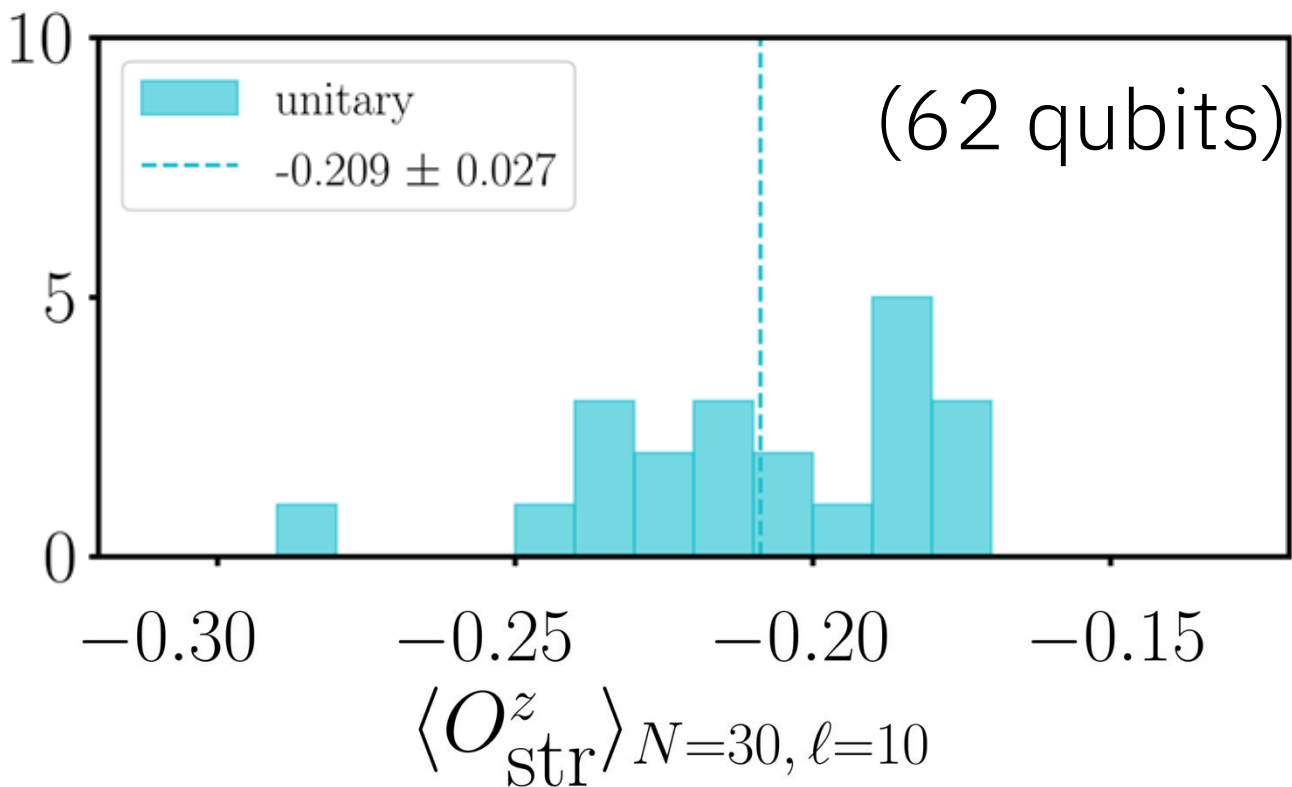


Validation of AKLT state preparation ($N_{\text{sites}} = 30$)

Measurement based (90 qubits)



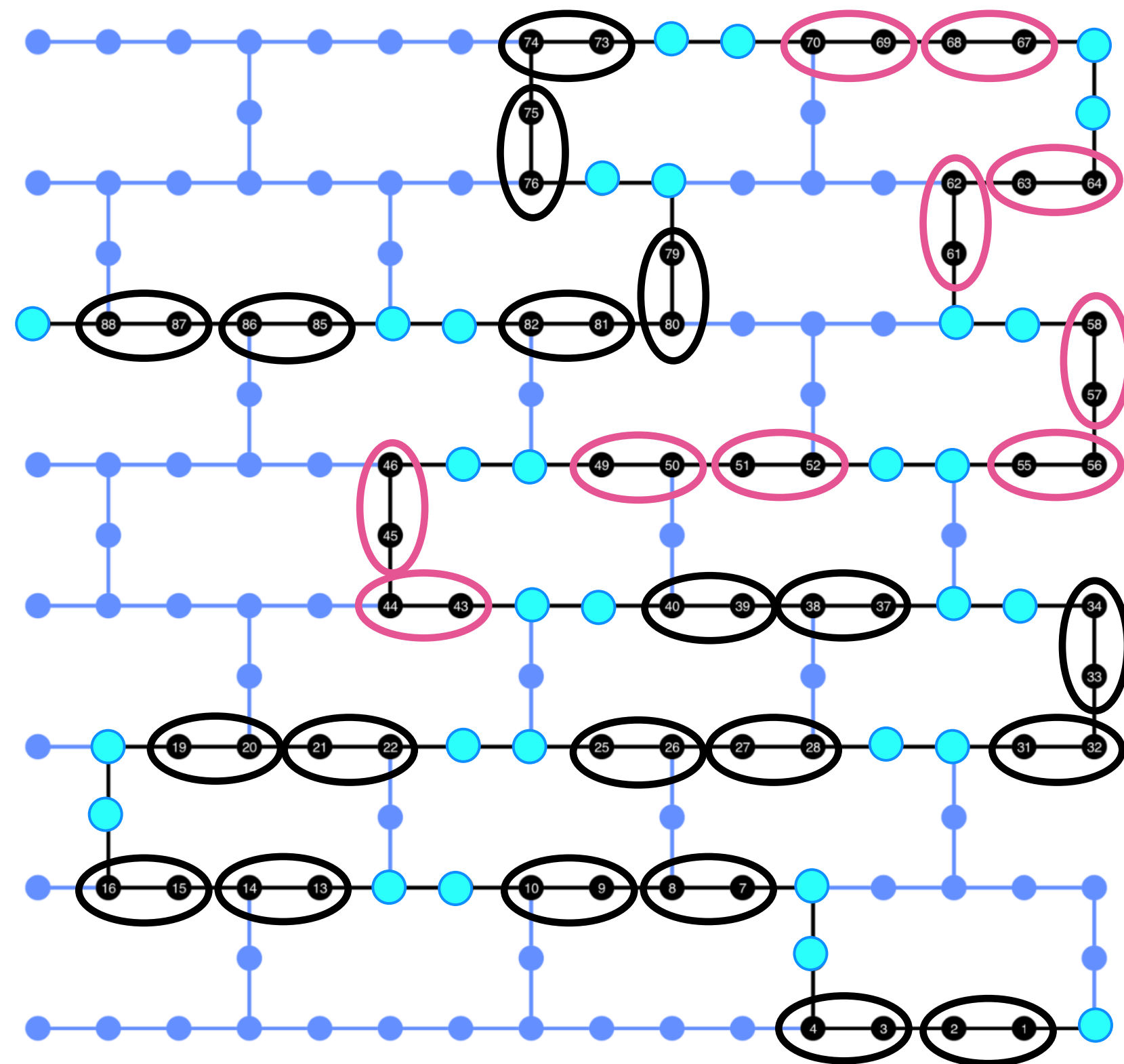
Circuit duration: $\sim 3.8\mu s$



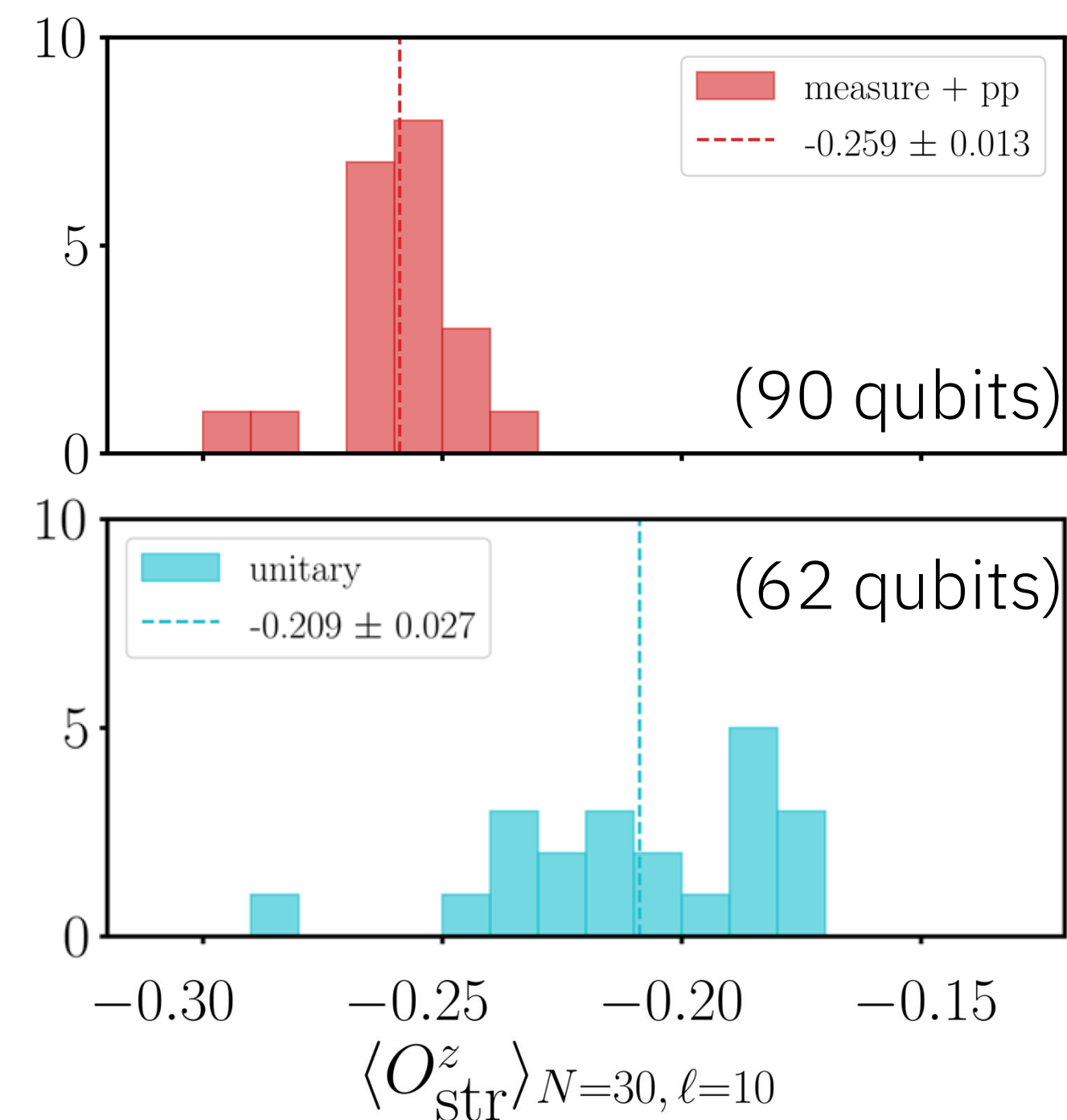
21 x length-10 chains

Validation of AKLT state preparation ($N_{\text{sites}} = 30$)

Measurement based (90 qubits) $l = 10$



Circuit duration: $\sim 3.8\mu\text{s}$

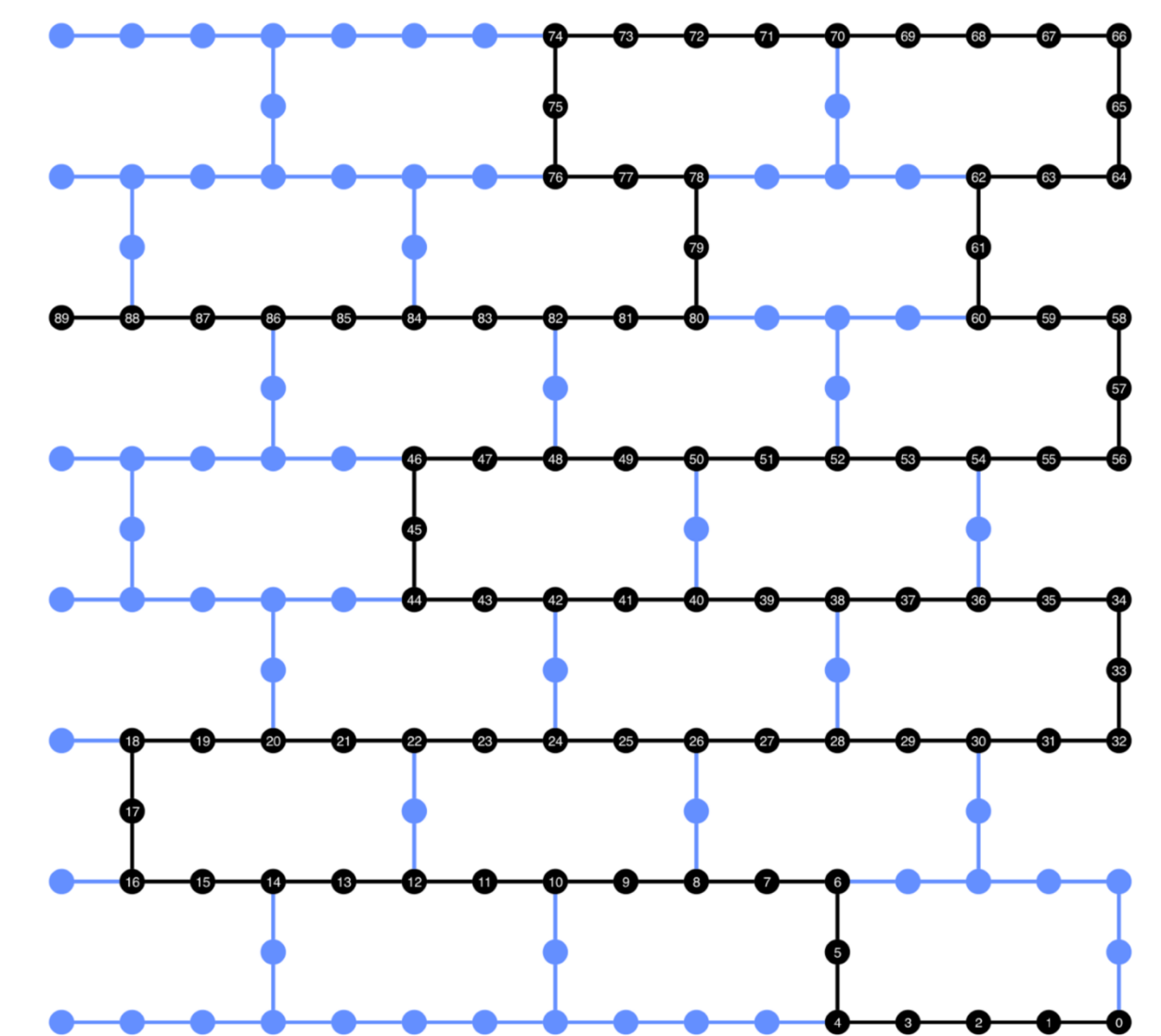


21 x length-10 chains

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90 qubits for $N_{\text{sites}} = 30$



Dynamic AKLT state preparation ($N_{\text{sites}} = 6$)

Total number of qubits needed

$$= N_{\text{sites}} \times 2 + \text{ancillas}$$

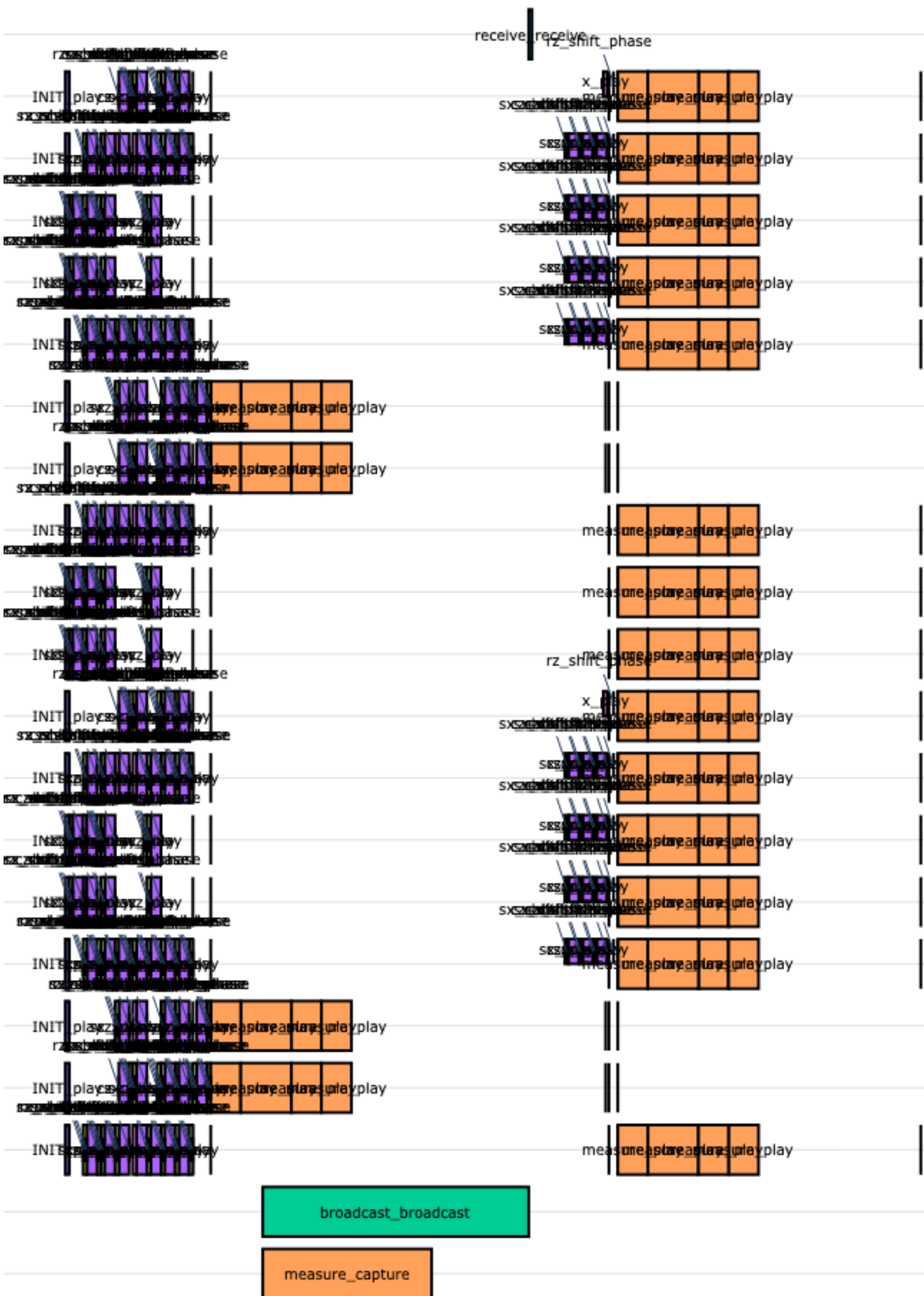
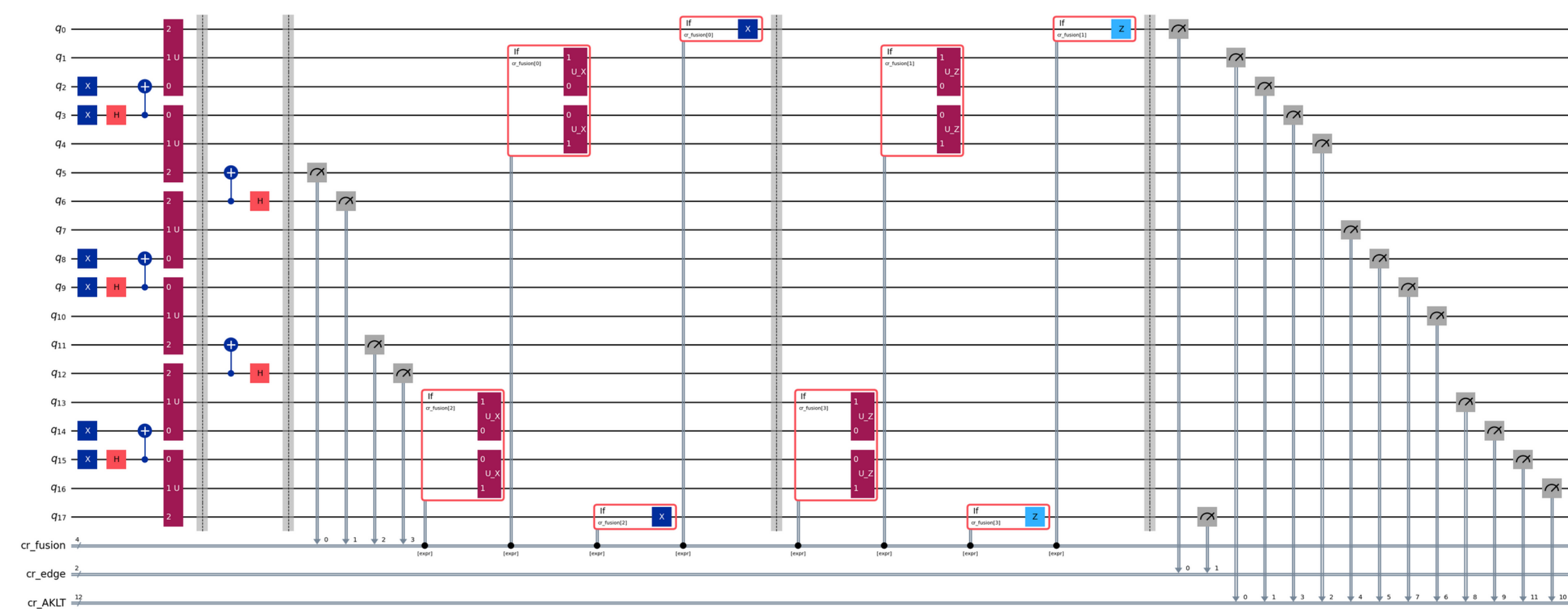
$$= N_{\text{sites}} \times 3 = 18$$



$Z(\theta)$'s plus:

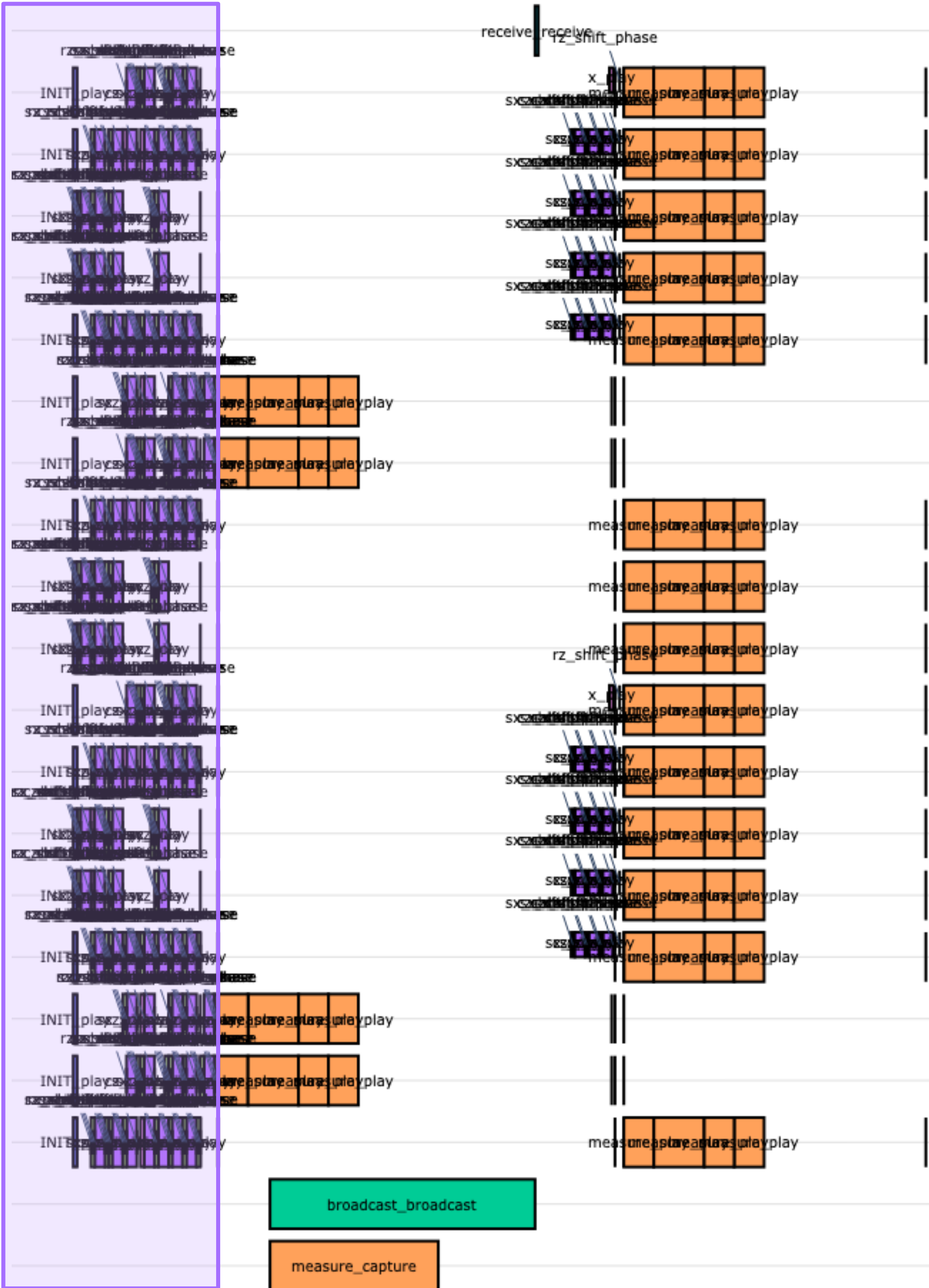
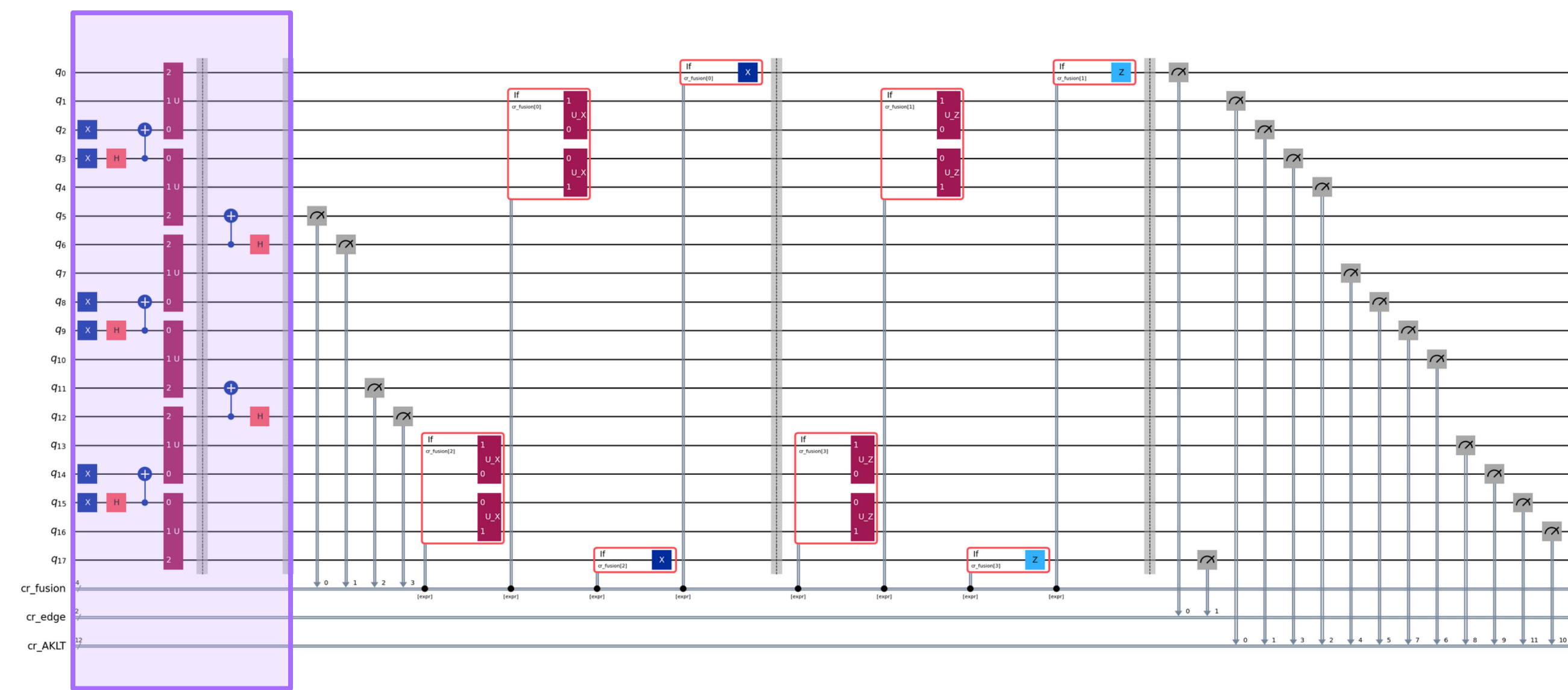
$$7CZ + 10Sx + SWAP (3CZ + 3Sx) + fwd$$

Dynamic scheduled circuit ($N_{\text{sites}} = 6$)



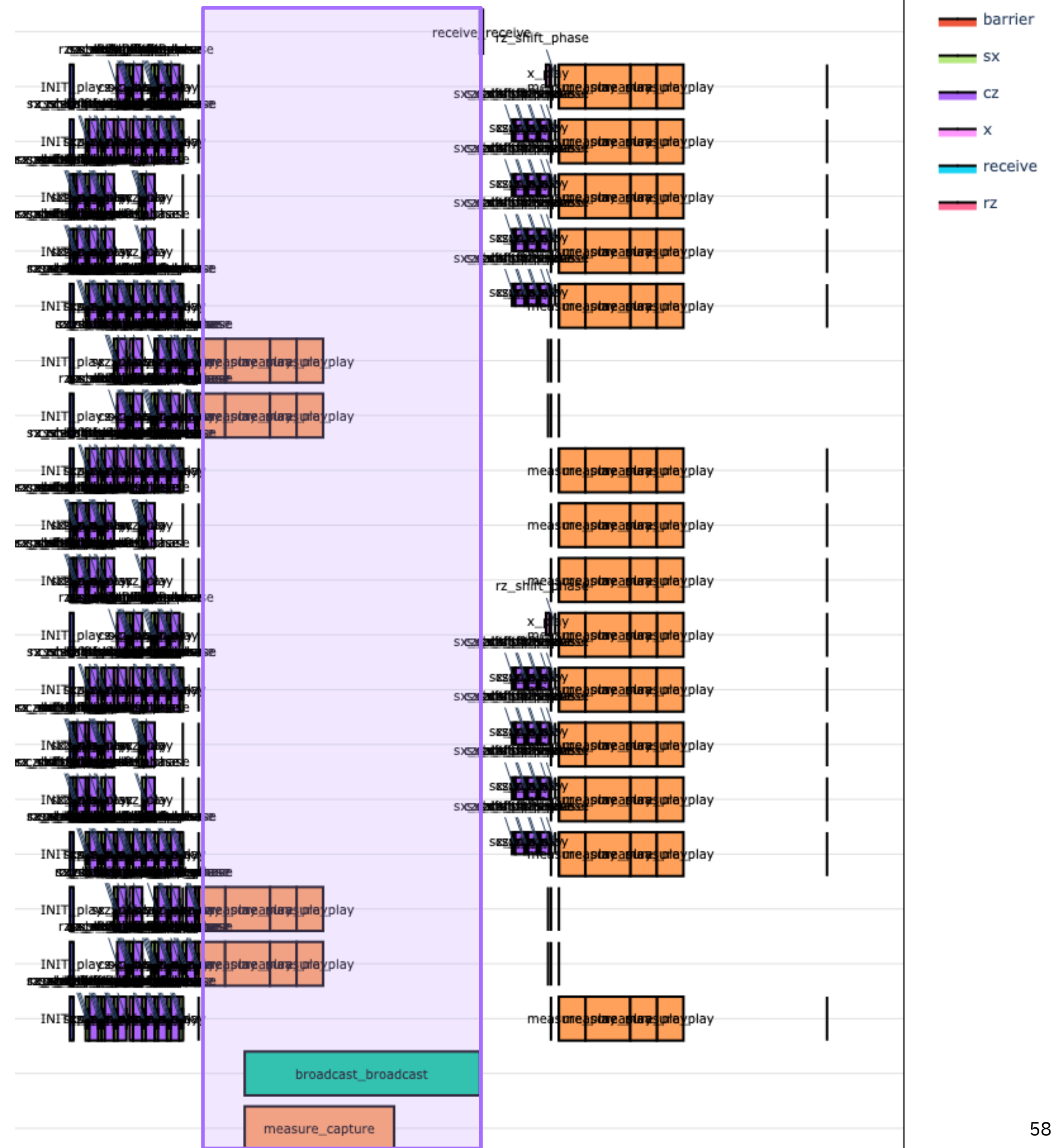
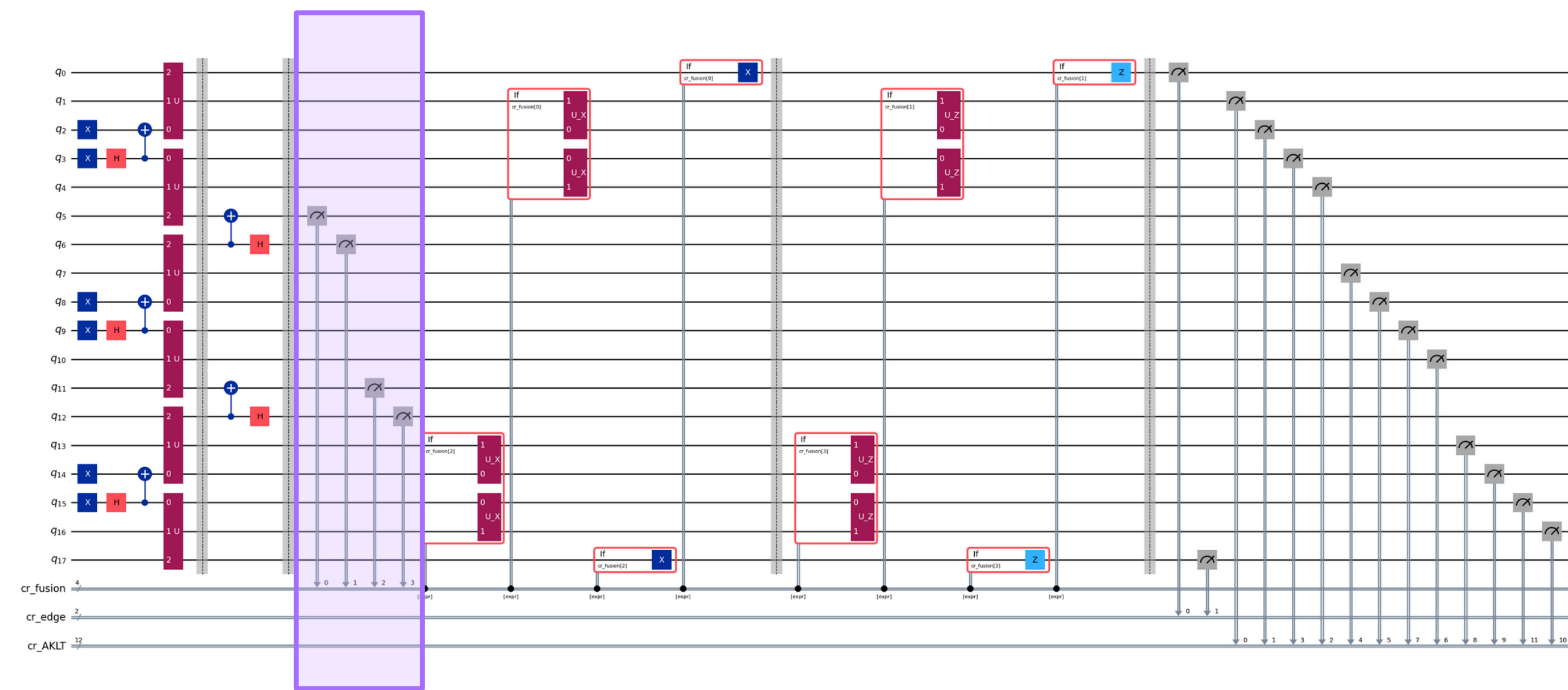
- measure
- broadcast
- barrier
- SX
- CZ
- X
- receive
- RZ

Dynamic scheduled circuit ($N_{\text{sites}} = 6$)

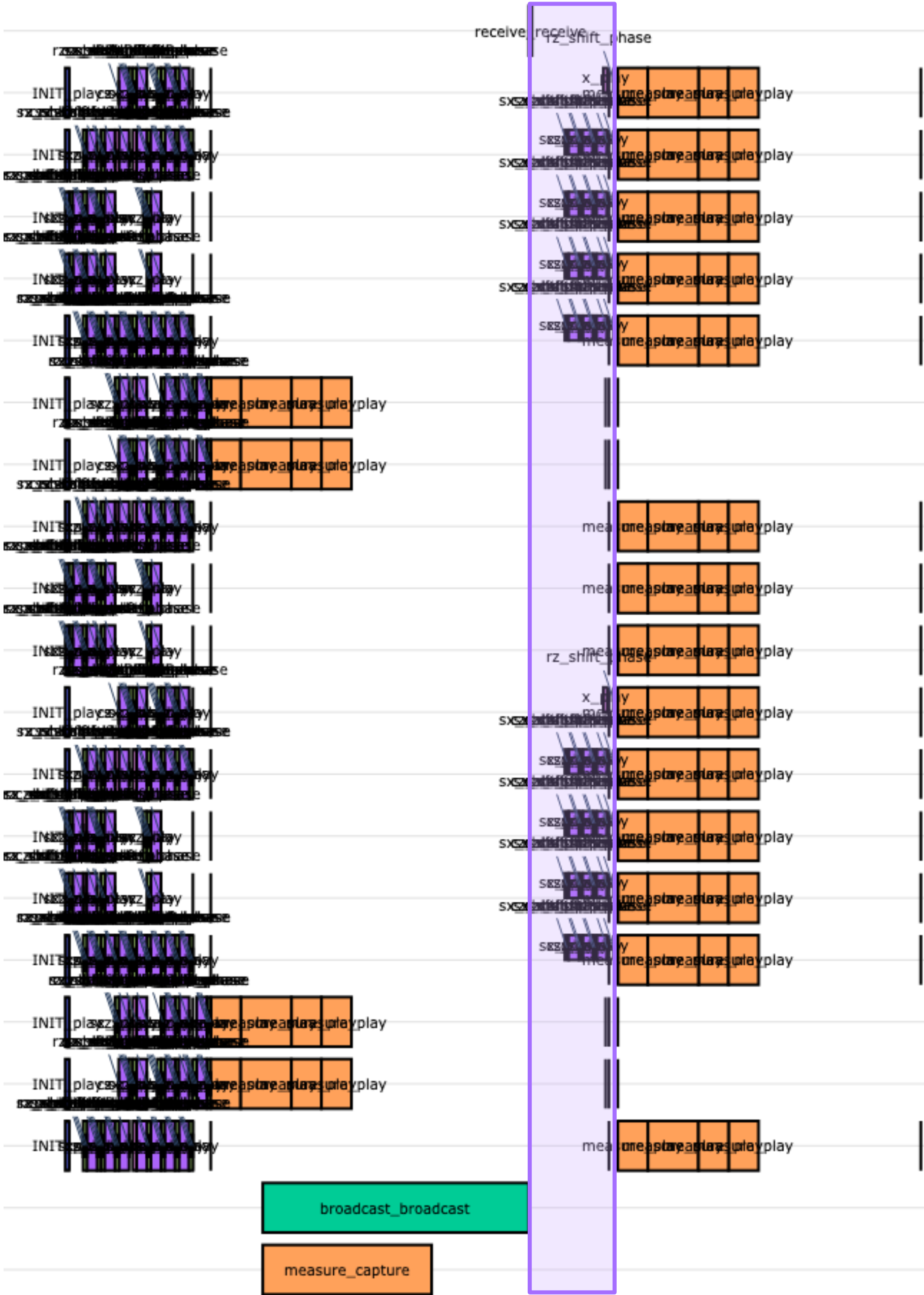
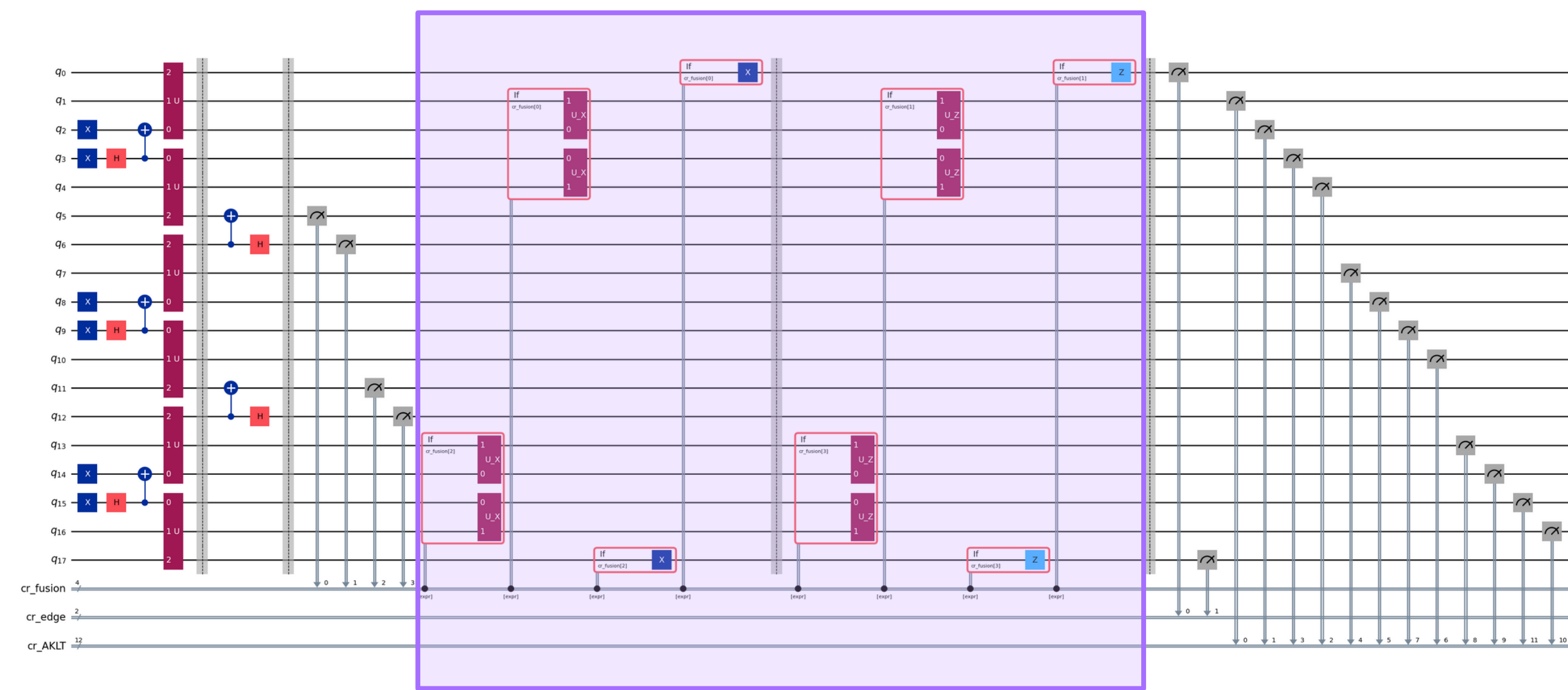


- measure
- broadcast
- barrier
- SX
- CZ
- X
- receive
- RZ

Dynamic scheduled circuit ($N_{\text{sites}} = 6$)

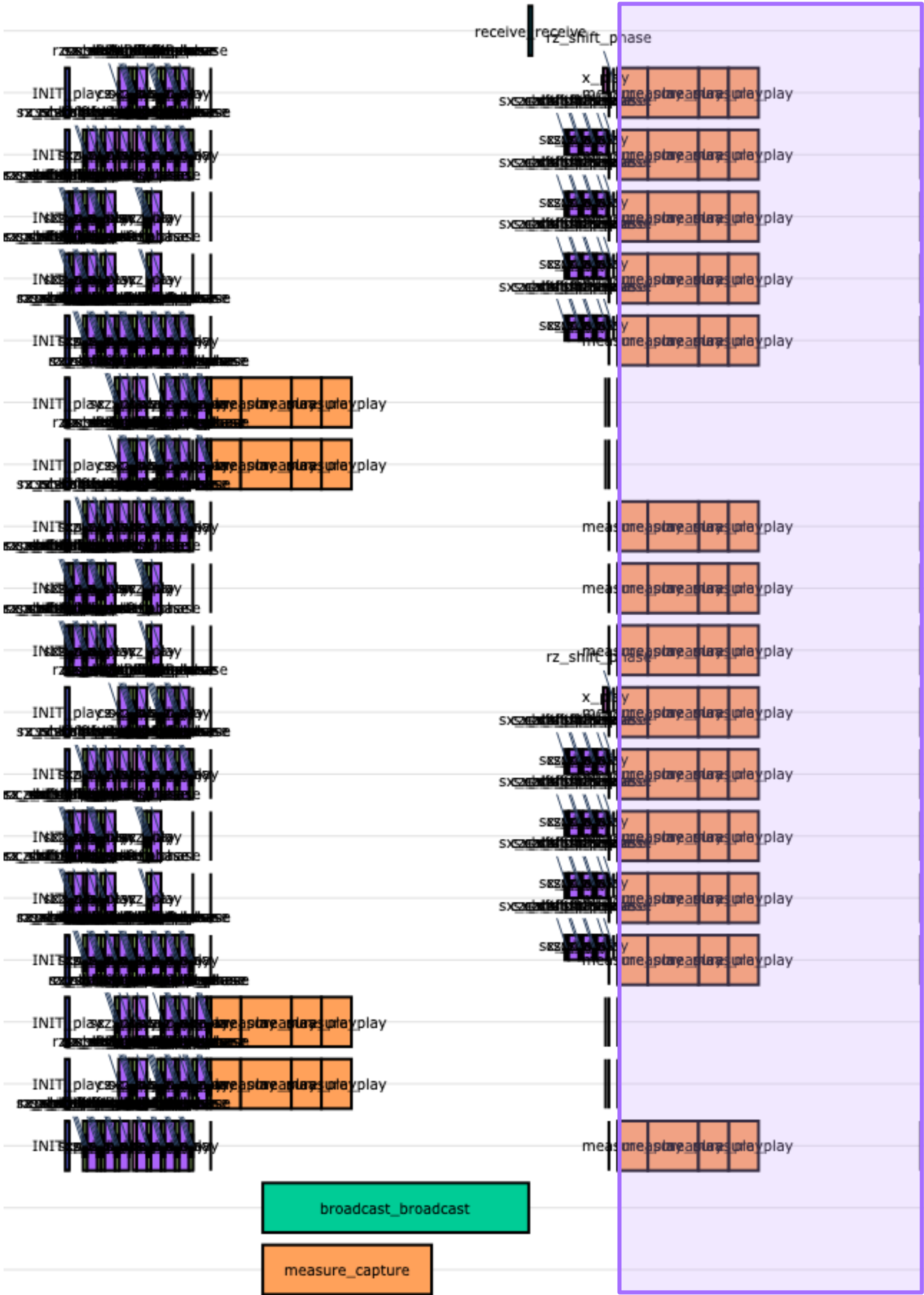
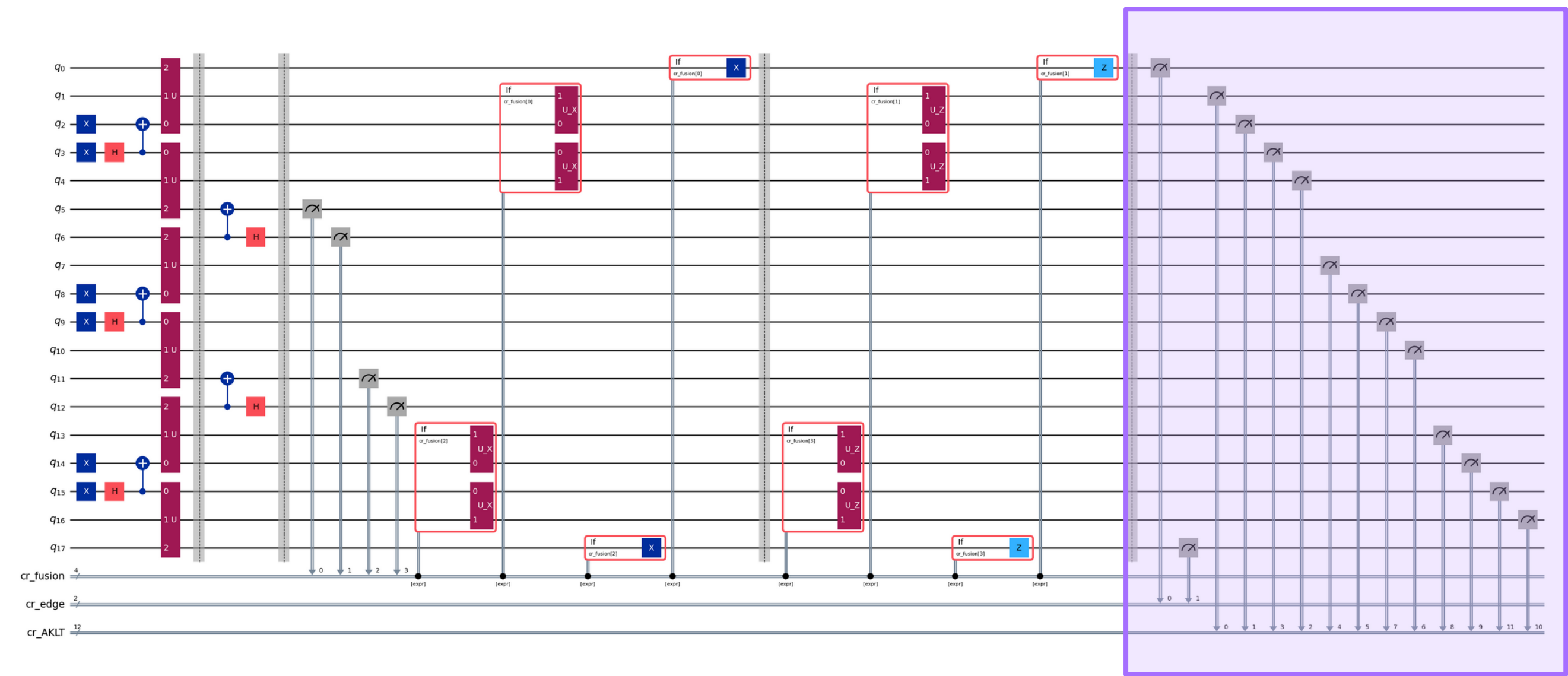


Dynamic scheduled circuit ($N_{\text{sites}} = 6$)



- measure
- broadcast
- barrier
- SX
- CZ
- X
- receive
- RZ

Dynamic scheduled circuit ($N_{\text{sites}} = 6$)

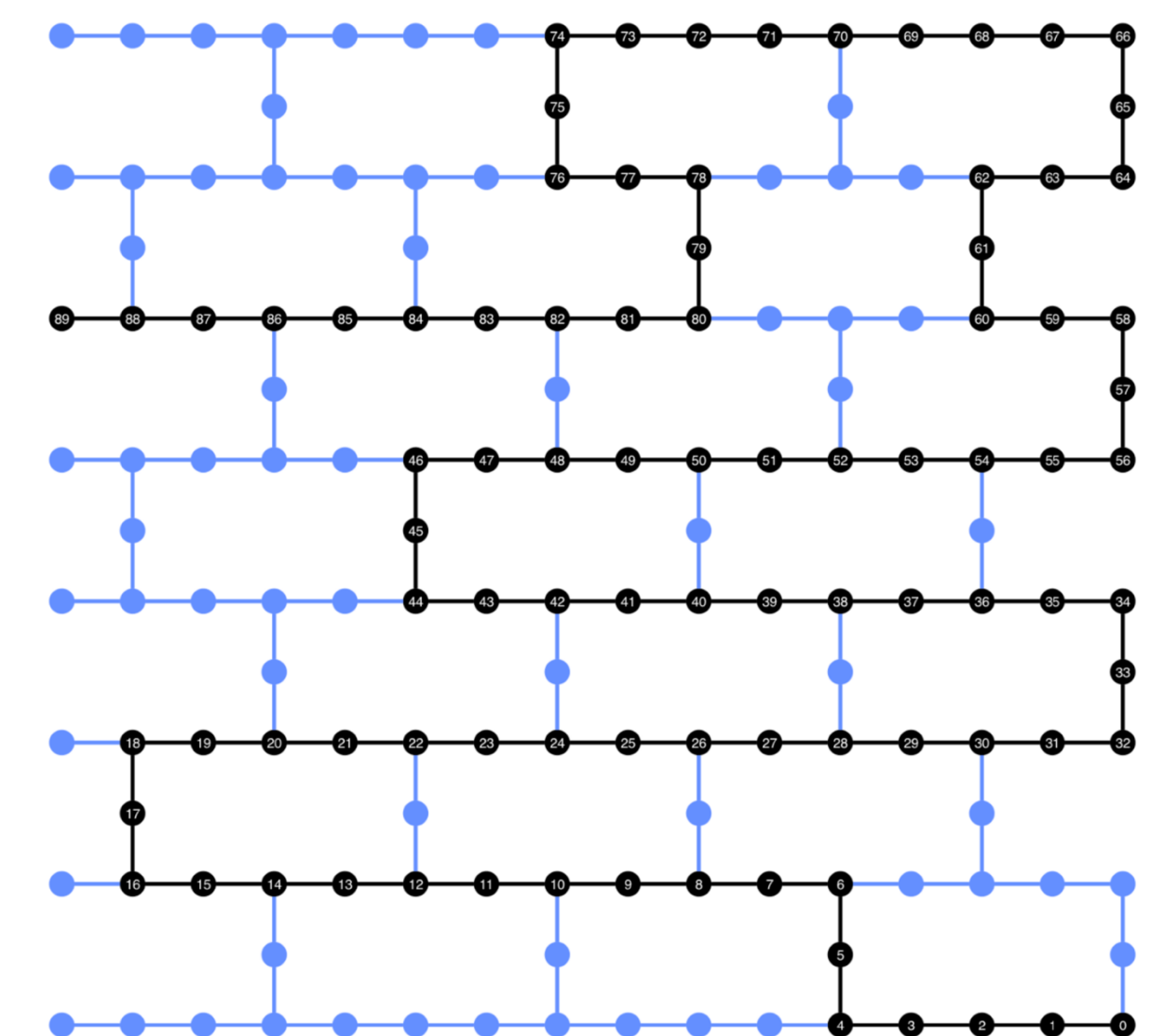


- measure
- broadcast
- barrier
- SX
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- X
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- RZ

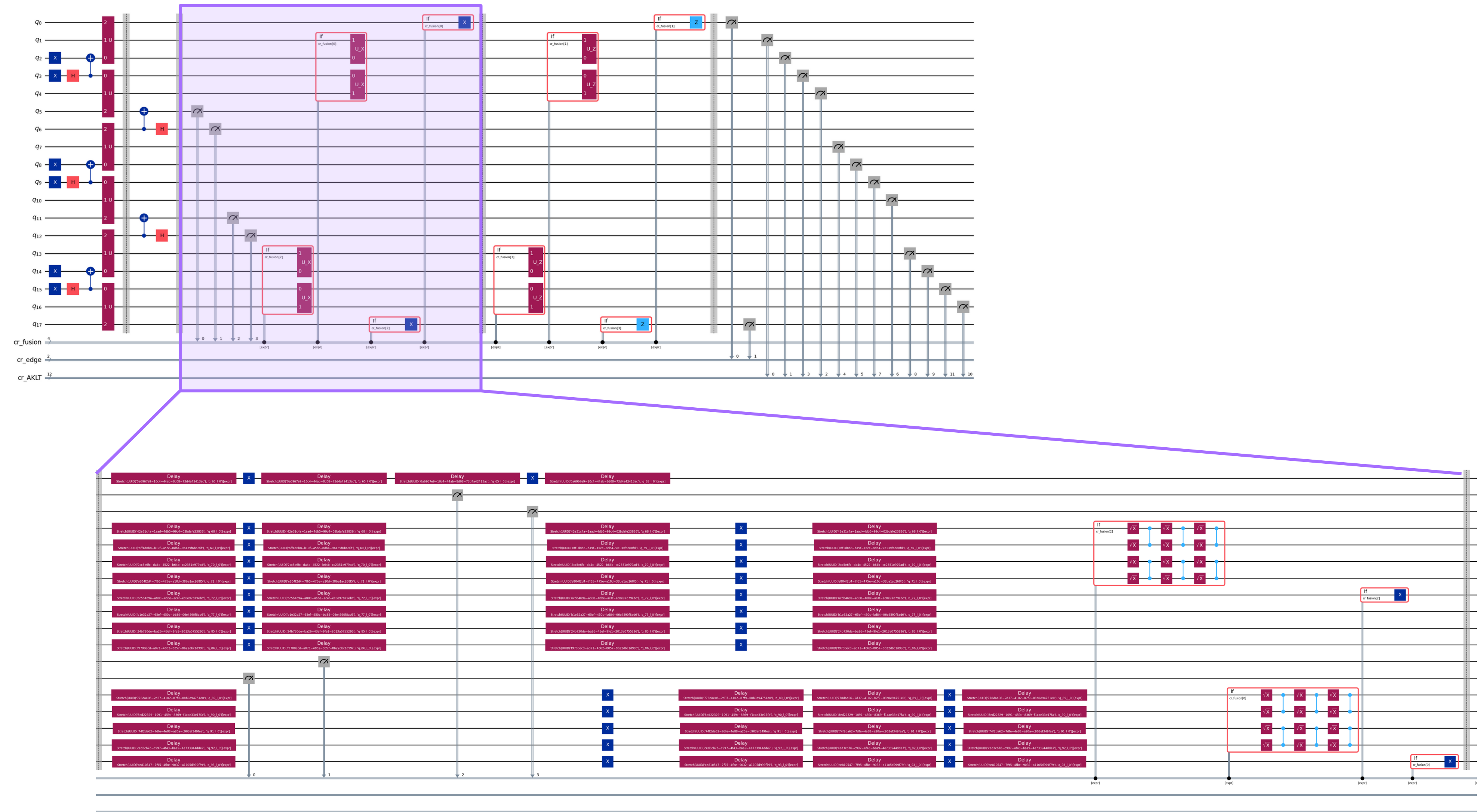
AKLT state preparation with utility scale dynamic circuits

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 4. Visualization (beta) for optimization and debugging

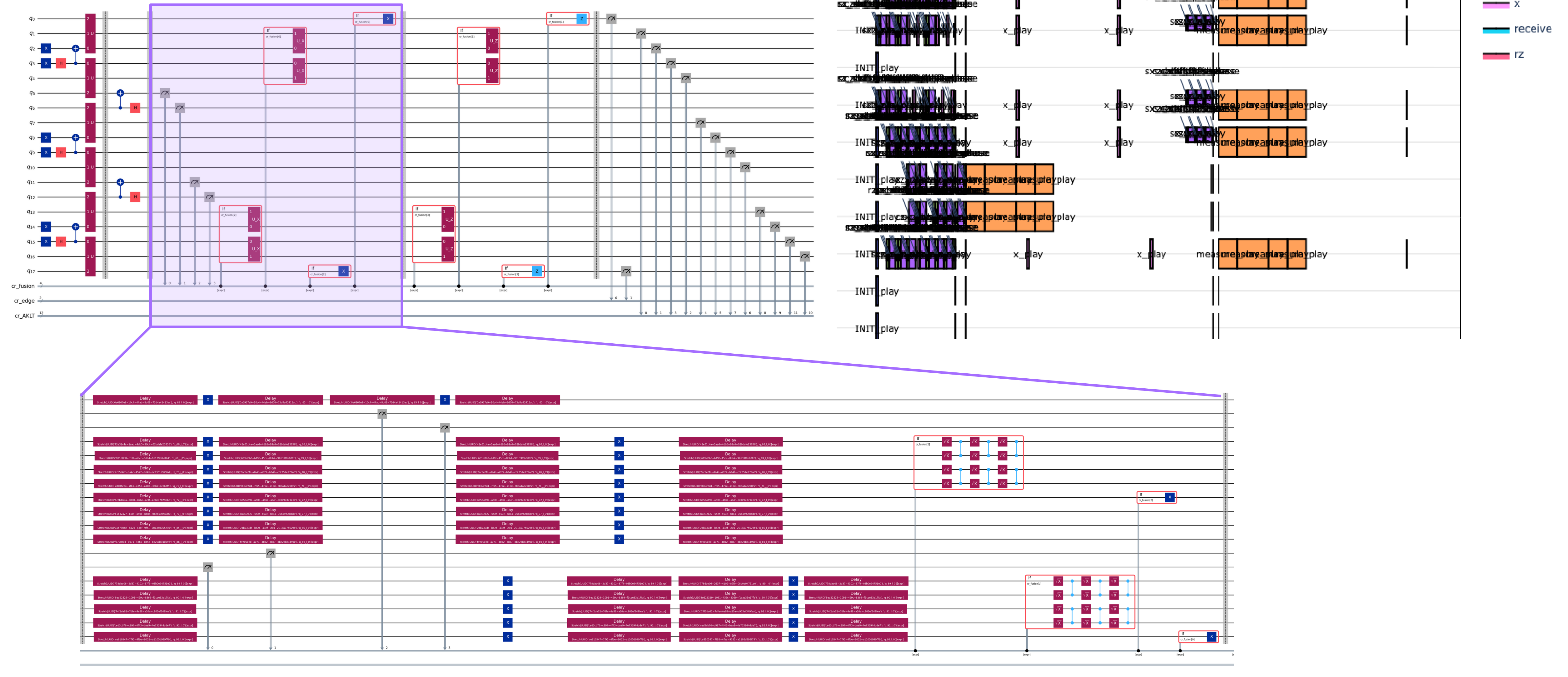
90 qubits for $N_{\text{sites}} = 30$



Dynamic with stretch scheduled circuit ($N_{\text{sites}} = 6$)



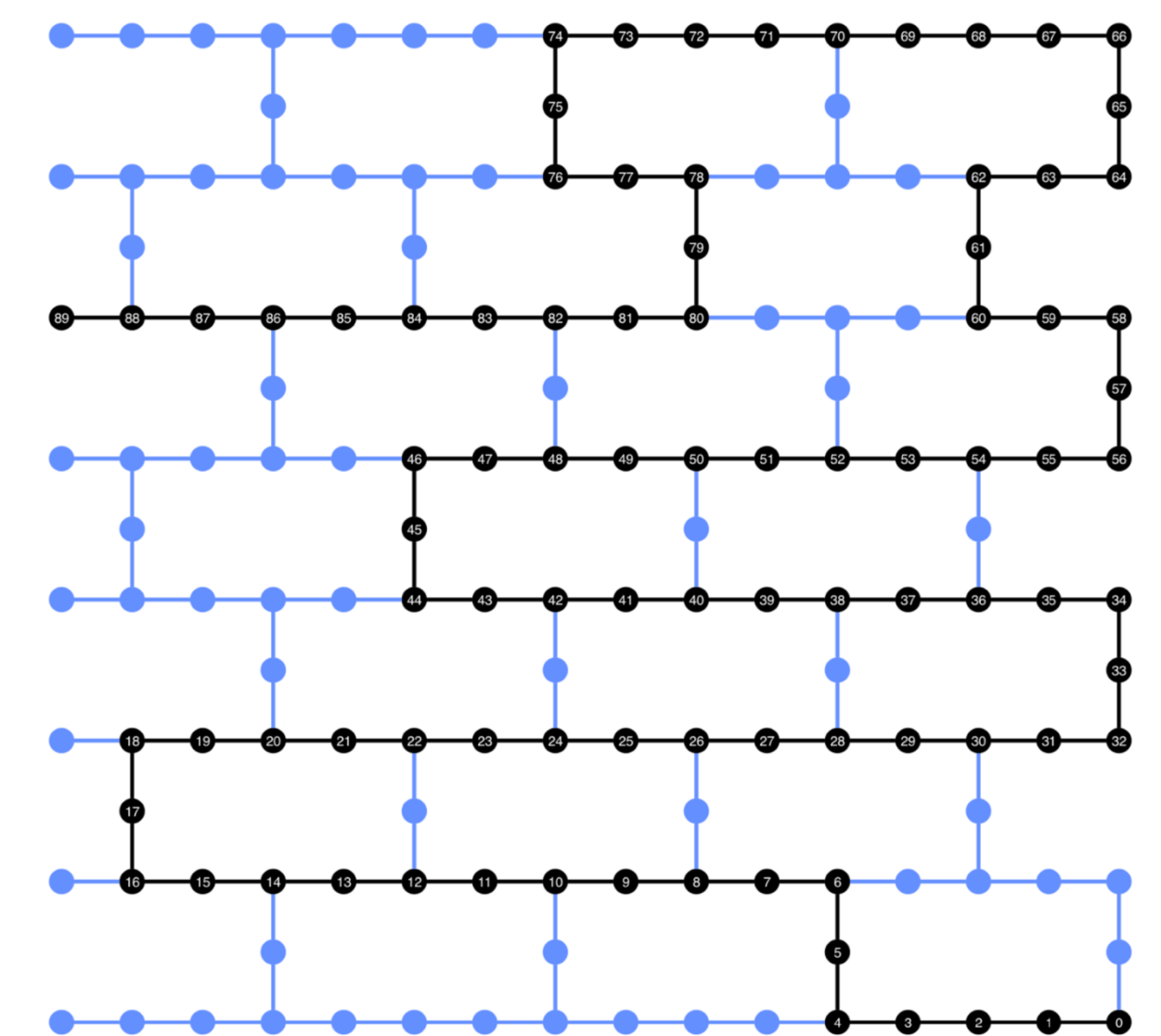
Dynamic with stretch scheduled circuit ($N_{\text{sites}} = 6$)



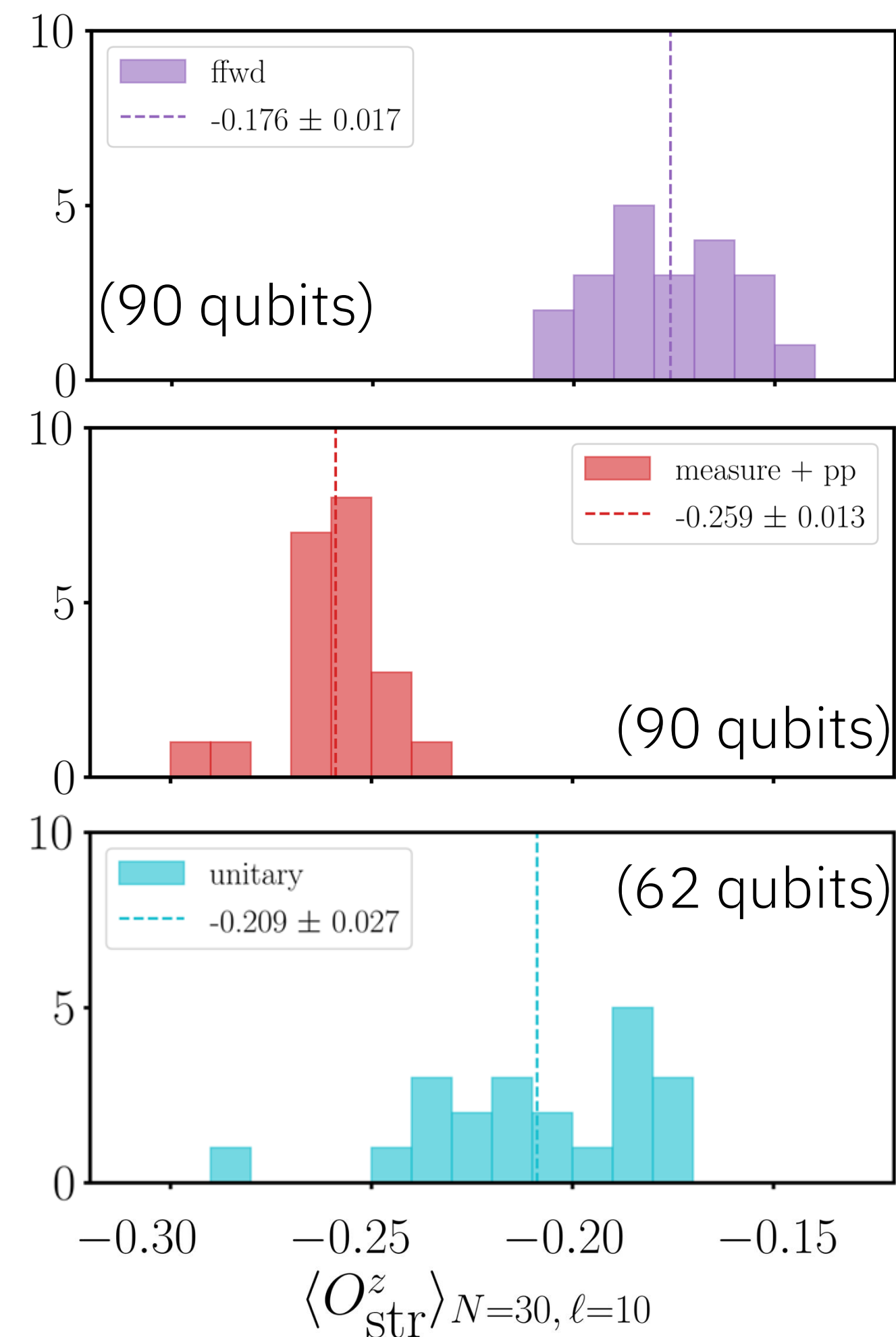
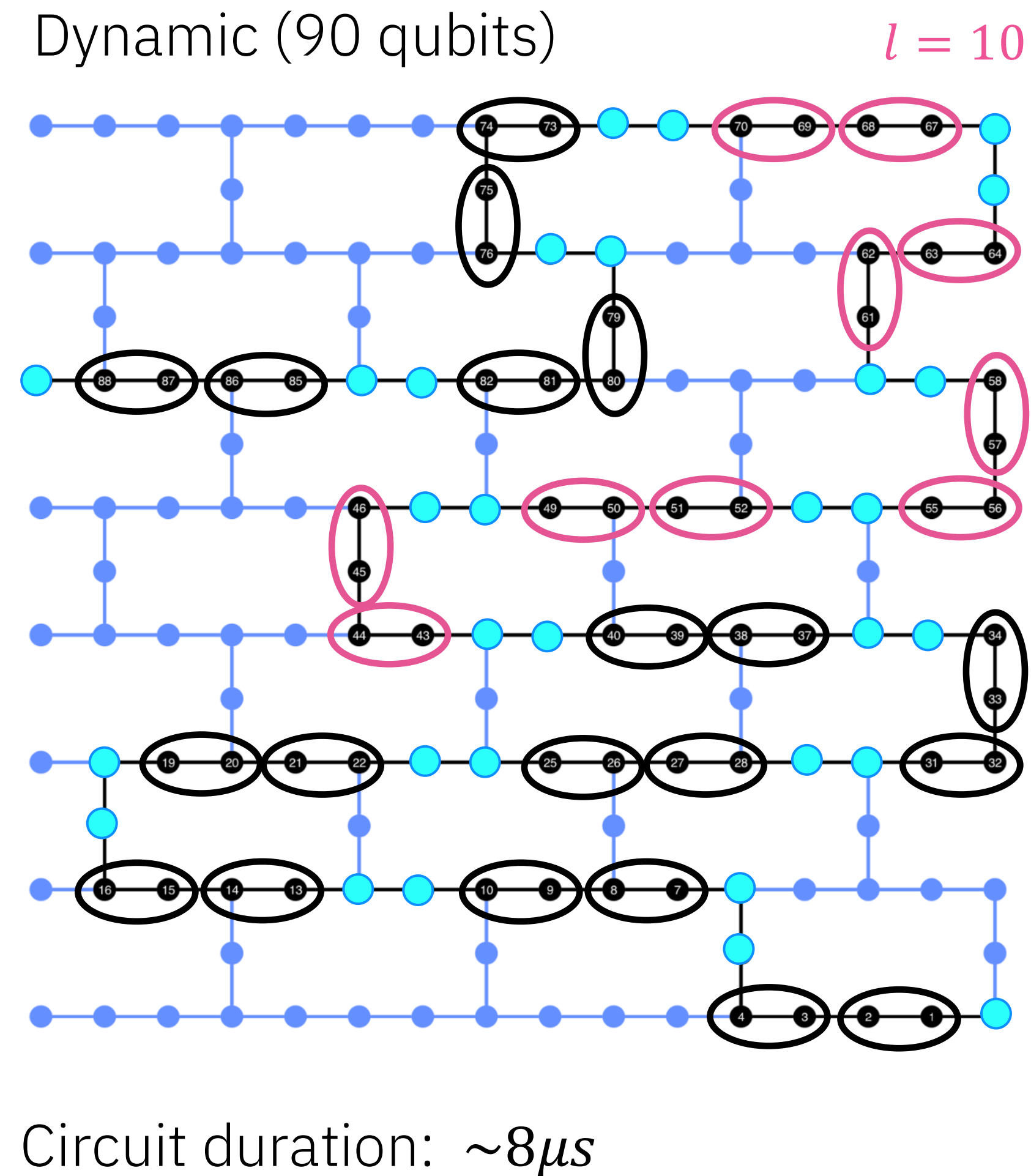
AKLT state preparation with utility scale dynamic circuits

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90 qubits for $N_{\text{sites}} = 30$



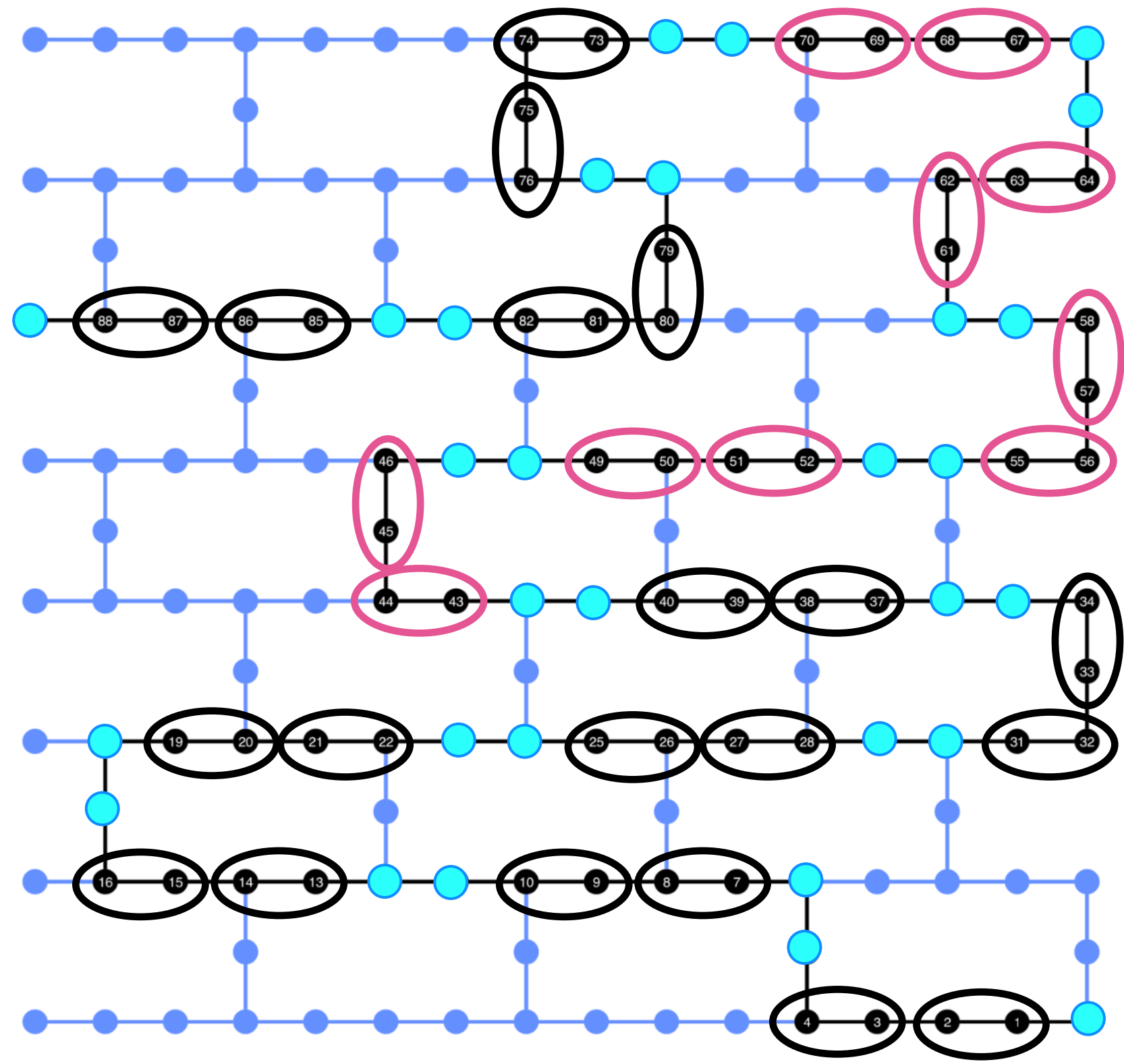
Validation of AKLT state preparation ($N_{\text{sites}} = 30$)



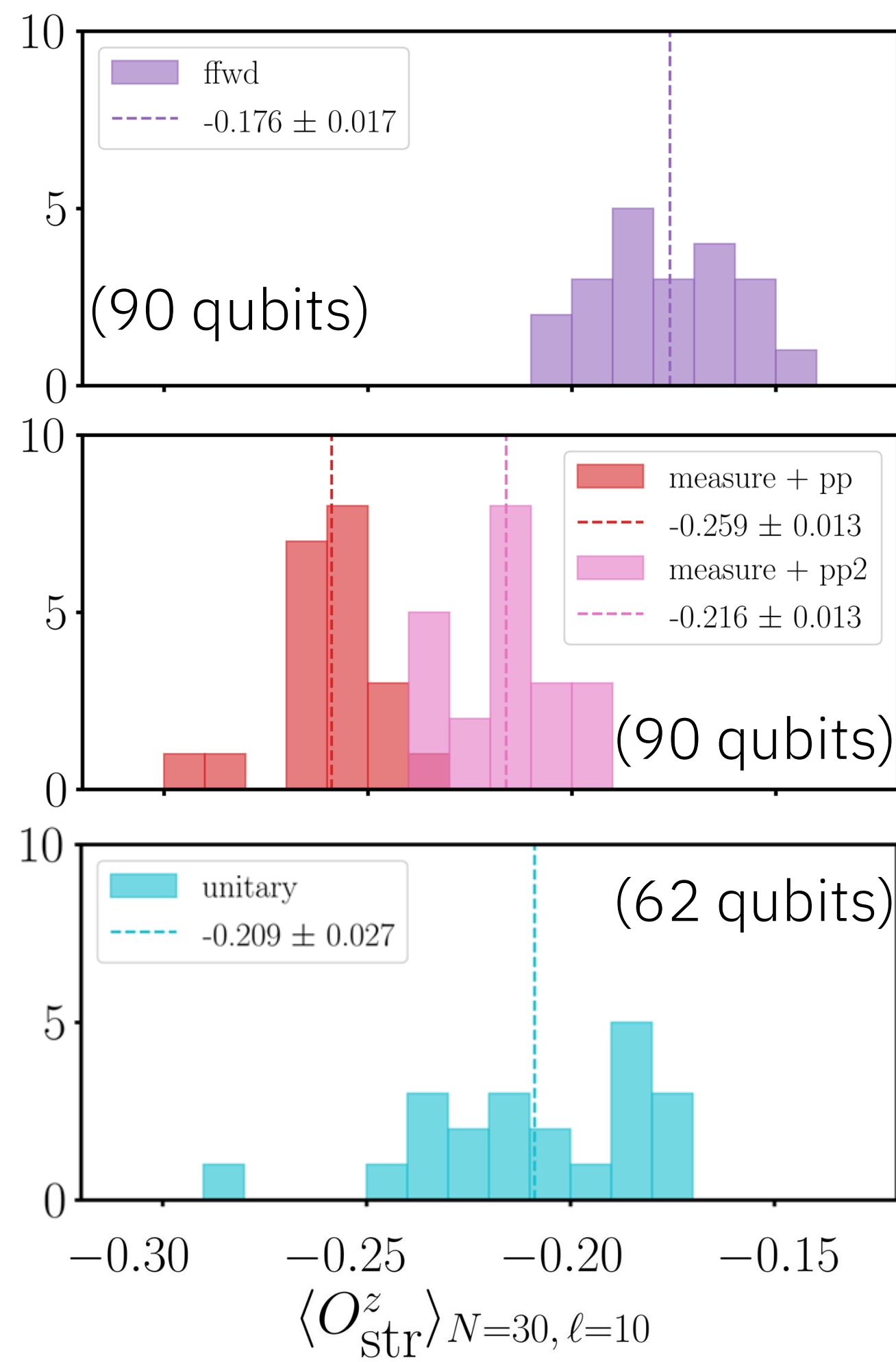
21 x length-10 chains

Validation of AKLT state preparation ($N_{\text{sites}} = 30$)

Dynamic (90 qubits) $l = 10$



Circuit duration: $\sim 8\mu s$

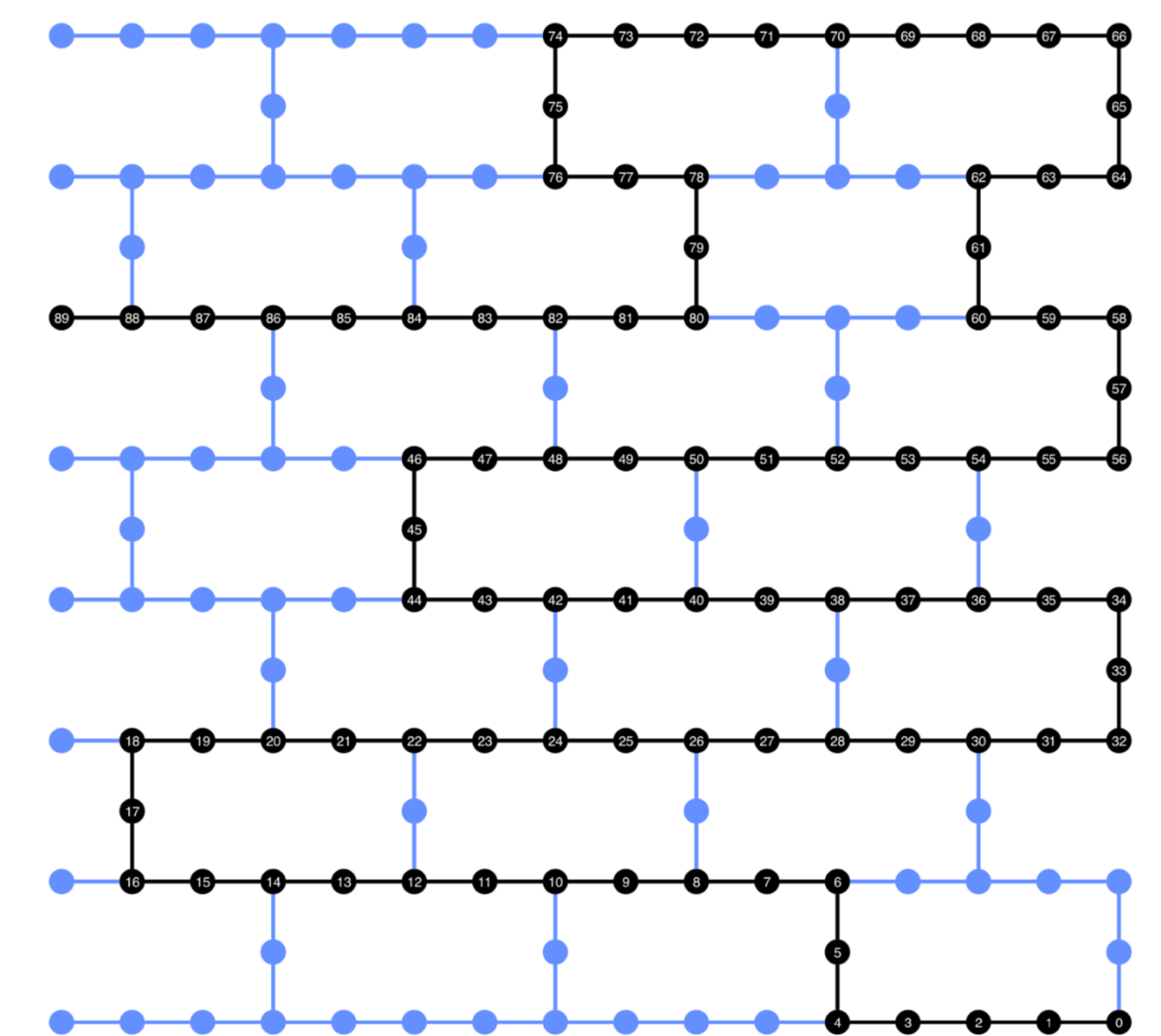


21 x length-10 chains

AKLT state preparation with utility scale dynamic circuits

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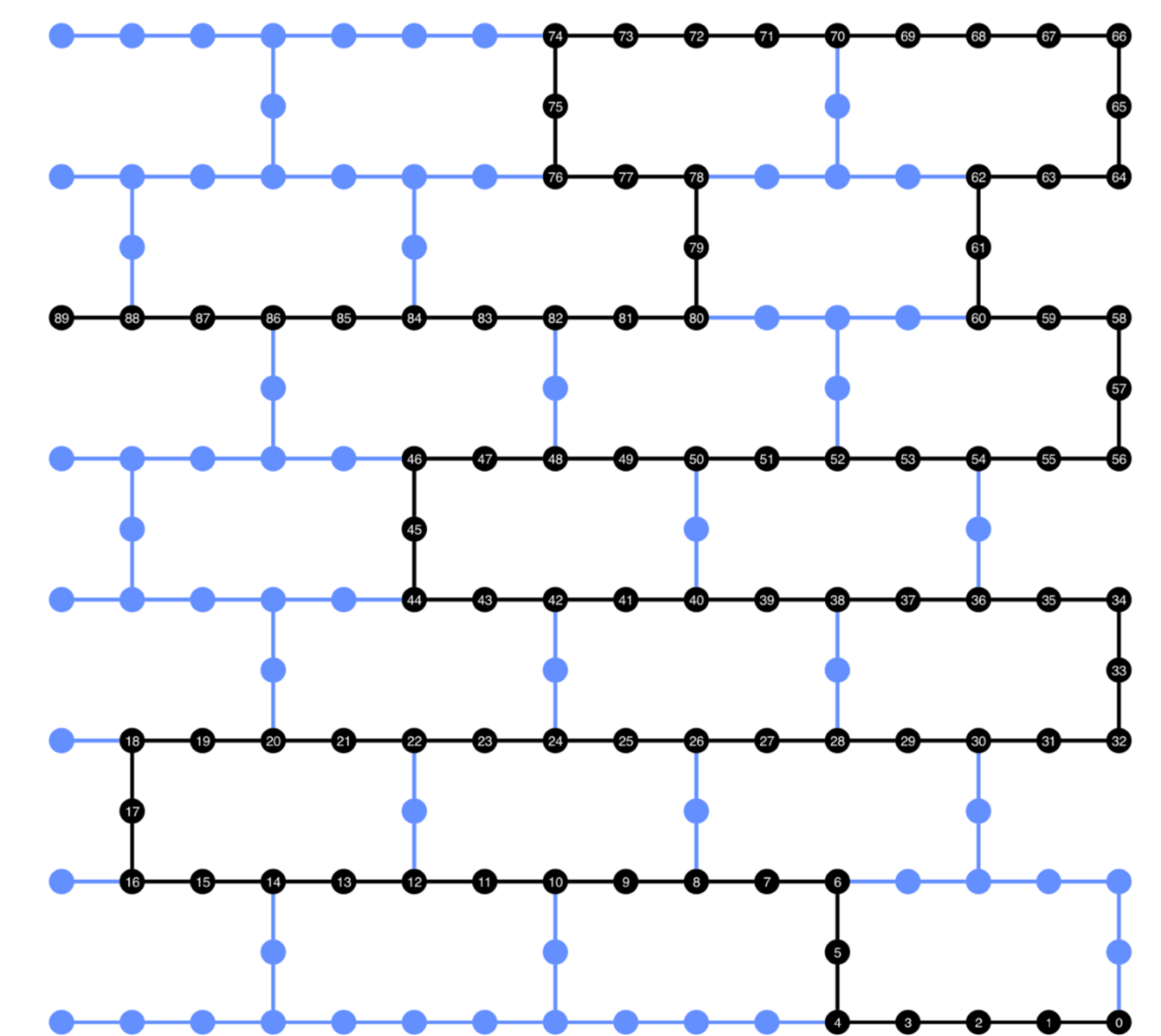
90 qubits for $N_{\text{sites}} = 30$



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90 qubits for $N_{\text{sites}} = 30$



Dynamic AKLT state preparation ($N_{\text{sites}} = 6$)

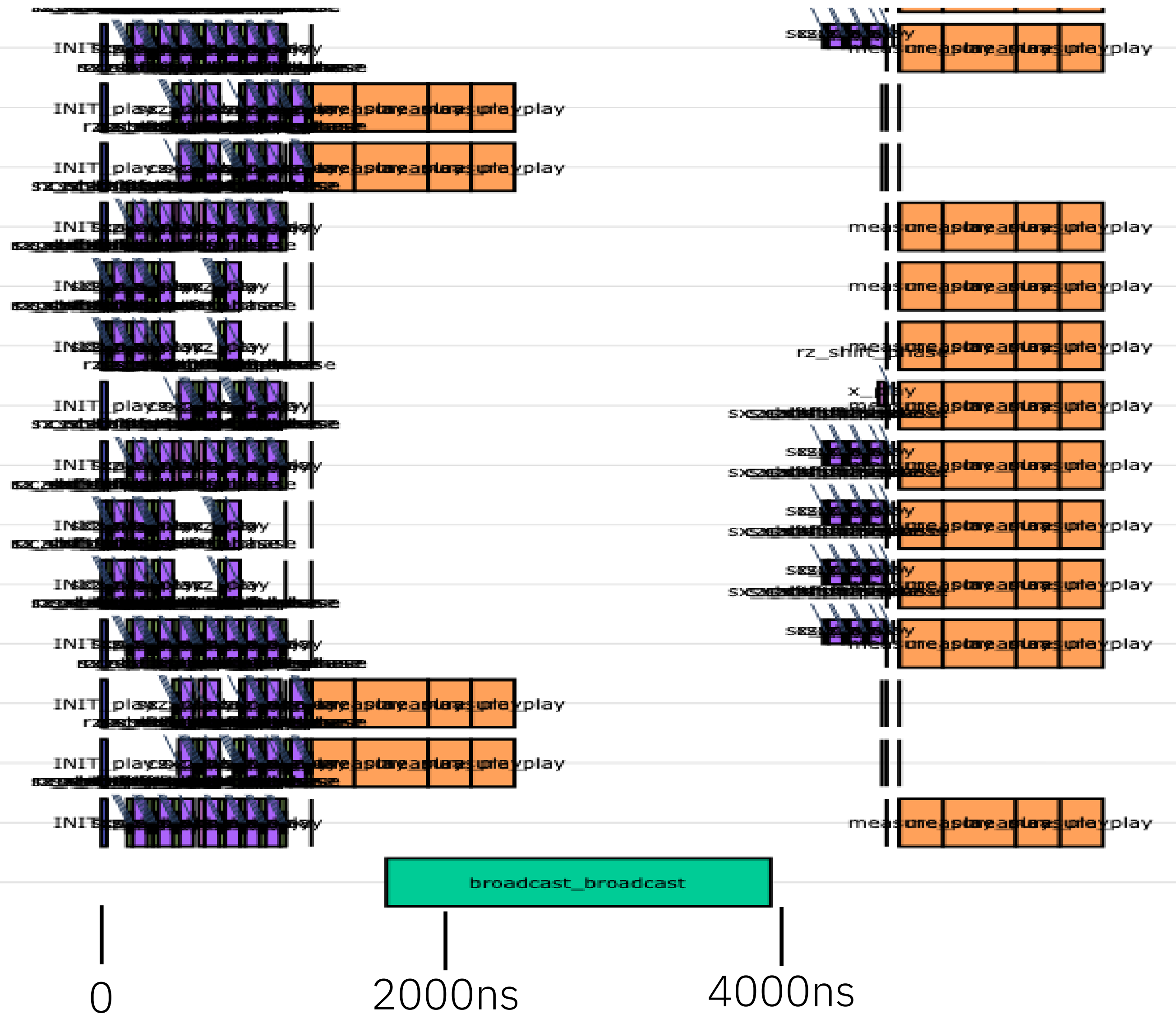


MidCircuitMeasure instruction

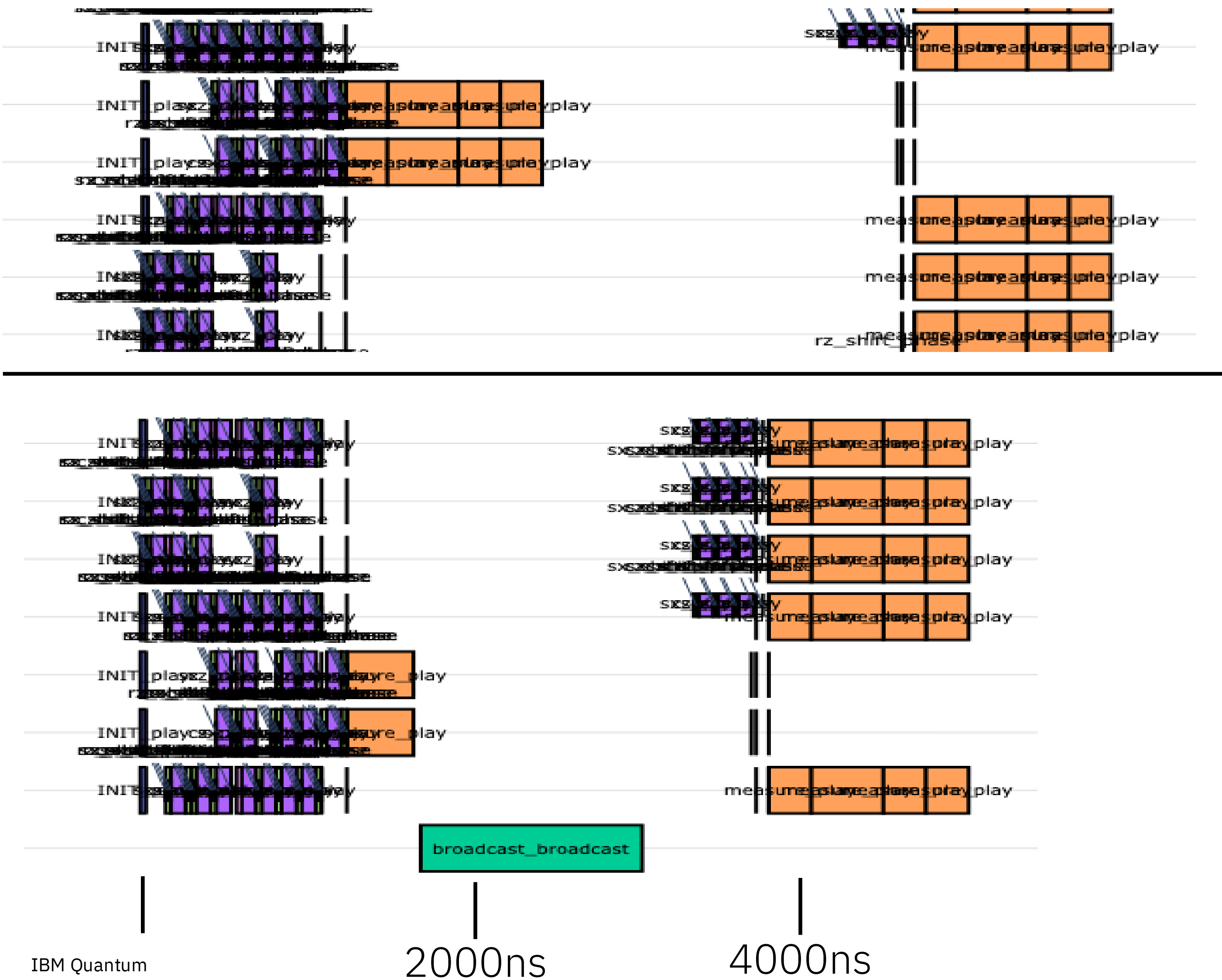
Dynamic with original *measure* scheduled circuit

```
measure
```

Capture length: 1440ns



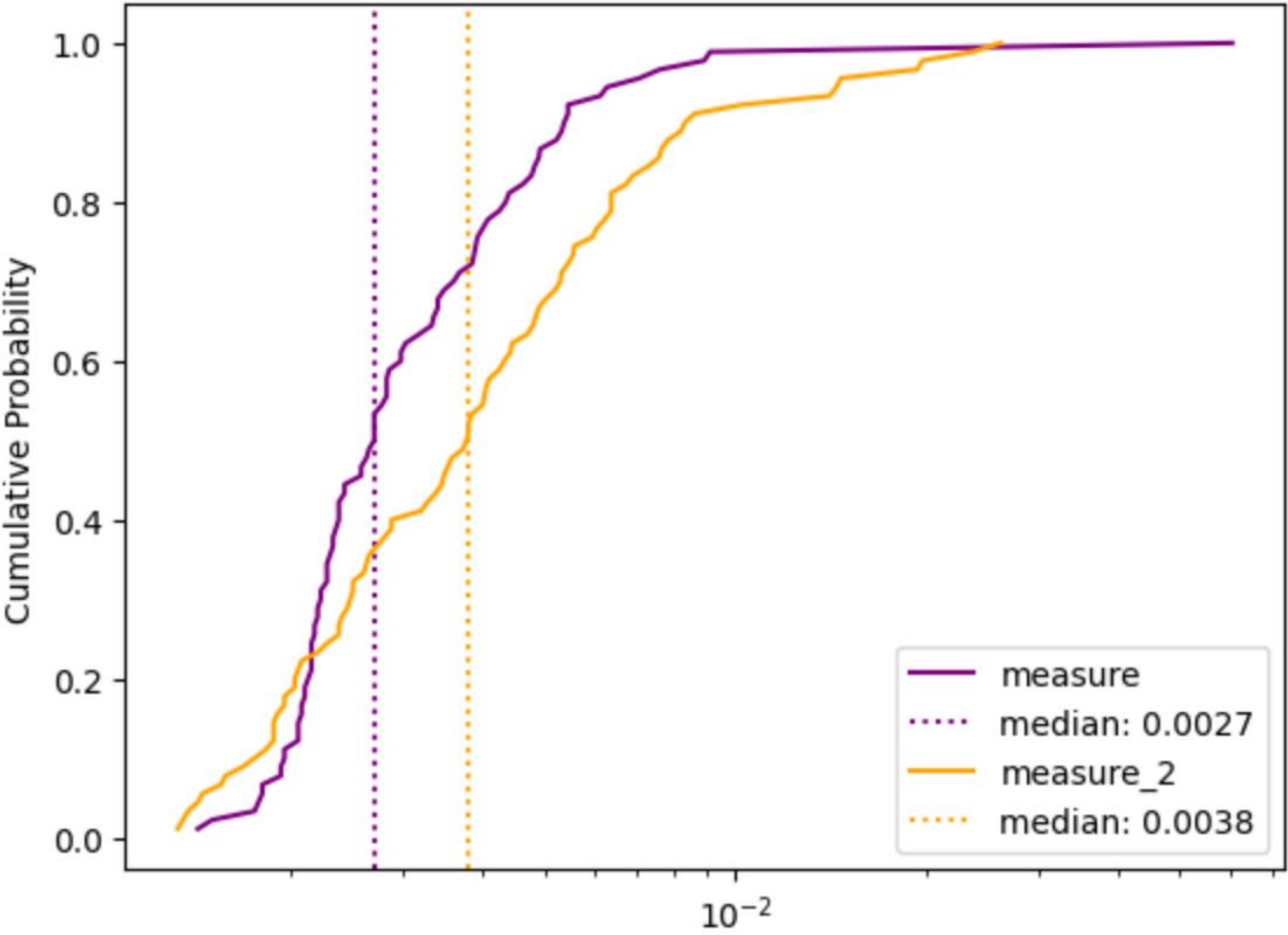
Dynamic with *measure_2* scheduled circuit



● measure

Capture length: 1440ns

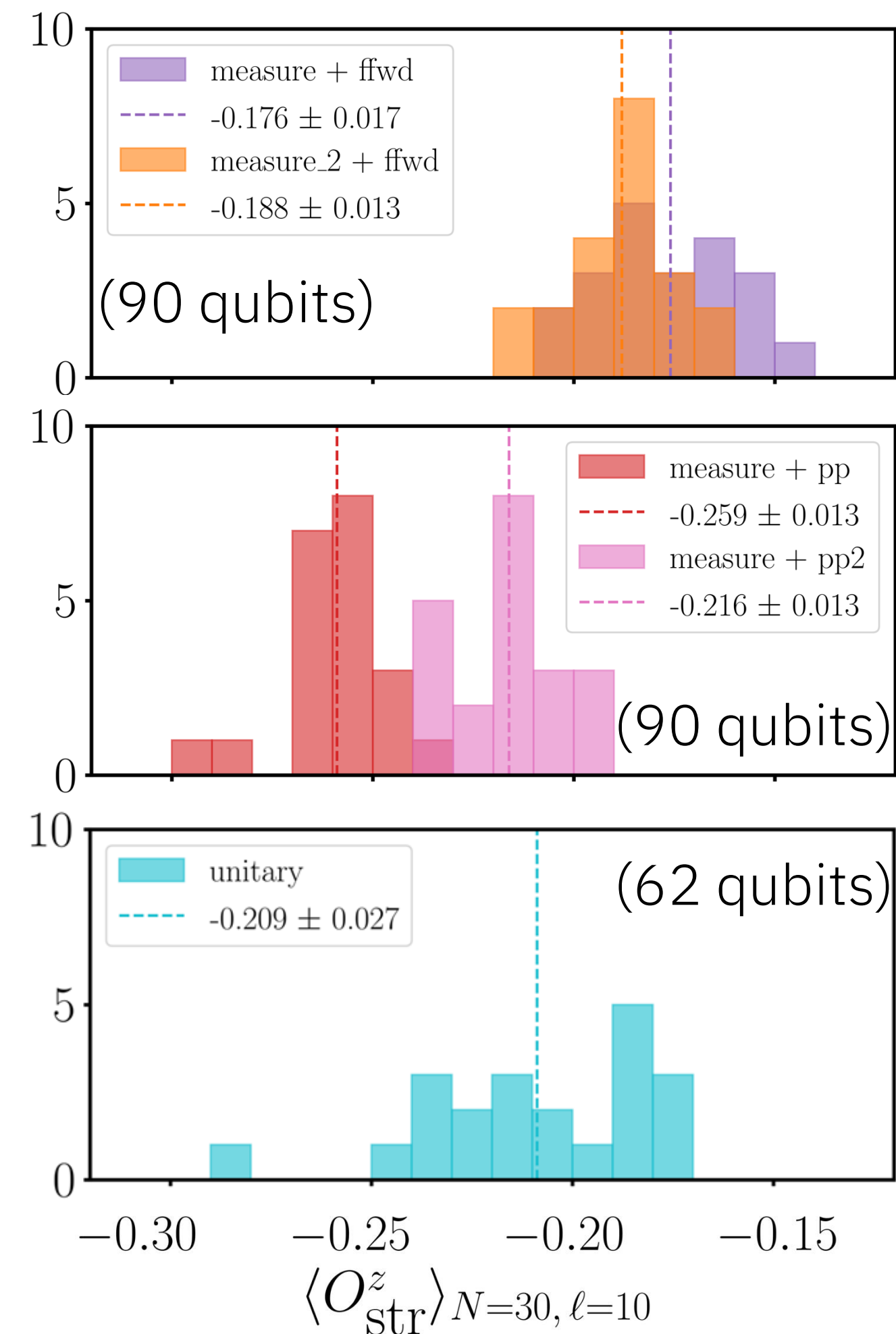
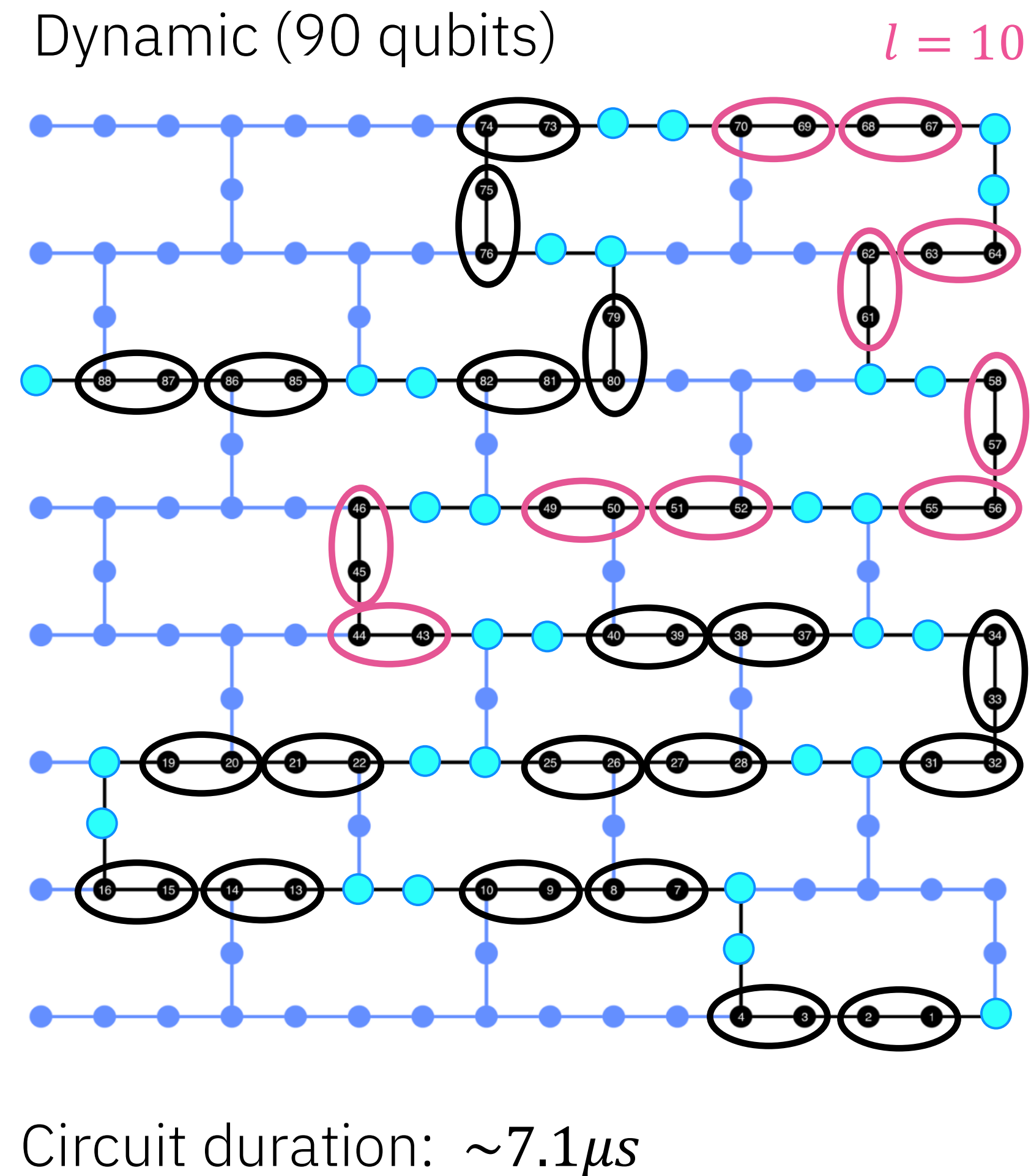
Readout error



● measure_2

Capture length: 500ns

Validation of AKLT state preparation ($N_{\text{sites}} = 30$)

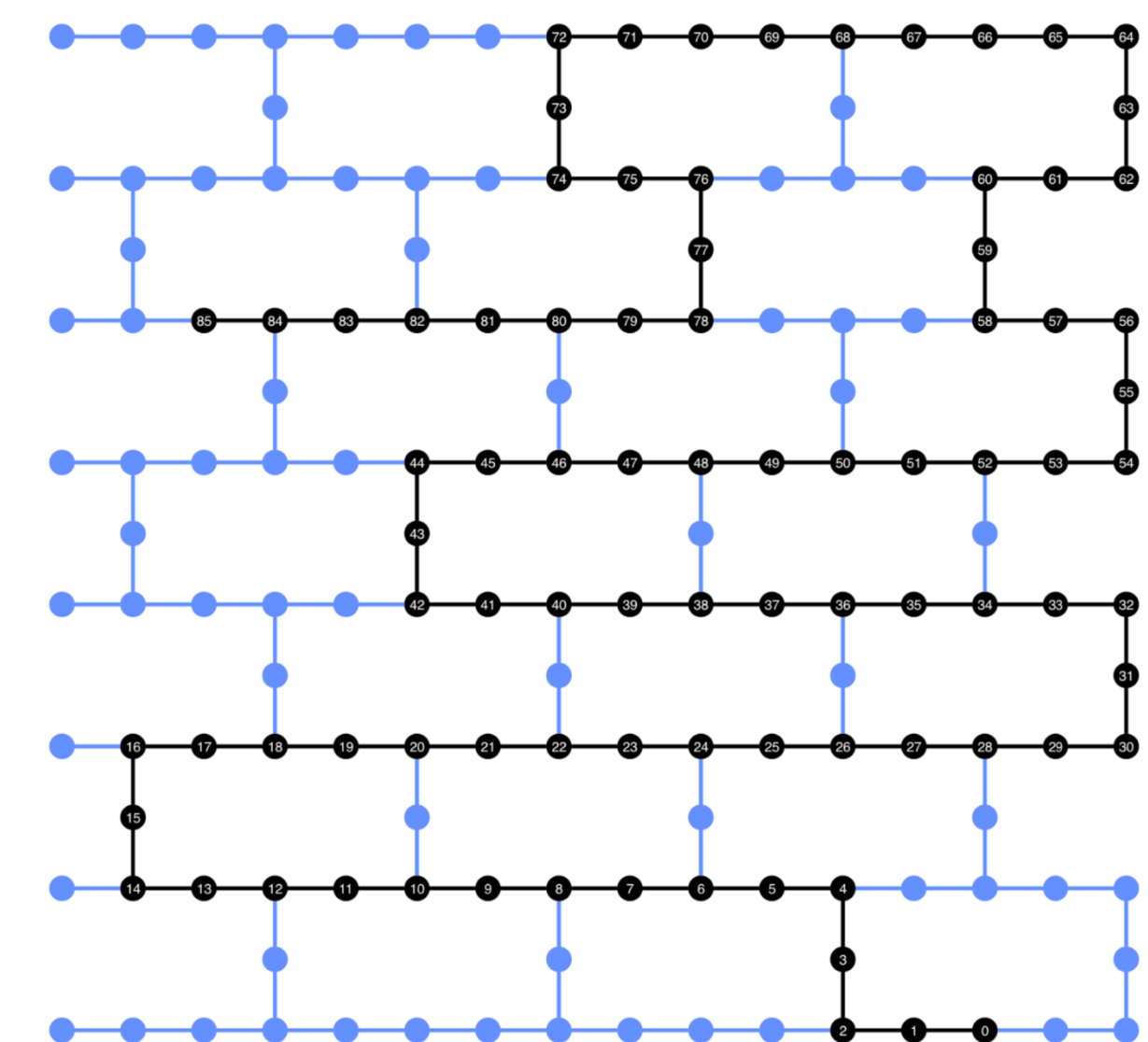


21 x length-10 chains

AKLT state preparation with utility scale dynamic circuits

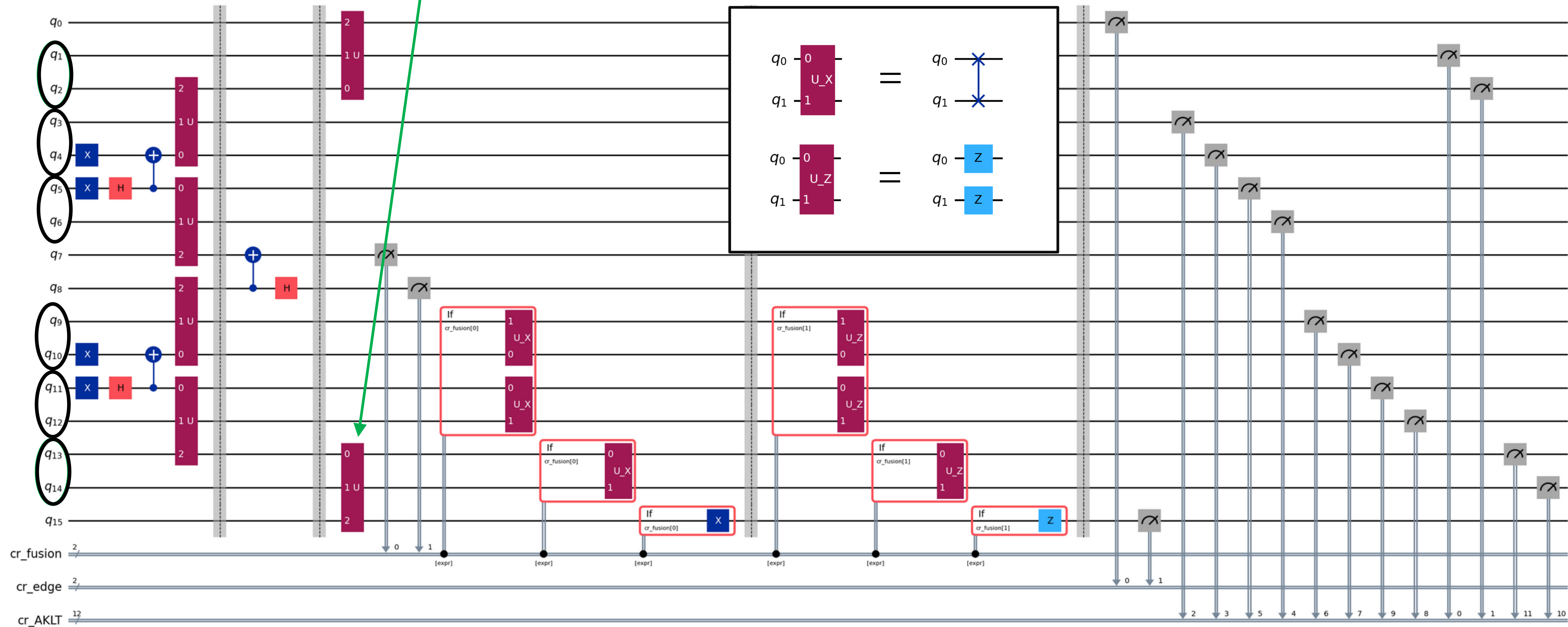
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86 qubits for $N_{\text{sites}} = 30$



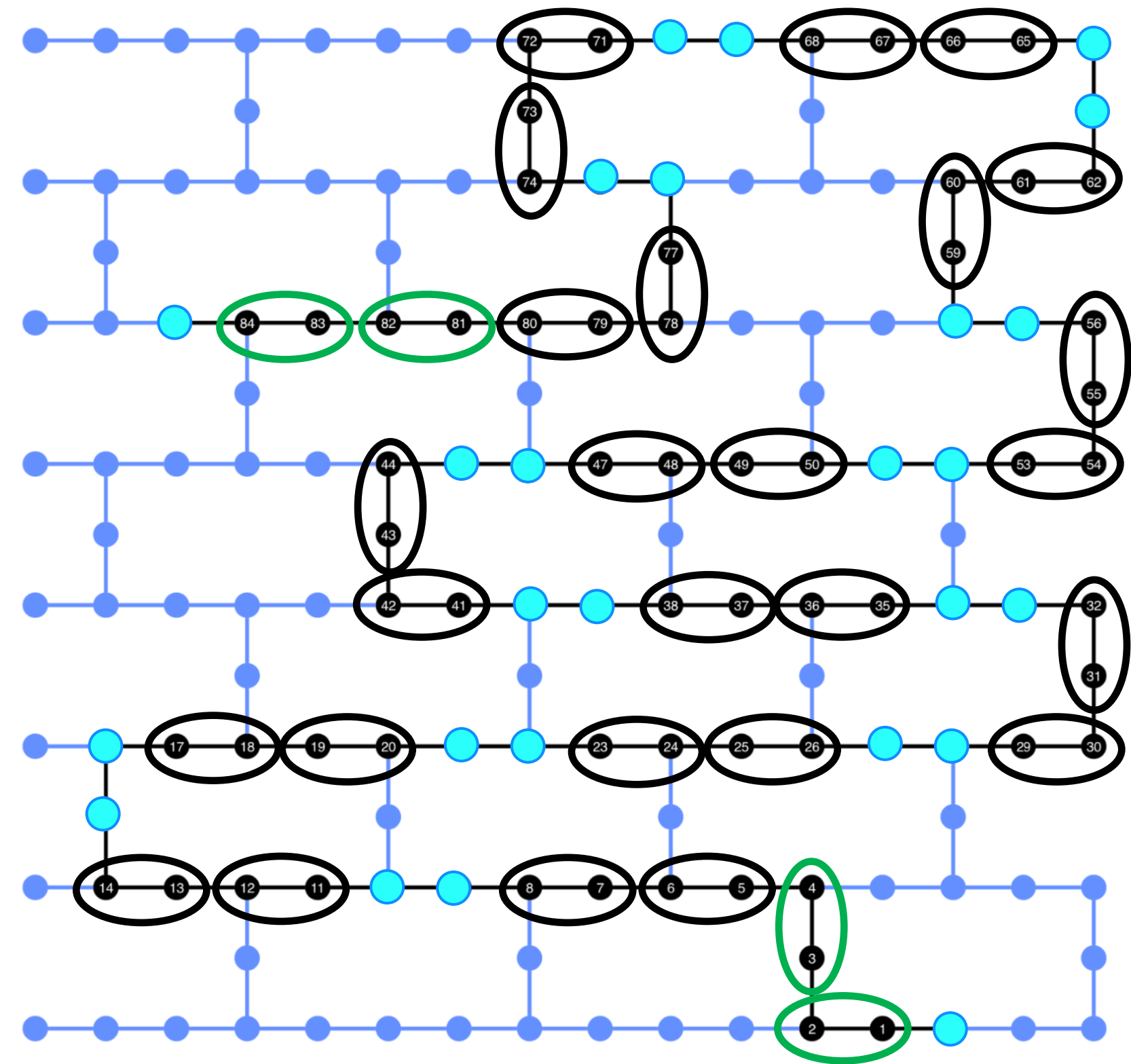
Dynamic ($N_{\text{sites}} = 4$) plus
unitary ($N_{\text{sites}} = 2$) extension

Total number of qubits needed
= $N_{\text{sites}} \times 2 + \text{ancillas}$
= $N_{\text{sites}} \times 2 + (N_{\text{sites}} - 2)$
= 16

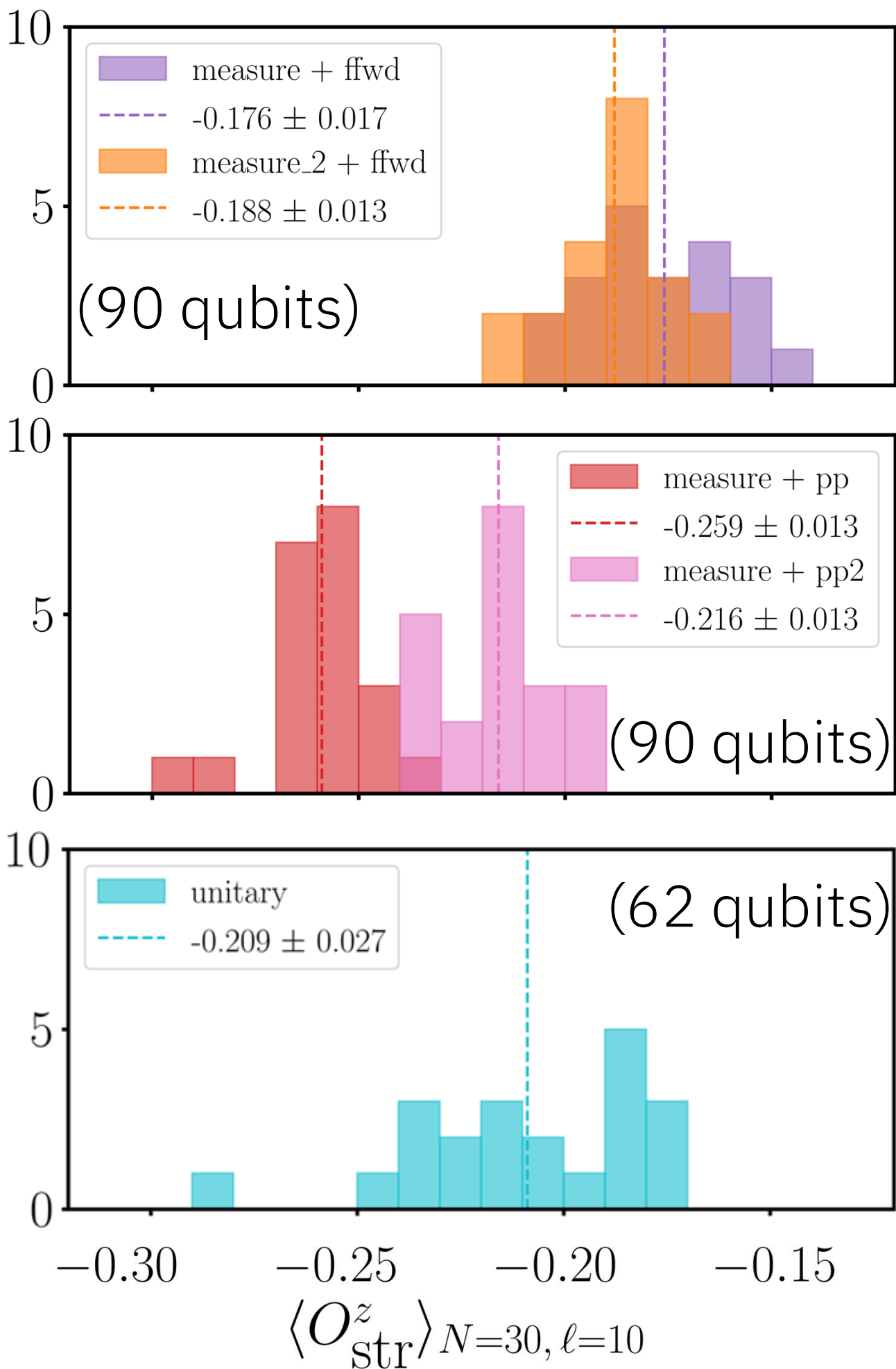


Validation of AKLT state preparation ($N_{\text{sites}} = 30$)

Dynamic + (L2+R2) unitary extension (86 qubits)



Circuit duration: $\sim 6.9\mu\text{s}$

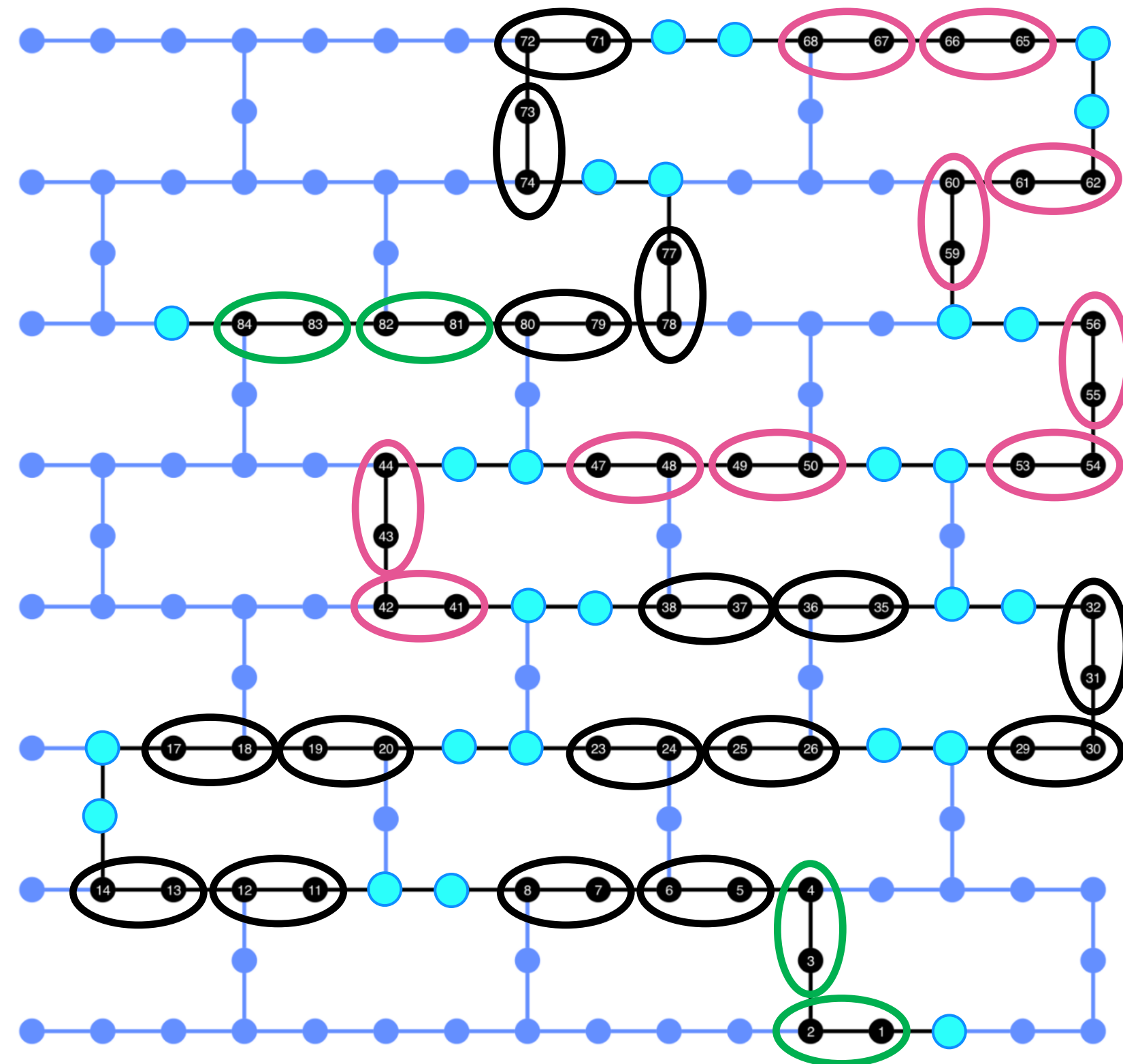


21 x length-10 chains

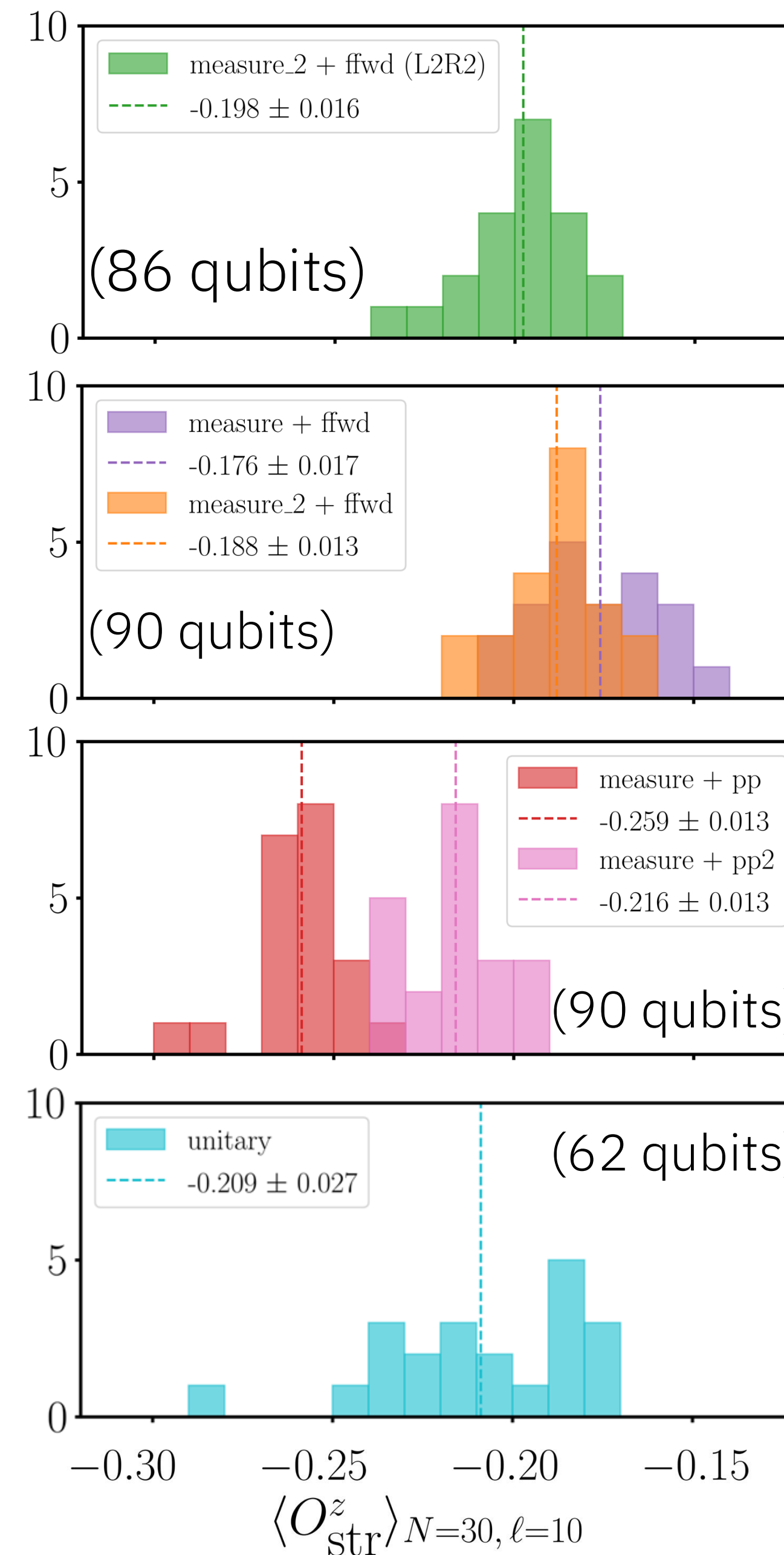
Validation of AKLT state preparation ($N_{\text{sites}} = 30$)

Dynamic + (L2+R2) unitary extension (86 qubits)

$l = 10$

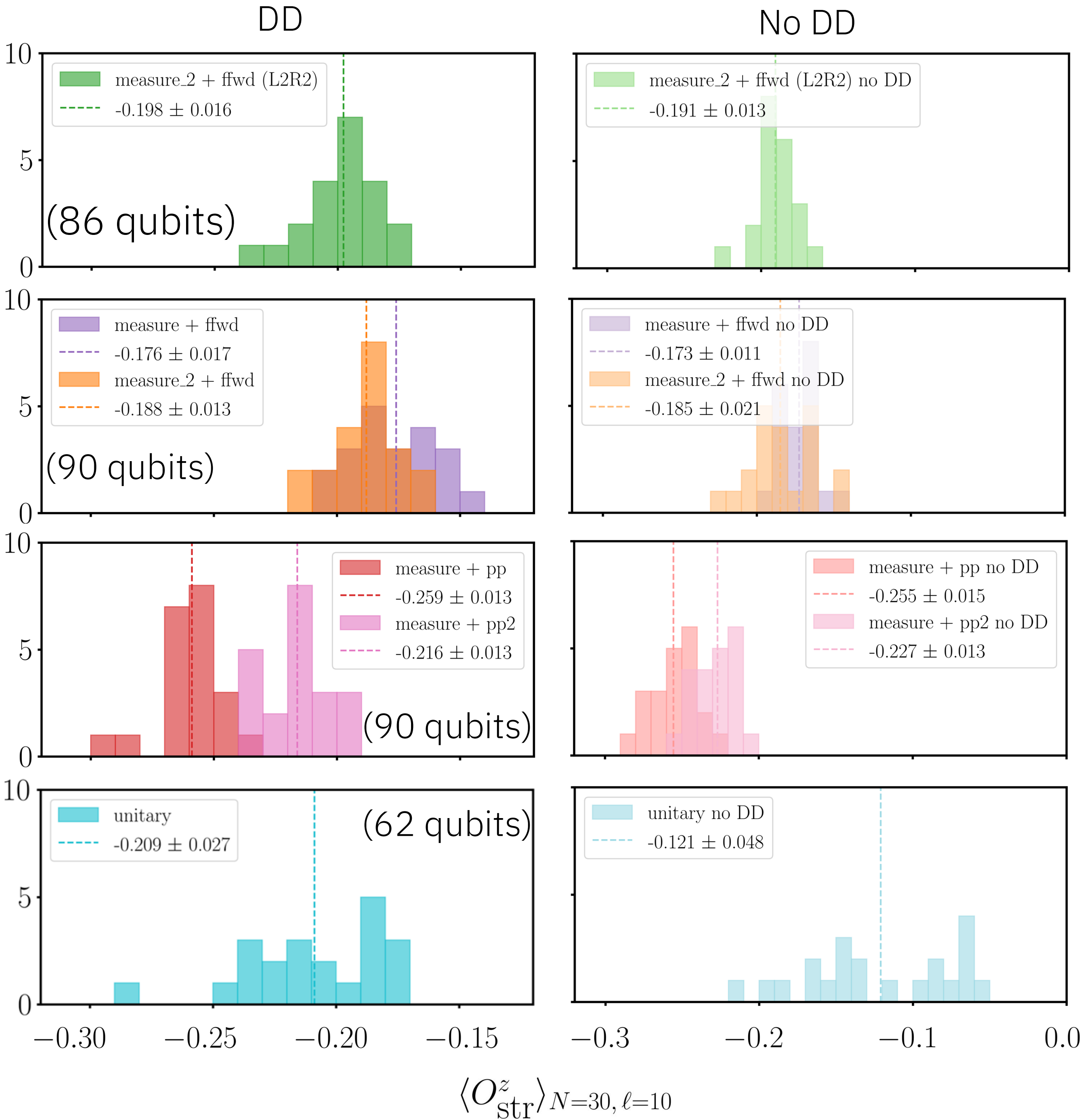


Circuit duration: $\sim 6.9\mu\text{s}$



21 x length-10 chains

DD or no DD???



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($N_{\text{sites}} = 30$)

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Summary

1. Parallel `ifs` enable utility scale experiments
2. More than one type of measurement enabled: `MidCircuitMeasure` instruction that maps to `measure_2`
3. Added `Stretch` as durations for `Delay` instructions for deferred timing resolution
4. Circuit timing visualization (beta) on dynamic circuits returned to users



